

Submitted to *Management Science*

Combating Conservativeness in Data-Driven Optimization under Uncertainty: A Solution Path Approach

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Abstract. In data-driven optimization, solution feasibility is often ensured through a “safe” reformulation of the uncertain constraints, such that an obtained data-driven solution is guaranteed to be feasible for the oracle formulation with high statistical confidence. Such approaches generally involve an implicit estimation of the whole feasible set that can scale rapidly with the problem dimension, in turn leading to over-conservative solutions. In this paper, we investigate a validation-based strategy to avoid set estimation by exploiting the intrinsic low dimensionality among all possible solutions output from a given reformulation. We demonstrate how our obtained solutions satisfy statistical feasibility guarantees with light dimension dependence, and how they are asymptotically optimal and thus regarded as the least conservative with respect to the considered reformulation classes. ~~We apply this strategy to several data-driven optimization paradigms including (distributionally) robust optimization, sample average approximation and scenario optimization. Numerical experiments show encouraging performances of our strategy compared to established benchmarks.~~ Our guarantees depend logarithmically on the number of candidate solutions on the solution path, yielding light dependence on the tuning-parameter dimension rather than the ambient decision dimension. Numerical experiments across various data-driven reformulations show improved objective values at the target feasibility confidence.

Key words: optimization under uncertainty, data-driven optimization, stochastic constraint, chance constraint, validation, dimension reduction

1. Introduction

We focus on optimization with stochastic or probabilistic constraints that, on a high level, can be written in the form

$$\min_x f(x) \text{ subject to } H(x) \geq \gamma \quad (1)$$

where $H(x)$ is an expectation of a random function of the decision variable x . Formulation (1) is ubiquitous in decision-making problems under multiple tradeoffs, where the constraint $H(x) \geq \gamma$ signifies a restriction

on the risk level or resource capacity (e.g., Atlason et al. (2004), Krokhmal et al. (2002)). Moreover, when the random function is an indicator of an event, formulation (1) is a so-called probabilistically constrained or chance-constrained problem (CCP) (Prékopa (2003)). This important formulation posits the decision to tolerate a small probability on catastrophic events such as system failures and big losses, and comprises a primary approach for safe decision-making when facing uncertainty.

We are interested in the situation where the probability distribution governing H is unknown but only observed through data. Finding good solutions under this setting has been studied prominently in the data-driven optimization literature, harnessing various tools from (distributionally) robust optimization (e.g., Bertsimas et al. (2011), Ben-Tal et al. (2009), Wiesemann et al. (2014)) to sample average approximation (e.g., Shapiro et al. (2009)) and scenario optimization (e.g., Campi and Garatti (2008)). From a statistical viewpoint, the problem challenge and the focus of these studies can be cast as a balancing between feasibility and optimality. Due to data noise, feasibility is at best guaranteed with a high statistical confidence, and accounting for this uncertainty incurs a price on the achieved objective value – resulting in conservativeness. This impact on optimality from ensuring feasibility depends heavily on the efficiency in assimilating statistical information into the data-driven formulation. In the following, we first explain how the established estimation frameworks can face severe “looseness” in this regard and lead to over-conservative solutions. This motivates our study that, on a high level, aims to investigate a strategy to substantially tighten the feasibility-optimality tradeoff compared to the previous methods.

1.1. Existing Frameworks and Motivation of Our Approach

To facilitate discussion, suppose for concreteness that the decision variable x lies in a d -dimensional deterministic space $\mathcal{X} \subset \mathbb{R}^d$. Denote $H(x) := \mathbb{E}_F[h(x, \xi)]$ where $\mathbb{E}_F[\cdot]$ is the expectation under $\xi \sim F$, and $h(\cdot, \cdot) : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ is a function of $x \in \mathcal{X}$ controlled by the randomness $\xi \in \mathbb{R}^m$. Also, since our focus is on handling uncertain constraints, we assume that the objective function f is deterministic (this can be relaxed with proper modifications of our subsequent discussion). Suppose we have i.i.d. observations $\xi_1, \dots, \xi_n$.

Let us first consider a natural idea to replace the unknown $H(\cdot)$ with some point estimate, say the sample average $\hat{H}(\cdot) = (1/n) \sum_{i=1}^n h(x, \xi_i)$, in the constraint. Though simple, this approach is typically inadequate to ensure feasibility in any statistical sense. To explain, suppose the “true” optimal solution x^* is at the boundary of the feasible region, i.e., $H(x^*) = \gamma$. If we use $(1/n) \sum_{i=1}^n h(x, \xi_i) \geq \gamma$ as the constraint, then, with significant probability an obtained solution $\hat{x}^*$ (conceivably also at the boundary of the data-driven constraint) can have $H(\hat{x}^*)$ below γ (when $(1/n) \sum_{i=1}^n h(\hat{x}^*, \xi_i) > \mathbb{E}_F[h(\hat{x}^*, \xi)]$), which is infeasible for the original problem. This issue may not arise if x^* or $\hat{x}^*$ is in the interior of the feasible region, but a priori we do not know our decision. In other words, the nature of constrained optimization enforces us to put some “safety” margin in addition to the point estimate, in order to achieve any reasonable confidence in

feasibility. Here, we can plausibly use a data-driven constraint $\hat{H}(x) - \epsilon(x) \geq \gamma$, where $\epsilon(x)$ is a properly chosen positive function such that $H(x) \geq \hat{H}(x) - \epsilon(x)$ for any x with high confidence (such as the scheme in Wang and Ahmed (2008), among others).

We place the above discussion in a more general framework. Let $\mathcal{F}$ be the (unknown) feasible region of (1). Given the data $\xi_1, \dots, \xi_n$, by a valid procedure we mean one that is able to output a solution $\hat{x}^*$ that is truly feasible with a given high confidence level, say $1 - \beta$ (e.g., 95%). That is,

$$P_{data}(\hat{x}^* \in \mathcal{F}) \geq 1 - \beta \quad (2)$$

where P_{data} refers to the probability with respect to the data. By a data-driven reformulation, we mean replacing $\mathcal{F}$ with $\hat{\mathcal{F}}$ that is constructed solely from the data $\xi_1, \dots, \xi_n$. This gives

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to } x \in \hat{\mathcal{F}} \quad (3)$$

which outputs solution $\hat{x}^*$. If we can choose $\hat{\mathcal{F}}$ such that

$$P_{data}(\hat{\mathcal{F}} \subset \mathcal{F}) \geq 1 - \beta \quad (4)$$

then we clearly have (2) since $P_{data}(\hat{x}^* \in \mathcal{F}) \geq P_{data}(\hat{\mathcal{F}} \subset \mathcal{F})$. In the example above, we have used $\hat{\mathcal{F}} = \{x \in \mathcal{X} : \hat{H}(x) - \epsilon(x) \geq \gamma\}$, in the hope that (4) holds in order to achieve (2).

We contend that most approaches in data-driven optimization rely on the above reasoning and are based on (4). In particular, (4) provides a convenient way to certify feasibility, by requiring that *all* solutions feasible for (3) are also feasible for (1) with high confidence. This set-level guarantee generally hinges on a simultaneous estimation task across all x in the decision space $\mathcal{X}$, for which a proper control of the statistical error can lead to a substantial shrinkage of the size of $\hat{\mathcal{F}}$ that exacerbates with problem dimension (either of the decision space or the probability space).

We provide several examples to illustrate the phenomenon above. Some of these examples apply most relevantly to CCP, where $H(x)$ is in the form $\mathbb{P}_F(G(x, \xi) \leq b)$ with $G(x, \xi) : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$.

EXAMPLE 1 (SAMPLE AVERAGE APPROXIMATION (SAA)). In the case of CCP, the SAA approach sets $\hat{\mathcal{F}} = \{x \in \mathcal{X} : \frac{1}{n} \sum_{i=1}^n \mathbf{1}(G(x, \xi_i) + \epsilon \leq b) \geq \gamma + \delta\}$, where ϵ and δ are suitably tuned parameters. For example, when G is Lipschitz continuous in x , selecting $\delta = \Omega(\sqrt{(d/n) \log(1/\epsilon)})$ can guarantee (4) (Luedtke and Ahmed (2008)), and similar relations also hold in discrete decision space (Luedtke and Ahmed (2008)) and expected value constraints (Wang and Ahmed (2008)). These estimates come from concentration inequalities in which union bounds are needed and give rise to the dependence on the dimension d . Note that the resulting margin δ scales in order $\sqrt{d}$, and to get any reasonably small δ , n must be of higher order than d . $\square$

EXAMPLE 2 (ROBUST OPTIMIZATION (RO) AND SAFE CONVEX APPROXIMATION (SCA)).

Focusing on CCP, RO sets

$$\hat{\mathcal{F}} = \{x \in \mathcal{X} : G(x, \xi) \leq b, \text{ for all } \xi \in \mathcal{U}\} \quad (5)$$

where $\mathcal{U}$ is known as the uncertainty set, and ξ in (5) is viewed as a deterministic unknown (Bertsimas et al. (2011), Ben-Tal et al. (2009)). A common example of $\mathcal{U}$ is an ellipsoidal set $\{\xi : (\xi - \hat{\mu})' \hat{\Sigma}^{-1} (\xi - \hat{\mu}) \leq \rho\}$ where $\hat{\mu} \in \mathbb{R}^d$, $\hat{\Sigma} \in \mathbb{R}^{d \times d}$ a positive semidefinite matrix, and $\rho \in \mathbb{R}$. Here the center $\hat{\mu}$ and shape $\hat{\Sigma}$ typically correspond to the mean and covariance of the data, and ρ controls the set size. A duality argument shows that, in the case of linear chance constraint in the form $G(x, \xi) = x' \xi$, (5) is equivalent to the quadratic constraint $\hat{\mu}' x + \sqrt{\rho} \|\hat{\Sigma}^{1/2} x\|_2 \leq b$. Using such type of convex constraints as inner approximations for intractable chance constraints is also known as SCA (e.g., Ben-Tal and Nemirovski (2000), Nemirovski (2003), Nemirovski and Shapiro (2006)).

It is known that, if for instance the random variable ξ has a known bounded support, the above approach guarantees an obtained solution has a satisfaction probability of order $1 - e^{-\rho/2}$ via Hoeffding's inequality, and ρ is chosen by matching this expression with the tolerance level γ . Although ρ calibrated this way may not explicitly depend on the problem dimension, its tightness varies heavily based on problem instance (due to the worst-case nature of concentration bounds), and its validity relies on a priori distributional information (e.g., support) rather than an efficient utilization of data. Another viewpoint that has been taken recently in data-driven RO (Bertsimas et al. (2018); Tulabandhula and Rudin (2014); Goldfarb and Iyengar (2003); Hong et al. (2017)) is to select $\mathcal{U}$ to be a set that contains γ -content of the distribution of ξ , i.e., $\mathbb{P}_F(\xi \in \mathcal{U}) \geq \gamma$, with a confidence level $1 - \beta$. In this case, any solution $\hat{x}$ feasible for (5) would satisfy $\mathbb{P}_F(G(\hat{x}, \xi) \leq b) \geq \mathbb{P}_F(\xi \in \mathcal{U}) \geq \gamma$ with at least $1 - \beta$ confidence, thus achieving (4) as well. Such generated uncertainty set however typically has a size that scales with the dimension of the probability space. For example, consider $G(x, \xi) = x' \xi$ with $\xi \in \mathbb{R}^m$ being standard multivariate Gaussian and the uncertainty set $\mathcal{U}$ is an ellipsoid with $\hat{\mu}$ and $\hat{\Sigma}$ being the true mean and covariance, i.e., $\mathcal{U} = \{\xi \in \mathbb{R}^m : \|\xi\|_2^2 \leq \rho\}$. Then, in order to make $\mathcal{U}$ a γ -content set the radius ρ has to be at least of order m since $\|\xi\|_2^2$ has a mean m , resulting in the robust counterpart $\sqrt{\rho} \|x\|_2 = \Theta(\sqrt{m}) \|x\|_2 \leq b$. However, the exact chance constraint in this case can be rewritten as $z_\gamma \|x\|_2 \leq b$, where z_γ is the γ -quantile of the univariate standard normal, which is independent of the dimension. $\square$

EXAMPLE 3 (DISTRIBUTIONALLY ROBUST OPTIMIZATION (DRO)). DRO sets

$$\hat{\mathcal{F}} = \{x \in \mathcal{X} : \inf_{Q \in \mathcal{U}} \mathbb{E}_Q[h(x, \xi)] \geq \gamma\} \quad (6)$$

where $\mathcal{U}$ is a set in the space of probability measures that is constructed from data, and is often known as the ambiguity set or uncertainty set. The rationale here is similar to RO, but views the uncertainty in terms of the distribution. If $\mathcal{U}$ is constructed such that it contains the true distribution F with high confidence, i.e.,

$P_{data}(F \in \mathcal{U}) \geq 1 - \beta$, then any solution $\hat{x}$ feasible for the DRO constraint (6) would satisfy $\mathbb{P}_F(G(\hat{x}, \xi) \leq b) \geq \gamma$ with at least $1 - \beta$ confidence so that (4) holds.

Popular choices of $\mathcal{U}$ include moment sets, i.e., specifying the moments of Q (to be within a range for instance) (Ghaoui et al. (2003); Delage and Ye (2010); Xu et al. (2012); Wiesemann et al. (2014); Goh and Sim (2010); Natarajan et al. (2008); Van Parys et al. (2016); Doan et al. (2015); Hanasusanto et al. (2015)), and distance-based sets, i.e., specifying Q in the neighborhood ball surrounding a baseline distribution, where the ball size is measured by a statistical distance such as ϕ -divergence (Petersen et al. (2000); Ben-Tal et al. (2013); Glasserman and Xu (2014); Lam (2016); Lam (2018); Hu and Hong (2013); Jiang and Guan (2016); Gotoh et al. (2018); Dupuis et al. (2016); Bayraksan and Love (2015)) or Wasserstein distance (Esfahani and Kuhn (2018); Blanchet and Murthy (2019); Gao and Kleywegt (2016); Xie (2018)).

Ensuring $P_{data}(F \in \mathcal{U}) \geq 1 - \beta$ means that $\mathcal{U}$ is a confidence region for F . In the moment set case, this boils down to finding confidence regions for the moments whose sizes in general scale with the probability space dimension. To explain, when only the mean $\mathbb{E}_F[\xi]$ is estimated, the confidence region constructed from, say the delta method (Marandi et al. (2019)), takes the form $\{\hat{\mu} + \hat{\Sigma}^{\frac{1}{2}}v : v \in \mathbb{R}^m, \|v\|_2^2 \leq \chi_{m,1-\beta}^2\}$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the sample mean and covariance and $\chi_{m,1-\beta}^2$ (which is of order m) is the $1 - \beta$ quantile of the χ^2 distribution with degree of freedom m , therefore the diameter of the confidence region scales as $\sqrt{m}$. When the mean and covariance are jointly estimated, the dimension dependence scales up further. In the distance-based set case, one needs to estimate statistical distances. If the Wasserstein distance is used to construct the ball surrounding the empirical distribution, results from measure concentration (Fournier and Guillin (2015)) indicate that the ball size needs to be of order $n^{-\frac{1}{m}}$ to ensure $P_{data}(F \in \mathcal{U}) \geq 1 - \beta$. Alternatively, if $\mathcal{U}$ is constructed as a ϕ -divergence ball surrounding some nonparametric kernel-type density estimate, results from kernel density estimation (see Section 4.3 in Wand and Jones (1994)) suggest that the estimation error is of order $n^{-\frac{4}{m+4}}$. In either case, the required size of the uncertainty set exhibits exponential dependence on the dimension. Recently, the empirical or the profile likelihood method has also been proposed to calibrate the ball size such that $\mathcal{U}$ can be (much) smaller than what is needed in being a confidence region for F , while at the same time (4) still holds (Lam and Zhou (2017); Duchi et al. (2016); Lam (2019); Blanchet and Kang (2016)). However, the ball size in this approach scales as the supremum of a so-called χ^2 -process over the decision space (e.g., Lam (2019)). An analysis using metric entropy (e.g., Example 2 in Section 14 in Lifshits (2013)) shows that the χ^2 -process supremum can scale linearly in the decision space dimension d , a much better but still considerable dependence on the dimension. $\square$

Finally, we discuss the only two exceptional paradigms, to our best knowledge, in providing guarantee (2) using (4). First, Gupta (2019) studies a Bayesian framework to define feasibility guarantees for (stochastic) constraints with unknown parameters, focusing on DRO formulations. The idea is to ensure the obtained data-driven solution satisfies the constraints with a high posterior probability on the unknown parameters. This definition of feasibility does not utilize the concept of experimental repetitions in the frequentist sense

as we have considered, but views the unknown parameters as random and considers the frequency of feasibility from the posterior belief, thus bypassing the set-level guarantee in (4). Indeed, Gupta (2019) shows that under suitable convexity assumption on the constraints (with respect to the unknown parameter) and discreteness of the underlying distribution, the size of the uncertainty set in DRO can be chosen lightly dependent on the problem dimension.

The second exceptional paradigm that we are aware of is scenario optimization (SO) (e.g., Calafiore and Campi (2005), Campi and Garatti (2008)), which applies to the case of CCP. In its basic form, this approach sets

$$\hat{\mathcal{F}} = \{x \in \mathcal{X} : G(x, \xi_i) \leq b \text{ for all } i = 1, \dots, n\}$$

i.e., using sampled constraints formed from the data. As the number of constraints increases, $\hat{\mathcal{F}}$ is postulated to populate the decision space in some sense and ensure the obtained solution $\hat{x}^*$ lies in $\mathcal{F}$. While the sample size required in the basic SO is linear in the decision dimension d , recent works reduce this dependence by an array of generalizations, including using regularization (Campi and Carè (2013)), tighter support rank estimates (Schildbach et al. (2013), Campi and Garatti (2018)) and sequential and validation-based schemes (Carè et al. (2014), Calafiore (2017)).

The approach that we propose in this paper aims to avoid using the set-level guarantee in (4) and the need to control its simultaneous estimation errors, which can cause over-conservativeness as discussed. Our approach operates under a frequentist framework, nonparametric assumptions on the underlying distributions, and applies to all the exemplified methods mentioned above (SAA, RO, DRO and SO). It is thus different from the Bayesian parametric framework in Gupta (2019). Our idea is closest to some of the validation-type schemes suggested for SO, but more general as it applies to stochastic constraints beyond CCP and to data-driven reformulations beyond SO. Akin to these SO studies, our main results concern the power of our validation procedures in guaranteeing feasibility, which informs the required sample size in relation to the problem dimension. Our results also introduce a notion of optimality with respect to the chosen reformulation class, and deduce joint optimality-feasibility guarantees. In these regards, one main contribution of our work can be viewed as a rigorous construction of the first general-purpose validation framework for data-driven constrained optimization to systematically reduce conservativeness.

1.2. Contributions and Organization

The main contributions of this paper are the following.

- We develop a rigorous construction of the first general-purpose validation framework for data-driven constrained optimization to systematically reduce conservativeness. The approach that we propose in this paper aims to avoid using the set-level guarantee in (4) and the need to control its simultaneous estimation errors, which can cause over-conservativeness. It operates under a frequentist framework, nonparametric assumptions on the underlying distributions, and applies to all the exemplified methods mentioned above

(SAA, RO, DRO and SO). It is thus different from the Bayesian parametric framework in Gupta (2019). Our approach is closest to some of the validation-type schemes suggested for SO, but more general as it applies to stochastic constraints beyond CCP and to data-driven reformulations beyond SO.

- We show that solutions chosen by our two-phase framework have asymptotic and finite-sample performance guarantees on feasibility, as well as asymptotic optimality within their reformulation class, under assumptions that are satisfied by typical data-driven reformulations of SAA, RO, DRO, and SO. Akin to SO studies, our main results concern the power of our validation procedures in guaranteeing feasibility, which inform the required sample size in relation to the problem dimension. Our results also introduce a notion of optimality with respect to the chosen reformulation class, and deduce joint optimality-feasibility guarantees.

- We present computational experiments which demonstrate that our framework provides better solutions than several benchmarks, in terms of objective values and tighter feasibility confidence, for different data-driven reformulations. We also compare our framework against other validation methods, such as cross-validation, across different reformulations and show that our framework performs similarly, if not better, while enjoying performance guarantees. These results shed light into... for the decision-maker.

The rest of the paper is organized as follows. Section 2 presents the two-phase solution-path framework. Section 3 develops the normalized Gaussian-supremum validator and its statistical guarantees. Section 5 illustrates how to instantiate the framework for several reformulation classes. Section 6 reports numerical experiments, and Section ?? concludes. The e-companion contains proofs and additional validator variants.

2. Overview of Our Framework and Rationale

Our framework, as discussed, aims to bypass the set-level guarantee in (4) and the need to control its simultaneous estimation errors. Our starting observation is the following. In all the described approaches above, the data-driven reformulation involves a key parameter that controls the level of conservativeness:

1. SAA: safety margin δ
2. RO and SCA: uncertainty set size ρ
3. DRO: divergence ball size or moment set size
4. SO: number of constraints

These parameters have the properties that setting it to one extreme (e.g., 0) would signal no uncertainty in the formulation, leading to a solution very likely infeasible, while setting it to another extreme (e.g., ∞) would cover the entire decision space, leading to a solution that is very conservative. In the established approaches, the parameter value is chosen to ensure (4), which tend to locate towards the latter extreme.

On the other hand, given a specific data-driven reformulation, it is easy to see that no matter how we choose this “conservativeness” parameter, the solution must lie in a low-dimensional manifold. More precisely, denote a given data-driven reformulation as

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to } x \in \hat{\mathcal{F}}(s) \quad (7)$$

where $s \in S$ denotes the conservativeness parameter, and we highlight the dependence of the data-driven feasible region $\hat{\mathcal{F}}(s)$ on s . We denote the obtained solution from (7) as $x^*(s)$. The *solution path* $\{x^*(s) : s \in S\}$ contains all possible obtainable solutions from the data-driven reformulation (7). Intuitively, any statement on feasibility suffices to focus on this solution path, instead of the whole decision space.

Nonetheless, besides the conservativeness parameter, a data-driven reformulation could have other parameters playing various roles (e.g., center and shape of an ellipsoidal uncertainty set in RO, baseline distribution in distance-based DRO etc.). The flexibility of these parameter values can enlarge the obtainable solution space and elevate its dimensionality. Suppose we want to contain this enlargement, and at the same time be able to select the optimal candidate within the low-dimensional manifold $\{x^*(s) : s \in S\}$. We propose the following two-phase framework to achieve this rigorously.

Algorithm 1 The Two-Phase Framework

Input: data $\xi_{1:n} = \{\xi_1, \dots, \xi_n\}$; numbers of data n_1, n_2 allocated to each phase ($n_1 + n_2 = n$); a confidence level $1 - \beta$; a given method to construct data-driven reformulation with a (possibly multi-dimensional) parameter $s \in S$; a discrete mesh $\{s_1, s_2, \dots, s_p\} \subseteq S$.

Phase one:

1. Use n_1 observations, which we index as $\{\xi_{n_2+1}, \dots, \xi_n\}$ for convenience, to construct the data-driven reformulation $OPT(s)$ in the form (7) parameterized by $s \in S$.
2. For each $j = 1, \dots, p$, compute the optimal solution $x^*(s_j)$ of $OPT(s_j)$.

Phase two:

Use a validator V to select $(\hat{s}^*, x^*(\hat{s}^*)) = V(\{\xi_1, \dots, \xi_{n_2}\}, \{x^*(s_1), \dots, x^*(s_p)\}, 1 - \beta)$, where $x^*(\hat{s}^*)$ is a solution and $\hat{s}^*$ is the associated parameter value.

Output: $x^*(\hat{s}^*)$.

Our procedure (Algorithm 1) splits the data into two groups. With the first group of data, we construct a given data-driven reformulation parametrized by a conservativeness parameter s that varies over a space S , which we call $OPT(s)$. We obtain the optimal solution $x^*(s)$ for a range of values $s = s_j, j = 1, \dots, p$. This step assumes the availability of an efficient solver for $OPT(s)$. Next, the second group of data is fed into a *validator* V that aims to identify the best feasible solution $x^*(\hat{s}^*)$ among $\{x^*(s_j) : j = 1, \dots, p\}$. The number of points p required to validate depends on the size of S , which is constructed to be low-dimensional. There are multiple ways to set up the validator V , each with its own benefits and requirements. In the next two sections, we will introduce two classes of validators, one we call *Gaussian supremum validator* (Section 3), and another one we call *univariate Gaussian validator* (Section 4). We will present their rationales, theoretical-statistical guarantees, and implications on the feasibility and optimality of the obtained solution.

Section 5 will then tie back the applicability of these validators to the exemplified approaches in Section 4.1.

Our procedure (Algorithm 1) splits the data into two groups. With the first group of data, we construct a chosen data-driven reformulation parametrized by a conservativeness parameter $s \in S$, which we denote by $OPT(s)$, and compute candidate solutions $x^*(s_j)$ on a discretized mesh $s = s_j$, $j = 1, \dots, p$. Next, the second group of data is fed into a *validator* that selects the best (lowest objective) candidate that is certified feasible with confidence level $1 - \beta$. The key point is that the number of candidate points p required for validation depends on the dimension of S , which is constructed to be low-dimensional. While there are multiple ways to set up the validator V , we focus on a single one: a *normalized Gaussian-supremum* margin calibrated over the finite candidate set (Section 3). Additional variants of V —along with their rationales, theoretical statistical guarantees, and implications on the feasibility and optimality of the obtained solution—are provided in the e-companion.

3. Validation via Multivariate Gaussian Supremum

Our first validator uses a simultaneous estimation of $H(x)$ in the constraint in (1) to assess feasibility over the discretized solution path of $x^*(s)$. More precisely, given the solution set $\{x^*(s_j) : j = 1, \dots, p\}$, we use a sample average with an appropriately calibrated safety margin, i.e., $\frac{1}{n_2} \sum_{i=1}^{n_2} h(x, \xi_i) - \epsilon$, to replace the unknown $H(\cdot)$ in (1) and output the best solution among the set. The margin ϵ is calibrated via the limiting distribution of $(\frac{1}{n_2} \sum_{i=1}^{n_2} h(x^*(s_j), \xi_i))_{j=1, \dots, p}$ which captures the estimation error of $H(\cdot)$ and is multivariate Gaussian. It contains a critical value $q_{1-\beta}$ that is the quantile of a Gaussian supremum. Algorithms 2 and 3 describe two variants of this validator, one unnormalized while another one normalized by the standard deviation at each s_j . In the following, we denote $N_p(0, \Sigma)$ as a p -dimensional Gaussian vector with mean zero and covariance Σ .

Algorithm 2 V : Unnormalized Gaussian Supremum Validator

Input: $\{\xi_1, \dots, \xi_{n_2}\}, \{x^*(s_1), \dots, x^*(s_p)\}, 1 - \beta$

1. For each $j = 1, \dots, p$ compute the sample mean $\hat{H}_j = (1/n_2) \sum_{i=1}^{n_2} h(x^*(s_j), \xi_i)$ and sample covariance matrix $\hat{\Sigma}$ with $\hat{\Sigma}(j_1, j_2) = (1/n_2) \sum_{i=1}^{n_2} (h(x^*(s_{j_1}), \xi_i) - \hat{H}_{j_1})(h(x^*(s_{j_2}), \xi_i) - \hat{H}_{j_2})$.
2. Compute $q_{1-\beta}$, the $(1 - \beta)$ -quantile of $\max\{Z_1, \dots, Z_p\}$ where $(Z_1, \dots, Z_p) \sim N_p(0, \hat{\Sigma})$, and let

$$\hat{s}^* = \operatorname{argmin} \left\{ f(x^*(s_j)) : \hat{H}_j \geq \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}}, 1 \leq j \leq p \right\}. \quad (8)$$

Output: $\hat{s}^*, x^*(\hat{s}^*)$.

The first Gaussian supremum validator (Algorithm 2) is reasoned from a joint central limit theorem (CLT) that governs the convergence of $\sqrt{n_2}(\hat{H}_1 - H(x^*(s_1)), \dots, \hat{H}_p - H(x^*(s_p)))$ to $N_p(0, \Sigma)$, where

Algorithm 3 V : Normalized Gaussian Supremum Validator**Input:** $\{\xi_1, \dots, \xi_{n_2}\}, \{x^*(s_1), \dots, x^*(s_p)\}, 1 - \beta$ **1.** Same as in Algorithm 2.**2.** Denote $\hat{\sigma}_j^2 = \hat{\Sigma}(j, j)$. Compute $q_{1-\beta}$, the $(1 - \beta)$ -quantile of $\max\{Z_j/\hat{\sigma}_j : \hat{\sigma}_j^2 > 0, 1 \leq j \leq p\}$ where $(Z_1, \dots, Z_p) \sim N_p(0, \hat{\Sigma})$, and let

$$\hat{s}^* = \operatorname{argmin} \left\{ f(x^*(s_j)) : \hat{H}_j \geq \gamma + \frac{q_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}}, 1 \leq j \leq p \right\}. \quad (9)$$

Output: $\hat{s}^*, x^*(\hat{s}^*)$.

$\Sigma(j_1, j_2) = \operatorname{Cov}_F(h(x^*(s_{j_1}), \xi), h(x^*(s_{j_2}), \xi))$. Using the sample covariance $\hat{\Sigma}$ from Step 1 of Algorithm 2 as an approximation of Σ , we have, by the continuous mapping theorem,

$$\max_{1 \leq j \leq p} \sqrt{n_2}(\hat{H}_j - H(x^*(s_j))) \approx \max_{1 \leq j \leq p} Z_j \text{ in distribution}$$

where $(Z_1, \dots, Z_p) \sim N_p(0, \hat{\Sigma})$. Therefore using the $1 - \beta$ quantile $q_{1-\beta}$ of the Gaussian supremum in the margin leads to

$$H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p, \text{ with probability } \approx 1 - \beta.$$

The second validator (Algorithm 3) uses an alternate version of the CLT that is normalized by the componentwise standard deviation σ_j , i.e., $\sqrt{n_2}((\hat{H}_1 - H(x^*(s_1)))/\sigma_1, \dots, (\hat{H}_p - H(x^*(s_p)))/\sigma_p)$ converges to $N_p(0, D\Sigma D)$, where D is a diagonal matrix of $1/\sigma_j, j = 1, \dots, p$. Note that the quantile $q_{1-\beta}$ in both validators can be computed to high accuracy via Monte Carlo.

Let us make the above reasoning precise. We present our results for two cases that need separate treatments: When $H(x) \geq \gamma$ is a “light-tailed” stochastic constraint, and when it is a chance constraint.

3.1. Performance Guarantees for General Stochastic Constraints

Recall that $H(x) = \mathbb{E}_F[h(x, \xi)]$. Denote

$$\sigma^2(x) := \operatorname{Var}_F(h(x, \xi))$$

as the variance of h for each decision $x \in \mathcal{X}$. We assume the following on optimization problem (1):

ASSUMPTION 1 (Light-tailedness). *There exists a constant $D_1 \geq 1$ such that for all $x \in \mathcal{X}$ with $\sigma^2(x) > 0$, we have*

$$\mathbb{E}_F \left[\exp \left(\frac{|h(x, \xi) - H(x)|^2}{D_1^2 \sigma^2(x)} \right) \right] \leq 2 \text{ and } \mathbb{E}_F \left[\left(\frac{|h(x, \xi) - H(x)|}{\sigma(x)} \right)^{2+k} \right] \leq D_1^k \text{ for } k = 1, 2.$$

This assumption stipulates that the distribution of $h(x, \xi)$ after being centered and normalized by its standard deviation is sufficiently light-tailed at each x . Note that no other regularity property, e.g., convexity or continuity, is assumed for the function h itself. We have the following finite-sample feasibility guarantees for the solution output by Algorithm 2 or 3:

THEOREM 1 (Finite-sample feasibility guarantee for unnormalized validator). *Suppose Assumption 1 holds. Let $\bar{H} = \max_{1 \leq j \leq p} H(x^*(s_j))$ and $\bar{\sigma}^2 = \max_{1 \leq j \leq p} \sigma^2(x^*(s_j))$. For every solution set $\{x^*(s_j) : 1 \leq j \leq p\}$, every n_2 , and $\beta \in (0, \frac{1}{2})$, the solution output by Algorithm 2 satisfies*

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq 1 - \beta - C \left(\left(\frac{D_1^2 \log^7(p n_2)}{n_2} \right)^{\frac{1}{6}} + \exp\left(-\frac{c n_2 \epsilon^2}{D_1^2 \bar{\sigma}^2}\right) + p \exp\left(-\frac{c n_2}{D_1^4}\right) \right)$$

with

$$\epsilon = \left(\bar{H} - \gamma - C \bar{\sigma} \sqrt{\frac{\log(p/\beta)}{n_2}} \right)_+ \quad (10)$$

where C and c are universal constants, and $P_{\xi_{1:n_2}}$ denotes the probability with respect to Phase two data $\{\xi_1, \dots, \xi_{n_2}\}$ and conditional on Phase one data $\{\xi_{n_2+1}, \dots, \xi_n\}$.

THEOREM 2 (Finite-sample feasibility guarantee for normalized validator). *Let $\bar{s} \in \arg \max\{H(x^*(s_j)) : j = 1, \dots, p\}$, i.e., $H(x^*(\bar{s})) = \bar{H}$. Under the same conditions of Theorem 1, the solution output by Algorithm 3 satisfies*

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq 1 - \beta - C \left(\left(\frac{D_1^2 \log^7(p n_2)}{n_2} \right)^{\frac{1}{6}} + \frac{D_1^2 \log^2(p n_2)}{\sqrt{n_2}} + \exp\left(-\frac{c n_2 \epsilon^2}{D_1^2 \sigma^2(x^*(\bar{s}))}\right) + p \exp\left(-\frac{c n_2^{2/3}}{D_1^{10/3}}\right) \right)$$

with

$$\epsilon = \left(\bar{H} - \gamma - C \sigma(x^*(\bar{s})) \sqrt{\frac{\log(p/\beta)}{n_2}} \right)_+ \quad (11)$$

where C and c are universal constants.

In both Theorems 1 and 2, the finite-sample coverage probability consists of two sources of errors. The first source comes from the CLT approximation that decays polynomially in the Phase 2 sample size n_2 . The second error arises from the possibility that none of the solutions $\{x^*(s_1), \dots, x^*(s_p)\}$ satisfies the criterion in (8) or (9), which vanishes exponentially fast. When ϵ in (10) or (11) is of constant order, the CLT error dominates. In this case the finite-sample error depends logarithmically on p , the number of candidate parameter values, and the bounds dictate a coverage tending to $1 - \beta$ when p is as large as $\exp(o(n_2^{1/7}))$.

The derivation of the logarithmic dependence on p in Theorem 1 builds on a high-dimensional CLT and an associated multiplier bootstrap approximation recently developed in Chernozhukov et al. (2017) (Appendix EC.2.1). The proof of Theorem 2 further requires a Hoeffding-type inequality for U-statistics

to control the errors of the sample variance estimates, as well as the so-called Nazarov's inequality, an anti-concentration inequality for multivariate Gaussian, to control the coverage errors when using estimated standard deviations in the margin (Appendix EC.2.2). Appendices EC.2.3 and EC.2.4 detail the proofs of Theorems 1 and 2 that put together the above mathematical developments.

We explain the implication on the dimensionality of the problem. Note that to sufficiently cover the whole solution path, p is typically exponential in the dimension of S , denoted $\dim(S)$ (this happens when we uniformly discretize the parameter space S). The discussion above thus implies a requirement that n_2 is of higher order than $\dim(S)^7$. Here the low dimensionality of S is crucial; for instance, a one-dimensional conservativeness parameter s would mean $\dim(S) = 1$, so that a reasonably small n_2 can already ensure adequate feasibility coverage. Moreover, the margin adjustments in Algorithms 2 and 3 both depend only on $\dim(S)$. Thus, choosing $\hat{s}^*$ relies only on $\dim(S)$, but not the dimension of the whole decision space. Note that Theorems 1 and 2 provide guarantee conditional on Phase one data. However, the universality of the involved constants means that analogous unconditional feasibility guarantees also hold if Assumption 1 can be verified uniformly or with high probability with respect to Phase one data, an observation that persists for other subsequent results.

Comparing between the two validators, we also see that the normalized one (Algorithm 3) is statistically more efficient than the unnormalized one (Algorithm 2) when the variance $\sigma^2(x)$ exhibits high variability across solutions. More specifically, in order to make the exponential error non-dominant, one needs at least $\epsilon > 0$. In the case of Algorithm 2, expression (10) suggests that, after ignoring the logarithmic factor $\log(p/\beta)$, this requires an n_2 to be of order $\bar{\sigma}^2/(\bar{H} - \gamma)^2$. In contrast, for Algorithm 3 this becomes $\sigma^2(x^*(\bar{s}))/(\bar{H} - \gamma)^2$, where the maximal variance is replaced with the variance at the solution that optimizes the H -value, which in general does not have the maximal variance.

Theorems 1 and 2 also give immediately the following asymptotic feasibility guarantee (proof in Appendix EC.2.4):

COROLLARY 1 (Asymptotic feasibility guarantee). *Suppose Assumption 1 holds. Let $\bar{H} = \max_{1 \leq j \leq p} H(x^*(s_j))$. For every solution set $\{x^*(s_j) : 1 \leq j \leq p\}$ such that $\bar{H} > \gamma$ and every $\beta \in (0, \frac{1}{2})$, the solution output by Algorithm 2 or 3 satisfies*

$$\liminf_{n_2 \rightarrow \infty \text{ and } p \exp(-n_2^{1/7}) \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq 1 - \beta.$$

3.2. Performance Guarantees for Chance Constraints

Underlying the finite-sample bounds in Theorems 1 and 2 is the light-tailedness condition in Assumption 1. However, in a CCP that takes the form

$$\min_{x \in \mathcal{X}} f(x) \text{ subject to } P(x) := \mathbb{P}_F((x, \xi) \in A) \geq 1 - \alpha \quad (12)$$

where $A \subseteq \mathbb{R}^d \times \mathbb{R}^m$ is a deterministic set and $1 - \alpha$ is a tolerance level for the satisfaction probability, the tail of the normalized indicator function $\mathbf{1}((x, \xi) \in A)$ can be arbitrarily heavy as the satisfaction probability approaches 0 or 1 and hence violates Assumption 1. Thus, instead, we present different finite-sample error bounds for (12) than Theorems 1 and 2 whose derivations rely on the Bernoulli nature of the underlying function:

THEOREM 3 (Finite-sample chance constraint feasibility guarantee for unnormalized validator).

Let $\bar{\alpha} = 1 - \max_{1 \leq j \leq p} P(x^*(s_j))$. For every solution set $\{x^*(s_j) : 1 \leq j \leq p\}$, every n_2 , and $\beta \in (0, \frac{1}{2})$, the solution output by Algorithm 2 satisfies

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (12)}) \geq 1 - \beta - C \left(\left(\frac{\log^7(pn_2)}{n_2\alpha} \right)^{\frac{1}{6}} + \exp(-cn_2 \min\{\epsilon, \frac{\epsilon^2}{\alpha}\}) \right)$$

with

$$\epsilon = \left(\alpha - \bar{\alpha} - C \sqrt{\frac{\log(p/\beta)}{n_2}} \right)_+ \quad (13)$$

where C and c are universal constants.

THEOREM 4 (Finite-sample chance constraint feasibility guarantee for normalized validator).

Under the same conditions of Theorem 3, the solution output by Algorithm 3 satisfies

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (12)}) \geq 1 - \beta - C \left(\left(\frac{\log^7(pn_2)}{n_2\alpha} \right)^{\frac{1}{6}} + \frac{\log^2(pn_2)}{\sqrt{n_2\alpha}} + \exp(-cn_2 \min\{\epsilon, \frac{\epsilon^2}{\alpha}\}) \right)$$

with

$$\epsilon = \left(\alpha - \bar{\alpha} - C \sqrt{\frac{(\bar{\alpha} + \log(n_2\alpha)/n_2) \log(p/\beta)}{n_2}} \right)_+ \quad (14)$$

where C and c are universal constants.

A comparison between Theorems 3 and 4 again reveals the higher statistical efficiency of Algorithm 3 than Algorithm 2 which, in the CCP context, applies to the case when the satisfaction probability is large (i.e., the common case). Suppose that $1 - \alpha$ approaches 1. In order to make $\epsilon > 0$ in (13), we need a sample size n_2 of order $(\alpha - \bar{\alpha})^{-2}$ (after ignoring the logarithmic factor $\log(p/\beta)$), whereas in (14) it can be seen to need only an n_2 of order $\alpha(\alpha - \bar{\alpha})^{-2}$, a much smaller size when $1 - \alpha$ is close to 1.

Lastly, we have the following asymptotic feasibility guarantee in the case of CCP in parallel to Corollary 1:

COROLLARY 2 (Asymptotic chance constraint feasibility guarantee). Let $\bar{\alpha} = 1 - \max_{1 \leq j \leq p} P(x^*(s_j))$. For every solution set $\{x^*(s_j) : 1 \leq j \leq p\}$ such that $\bar{\alpha} < \alpha$ and every $\beta \in (0, \frac{1}{2})$, the solution output by Algorithm 2 or 3 satisfies

$$\liminf_{n_2 \rightarrow \infty \text{ and } p \exp(-n_2^{1/7}) \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (12)}) \geq 1 - \beta.$$

Appendix EC.2.4 details the proofs of Theorem 3, Theorem 4 and Corollary 2.

To close this section, we note that our Gaussian supremum validators also enjoy a notion of asymptotic solution-path optimality under additional assumptions. To streamline our presentation, we defer this discussion to the next section and combine it with the discussion of our next validator.

4. Validation via Univariate Gaussian Margin

We offer an alternate validator that can perform more efficiently than Algorithms 2 and 3, provided that further regularity assumptions are in place. This is a scheme that simply uses a standard univariate Gaussian critical value to calibrate the margin (Algorithm 4).

Algorithm 4 outputs a solution with objective value no worse than Algorithms 2 and 3. Comparing the criteria to choose $\hat{s}^*$, we see that, due to the stochastic dominance of the maximum among a multivariate Gaussian vector over each of its individual components, the margin in (8) satisfies $q_{1-\beta} \geq z_{1-\beta} \hat{\sigma}_j$ for all j , and similarly the margin in (9) satisfies $q_{1-\beta} \hat{\sigma}_j \geq z_{1-\beta} \hat{\sigma}_j$, so that both are bounded from below by the margin in (15). Consequently the solution from (15) achieves an objective value no worse than the other two.

Algorithm 4 *V*: Univariate Gaussian Validator

Input: $\{\xi_1, \dots, \xi_{n_2}\}, \{x^*(s_1), \dots, x^*(s_p)\}, 1 - \beta$

1. For each $j = 1, \dots, p$ compute the sample mean $\hat{H}_j = (1/n_2) \sum_{i=1}^{n_2} h(x^*(s_j), \xi_i)$ and sample variance $\hat{\sigma}_j^2 = (1/n_2) \sum_{i=1}^{n_2} (h(x^*(s_j), \xi_i) - \hat{H}_j)^2$.

2. Compute

$$\hat{s}^* = \operatorname{argmin} \left\{ f(x^*(s_j)) \mid \hat{H}_j \geq \gamma + \frac{z_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}}, 1 \leq j \leq p \right\} \quad (15)$$

where $z_{1-\beta}$ is the $1 - \beta$ quantile of the standard Gaussian distribution.

Output: $\hat{s}^*, x^*(\hat{s}^*)$.

The univariate Gaussian critical value used in the margin in Algorithm 4 hints that feasibility needs to be validated at only one value of s instead of the solution path S . The validity of this procedure is based on the statistical consistency of the obtained solution $x^*(\hat{s}^*)$ to some limiting solution (correspondingly $\hat{s}^*$ to some limiting optimal parameter value) as n_2 increases. Intuitively, this implies that with sufficient sample size one can focus feasibility validation on a small neighborhood of $\hat{s}^*$, which further suggests that we need to control only the statistical error at effectively one solution parametrized at $\hat{s}^*$. For this argument to hold, however, we would need several additional technical assumptions including a low functional complexity of h , and a different line of derivations.

4.1. Asymptotic Performance Guarantees

We present the statistical guarantees of Algorithm 4 as Phase two data size $n_2 \rightarrow \infty$. We assume continuity for the objective of (1):

ASSUMPTION 2 (Continuous objective). *The objective function $f(x)$ is continuous on $\mathcal{X}$.*

For the constraint, we assume the following:

ASSUMPTION 3 (Functional complexity). *The function class $\mathcal{F} := \{h(x, \cdot) | x \in \mathcal{X}\}$ is F -Donsker.*

ASSUMPTION 4 (L_2 -boundedness). $\mathbb{E}_F[\sup_{x \in \mathcal{X}} |h(x, \xi) - H(x)|^2] < \infty$.

ASSUMPTION 5 (L_2 -continuity). *For any fixed $x \in \mathcal{X}$ and another $x' \in \mathcal{X}$, we have $\lim_{x' \rightarrow x} \mathbb{E}_F[(h(x', \xi) - h(x, \xi))^2] = 0$.*

To give a sense of the generality of the above assumptions, we identify two general classes of constraints for which these assumptions are guaranteed to hold, one suitable for general h , and another one for CCPs:

PROPOSITION 1. *Assumptions 3-5 hold in each of the following two cases:*

- i. There exists some $M(\xi)$ such that $\mathbb{E}_F[M(\xi)^2] < \infty$ and $|h(x_1, \xi) - h(x_2, \xi)| \leq M(\xi) \|x_1 - x_2\|$ for all $x_1, x_2 \in \mathcal{X}$, there exists some $\tilde{x} \in \mathcal{X}$ such that $\mathbb{E}_F[h(\tilde{x}, \xi)^2] < \infty$, and the decision space $\mathcal{X}$ is compact;*
- ii. $h(x, \xi) = \mathbf{1}(a'_k A_k(x) \leq b_k)$ for $k = 1, \dots, K$ for some $K < \infty$, where each $A_k(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^{m_k}$ is a continuous mapping and each $a_k \in \mathbb{R}^{m_k}, b_k \in \mathbb{R}$ satisfies either (i) a_k has a density on $\mathbb{R}^{m_k}$ and b_k is a non-zero constant or (ii) (a_k, b_k) has a density on $\mathbb{R}^{m_k+1}$.*

Case (i) in Proposition 1 follows from standard results in empirical process theory, including in particular the Jain-Marcus Theorem. The proof of Case (ii) involves checking the finite Vapnik-Chervonenkis (VC) dimension and pointwise separability of the function class in order to verify F -Donskerness. Appendix EC.3 details the proof of Proposition 1.

We impose one more assumption on the constraint function regarding its variance:

ASSUMPTION 6 (Non-degeneracy of the variance on the boundary). $\sigma^2(x) > 0$ for all $x \in \mathcal{X}$ such that $H(x) = \gamma$.

In Assumption 6, non-zero variance is assumed only for those x 's at which the stochastic constraint is satisfied with equality, but not necessarily for other x . This is significant in the case of CCP (12). While there could exist 100% or 0% safe solutions, i.e., x such that $P(x) = 1$ or 0, and hence non-degeneracy may not be satisfied over the whole $\mathcal{X}$, it holds for those x 's with $P(x) = 1 - \alpha$ that have (the same) non-zero variance $\alpha(1 - \alpha)$.

Now we present our assumptions on the data-driven reformulation $OPT(s), s \in S$. We focus on formulations with a single parameter (A separate set of results for formulations with multiple parameters can be found in Appendix EC.6). We first assume that the solution path is piecewise continuous:

ASSUMPTION 7 (Piecewise continuous solution path). *The parameter space S is a finite interval $[s_l, s_u]$. The optimal solution $x^*(s)$ of $OPT(s)$ exists and is unique except for a finite number of parameter values $\tilde{s}_i, i = 1, \dots, M - 1$ such that $s_l = \tilde{s}_0 < \tilde{s}_1 < \dots < \tilde{s}_{M-1} < \tilde{s}_M = s_u$, and the parameter-to-solution mapping $x^*(s)$ is uniformly continuous on each piece $[\tilde{s}_0, \tilde{s}_1]$, $(\tilde{s}_{M-1}, \tilde{s}_M]$, and $(\tilde{s}_{i-1}, \tilde{s}_i)$ for $i = 2, \dots, M - 1$.*

Continuity of the solution path allows approximating the whole solution curve by discretizing the parameter space S . Also note that under Assumption 7 the solution $x^*(s)$ exists and is unique for almost surely every $s \in S$ with respect to the Lebesgue measure. Therefore, if one discretizes the parameter space by randomizing via a continuous distribution over S , then with probability one the solution $x^*(s)$ is unique at all sampled parameter values. This provides an easy way to ensure the assumption that none of the parameter values $\{s_1, \dots, s_p\}$ used in Phase one of Algorithm 1 belongs to the discontinuity set $\{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$.

To explain the superior performance of Algorithm 4, we introduce a notion of optimality within the solution path $\{x^*(s) : s \in S\}$. First, since the parameter-to-solution mapping $x^*(s)$ is not defined at the discontinuities under Assumption 7, we need to fill in these holes in the solution path. Thanks to uniform continuity, the mapping $x^*(s)$ on each piece $(\tilde{s}_{i-1}, \tilde{s}_i)$ can be continuously extended to the closure $[\tilde{s}_{i-1}, \tilde{s}_i]$ by taking left and right limits. Specifically, we define:

DEFINITION 1. Under Assumption 7, the parameter-to-solution mapping $x^*(\cdot)$ at each discontinuity $\tilde{s}_i, i = 1, \dots, M - 1$ is defined in an extended fashion as

$$x^*(\tilde{s}_i) = \{x^*(\tilde{s}_i-), x^*(\tilde{s}_i+)\} \text{ where } x^*(\tilde{s}_i-) := \lim_{s \rightarrow \tilde{s}_i-} x^*(s) \text{ and } x^*(\tilde{s}_i+) := \lim_{s \rightarrow \tilde{s}_i+} x^*(s).$$

Note that the two solutions $x^*(\tilde{s}_i-)$ and $x^*(\tilde{s}_i+)$ are different if the i -th and $(i + 1)$ -th pieces are disconnected. With the extended parameter-to-solution mapping $x^*(\cdot)$, we now introduce the notions of optimal solution and optimal parameter associated with the solution path:

DEFINITION 2. Associated with the solution path $\{x^*(s) : s \in S\}$, the optimal solution set is

$$\mathcal{X}_S^* := \operatorname{argmin}\{f(x) : H(x) \geq \gamma, x = x^*(s) \text{ for } s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} \text{ or } x \in x^*(\tilde{s}_i) \text{ for some } i = 1, \dots, M - 1\} \quad (16)$$

and the optimal parameter set is

$$S^* := \{s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} : x^*(s) \in \mathcal{X}_S^*\} \cup \{\tilde{s}_i : x^*(\tilde{s}_i) \cap \mathcal{X}_S^* \neq \emptyset, i = 1, \dots, M - 1\}. \quad (17)$$

We need several additional technical assumptions. The first is that the stochastic constraint is not binding at the endpoints of each piece of the solution path:

ASSUMPTION 8. $H(x^*(\tilde{s}_i-)) \neq \gamma$ and $H(x^*(\tilde{s}_i+)) \neq \gamma$ for all $i = 1, \dots, M - 1$, $H(x^*(s_l)) \neq \gamma$, $H(x^*(s_u)) \neq \gamma$, and $\sup_{s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}} H(x^*(s)) > \gamma$.

Since the solution path $\{x^*(s) : s \in S\}$ depends on Phase one data $\xi_{n_2+1:n}$, the path and hence the endpoints $x^*(\tilde{s}_i-), x^*(\tilde{s}_i+)$ are random objects, and so the first part of Assumption 8 is expected to hold almost surely provided that the set $\{x \in \mathcal{X} : H(x) = \gamma\}$ is a null set under the Lebesgue measure on $\mathbb{R}^d$. The second part states that the solution path contains a strictly feasible solution which in turn ensures that the optimal solution set $\mathcal{X}_S^*$ is non-empty. Note that this can typically be achieved by simply including very conservative parameter values in S .

Another property we assume regards the monotonicity of the feasible set size with respect to the parameter s in the reformulation $OPT(s)$:

ASSUMPTION 9. Denote by $\text{Sol}(s) := \mathcal{X} \cap \hat{\mathcal{F}}(s)$ the feasible set of $OPT(s)$. Assume $\text{Sol}(s)$ is a closed set for all $s \in S$ and $\text{Sol}(s_2) \subseteq \text{Sol}(s_1)$ for all $s_1, s_2 \in S$ such that $s_1 < s_2$.

Assumption 9 holds for all common reformulations (all examples in the beginning of Section 2) as s controls the conservativeness level. For instance, in RO with ellipsoidal uncertainty set, the RO feasible region shrinks with the radius of the ellipsoid, and similar relations hold for DRO, SAA, and SO. A straightforward consequence of Assumption 9 is the monotonicity of the parameter-to-objective mapping

$$v(s) := \inf\{f(x) : x \in \mathcal{X} \cup \hat{\mathcal{F}}(s)\}$$

as described in the following proposition (proof in Appendix EC.3):

PROPOSITION 2. Suppose Assumptions 7 and 9 hold. For all $s_1, s_2 \in S$ such that $s_1 < s_2$ it holds $v(s_1) \leq v(s_2)$, and if additionally $s_1, s_2 \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ then $v(s_1) < v(s_2)$ if and only if $x^*(s_1) \neq x^*(s_2)$.

The assumptions we have made for the formulation $OPT(s)$ give rise to the following uniqueness characterization of the optimal solution set $\mathcal{X}_S^*$ and the optimal parameter set S^* within the solution path, which would be used to establish the feasibility guarantees for Algorithm 4.

PROPOSITION 3 (Structure of solution-path optima). Under Assumptions 2, 5, and 7-9, the optimal solution set $\mathcal{X}_S^*$ is a singleton $\{x_S^*\}$ and the optimal parameter set S^* is a closed interval $[s_l^*, s_u^*]$ for $s_l^*, s_u^* \in S$. In addition, if $v(s)$ is strictly increasing on S , then S^* is a singleton $\{s^*\}$.

The proof of Proposition 3, which is in Appendix EC.3, involves an exhaustion of all possible structures of the set $\mathcal{X}_S^*$ that contain more than one solution, and showing each of them contradict with our assumptions (especially Assumption 9).

Lastly, we assume the following technical assumption for the set of optima:

ASSUMPTION 10. For any $\epsilon > 0$ there exists an $s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ such that $H(x^*(s)) > \gamma$ and $\|x^*(s) - x_S^*\|_2 < \epsilon$, where x_S^* is the unique optimal solution from Proposition 3.

This assumption trivially holds if $\mathcal{X}_S^* = \{x_S^*\}$ as described in Proposition 3 and $H(x_S^*) > \gamma$. Otherwise, if $H(x_S^*) = \gamma$, it rules out the case that the solution path $x^*(s)$ passes through x_S^* without entering the interior of the feasible set of (1). The latter exceptional case typically happens with zero probability, in view of the fact that the solution path is itself random with respect to Phase one data.

Now we are ready to present the asymptotic performance guarantee for Algorithm 4:

THEOREM 5 (Asymptotic joint feasibility+optimality guarantee). *Suppose Assumptions 2-6 hold for (1). Also suppose Assumptions 7-10 hold for the reformulation $OPT(s)$ constructed in Algorithm 1, and $\{s_1, \dots, s_p\} \cap \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} = \emptyset$. Denote by $\epsilon_S = \sup_{s \in S} \inf_{1 \leq j \leq p} |s - s_j|$ the mesh size, and by x_S^* the unique optimal solution from Proposition 3. Then, with respect to $\{\xi_1, \dots, \xi_{n_2}\}$, the solution and parameter output by Algorithm 4 satisfy*

$$\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} x^*(\hat{s}^*) = x_S^* \text{ and } \lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} d(\hat{s}^*, S^*) = 0 \quad (18)$$

almost surely. Moreover, if $H(x_S^) = \gamma$ we have*

$$\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq 1 - \beta, \quad (19)$$

otherwise if $H(x_S^) > \gamma$ we have*

$$\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) = 1. \quad (20)$$

Theorem 5 states that as the mesh $\{s_1, \dots, s_p\}$ gets increasingly fine and the data size grows, the solution given by Algorithm 4 enjoys performance guarantees concerning both feasibility and solution-path optimality. In particular, the estimated solution and the conservativeness parameter converge to the optimal solution x_S^* and the optimal parameter set S^* respectively, while simultaneously the obtained solution is feasible with the desired confidence level $1 - \beta$.

The proof of Theorem 5 is in Appendix EC.3. The consistency result in (18) is shown via a dense approximation of the discrete parameter set $\{s_1, \dots, s_p\}$ on the continuum S , through the continuity of the solution path and a uniform law of large numbers. Then, based on this consistency, the feasibility guarantee (19) is established by showing $P(H(x^*(\hat{s}^*)) \geq \gamma) \geq P(H(x^*(\hat{s}^*)) \geq \hat{H}(x^*(\hat{s}^*)) - z_{1-\beta} \hat{\sigma}(x^*(\hat{s}^*)) / \sqrt{n_2}) \approx P(H(x_S^*) \geq \hat{H}(x_S^*) - z_{1-\beta} \hat{\sigma}(x_S^*) / \sqrt{n_2}) \rightarrow 1 - \beta$, where the “ $\geq$ ” follows from our validation criterion (15) whereas the “ $\approx$ ” comes from the asymptotic tightness of the empirical process $\{\sqrt{n_2}(\hat{H}(x^*(s)) - H(x^*(s))) : s \in S\}$ and the L_2 continuity of the constraint function $h(x, \xi)$.

Furthermore, under additional smoothness conditions on the constraint function h and the solution path $\{x^*(s) : s \in S\}$, we also establish the finite-sample counterparts for the optimality guarantee (18) and feasibility guarantee (19) for Algorithm 4. These are presented in Appendix EC.5.

Note that the confidence level (19) at which Algorithm 4 outputs a feasible solution (and also Algorithms 2 and 3, i.e., Corollaries 1 and 2) is generally not tight, i.e., a lower bound instead of an equality is guaranteed. However, with a strict monotonicity condition on the reformulation $OPT(s)$ and a finer discretization mesh for the conservativeness parameter, Algorithm 4 can give a tight confidence guarantee:

THEOREM 6 (Asymptotically tight feasibility guarantee). *In addition to the conditions of Theorem 5, further assume that the parameter-to-objective mapping $v(s)$ is strictly increasing on S , and consider the case that $H(x_S^*) = \gamma$. If the mesh $\{s_1, \dots, s_p\}$ is fine enough so that*

$$\max_{i=1, \dots, M} \max_{j=1, \dots, p_i-1} |H(x^*(s_j^i)) - H(x^*(s_{j+1}^i))| = o\left(\frac{1}{\sqrt{n_2}}\right) \quad (21)$$

where $s_1^i < \dots < s_{p_i}^i$ are the parameter values $\{s_j : s_j \in (\tilde{s}_{i-1}, \tilde{s}_i), j = 1, \dots, p\}$ (so that $\sum_{i=1}^M p_i = p$), then we must have

$$\lim_{n_2 \rightarrow \infty \text{ and } \epsilon_S \rightarrow 0 \text{ s.t. (21) holds}} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) = 1 - \beta$$

for the solution output by Algorithm 4.

Roughly speaking, the loose confidence guarantee in (19) can be attributed to the one-sided nature of the inequality criterion used in (15). The monotonicity of $v(s)$ and the mesh condition (21) give rise to a tight confidence guarantee by strengthening this inequality criterion to an equality (with a negligible error) at the chosen parameter value $\hat{s}^*$. Note that, when the expected constraint value $H(x^*(s))$ is Lipschitz continuous in the parameter, the mesh condition (21) is guaranteed if $\epsilon_S = o\left(\frac{1}{\sqrt{n_2}}\right)$ or if $\frac{p}{\sqrt{n_2}} \rightarrow \infty$ and the mesh is equispaced. The proof of Theorem 6 is in Appendix EC.3.

Relatedly, the following shows that, like Algorithm 4, the supremum-based validators in Algorithms 2 and 3 also exhibit joint asymptotic feasibility and solution-path optimality guarantees. However, their confidence guarantees for feasibility are not as tight. This result complements our discussions at the end of Section 3 regarding the optimality property of the supremum-based validators, and also at beginning of Section 4 regarding the better objective value of the solution obtained by Algorithm 4, which is consistent with its tighter achievement of the feasibility confidence level.

THEOREM 7 (Asymptotic joint feasibility+optimality guarantee with Gaussian supremum validator). *Under the same conditions as Theorem 5, the solution and parameter from Algorithm 2 satisfy the consistency guarantee (18). In the case $H(x_S^*) = \gamma$ it holds*

$$\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq \Phi\left(\frac{\bar{q}_{1-\beta}}{\sigma(x_S^*)}\right) \geq 1 - \beta \quad (22)$$

where $\bar{q}_{1-\beta}$ is the $1 - \beta$ quantile of the supremum of the Gaussian process indexed by $s \in S \setminus \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ with the covariance structure $\text{Cov}(s, s') = \text{Cov}_F(h(x^*(s), \xi), h(x^*(s'), \xi))$, and Φ is the distribution function of the standard normal.

If it is further assumed that $\inf_{x \in \mathcal{X}} \sigma^2(x) > 0$, then (18) also holds for Algorithm 3, and in the case $H(x_S^*) = \gamma$ we have

$$\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq \Phi(\tilde{q}_{1-\beta}) \geq 1 - \beta \quad (23)$$

where $\tilde{q}_{1-\beta}$ is the $1 - \beta$ quantile of the supremum of the Gaussian process on $S \setminus \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ with covariance $\text{Cov}(s, s') = \text{Cov}_F(h(x^*(s), \xi), h(x^*(s'), \xi)) / (\sigma(x^*(s))\sigma(x^*(s')))$.

In general, when the Gaussian processes involved in (22) and (23) have non-constant covariance structures, the asymptotic confidence levels rendered by Algorithms 2 and 3 are strictly higher than the nominal level $1 - \beta$. This suggests that supremum-based margins tend to generate more conservative solutions than the univariate Gaussian margin does, although they all approach the same optimal solution x_S^* in the limit.

The proof of Theorem 7 (in Appendix EC.3) involves steps similar to that of Theorem 5, but furthermore showing the statistical consistency of the critical value $q_{1-\beta}$ calibrated in Algorithm 2 or 3. The latter utilizes the separability of the limiting Gaussian process and a control of errors coming from the associated multiplier bootstrap approximation.

5. Applying Our Framework in Data-Driven Reformulations

In this section we showcase various data-driven reformulations of (1) or (12) to which our proposed framework can be applied. We first comment that our Gaussian supremum validators (Algorithms 2 and 3) are applicable to all formulations considered here, as long as the constraint function $h(x, \xi)$ is sufficiently light-tailed as described in Assumption 1 or the constraint is a chance constraint. That is,

THEOREM 8 (Applicability of Gaussian supremum validators). *Consider the general stochastically constrained problem (1) that satisfies Assumption 1, or CCP (12). All the data-driven reformulations $OPT(s)$ presented below, namely SAA, DRO with ϕ -divergence, Wasserstein and moment-based uncertainty sets, RO with polyhedral and ellipsoidal uncertainty sets, and SO (the last two approaches are for CCP only), can be validated by the Gaussian supremum validators in Algorithms 2 and 3 and elicit the conclusions in all theorems and corollaries in Section 3.*

The tighter univariate Gaussian validator (Algorithm 4) however requires some extra regularity conditions from the data-driven formulation $OPT(s)$, but still works for many common formulations. We consider decision space $\mathcal{X}$ that has the form:

ASSUMPTION 11. $\mathcal{X} = \{x \in \mathbb{R}^d : f_r(x) \leq 0 \text{ for } r = 1, \dots, R \text{ and } Wx \leq z\}$, where each f_r is continuous and convex, and $W = [w_1, w_2, \dots, w_L]' \in \mathbb{R}^{L \times d}$, $z \in \mathbb{R}^L$.

We consider optimization formulations that satisfy the following two assumptions:

ASSUMPTION 12 (Slater's condition). *Slater's condition holds for $OPT(s_u) := \min\{f(x) : x \in \mathcal{X} \cap \hat{\mathcal{F}}(s_u)\}$ where s_u is the maximal parameter value.*

ASSUMPTION 13 (Non-empty and bounded level set). *There exists a constant c such that $\mathcal{X} \cap \hat{\mathcal{F}}(s_l) \cap \{x : f(x) \leq c\}$ is bounded and $\mathcal{X} \cap \hat{\mathcal{F}}(s_u) \cap \{x : f(x) \leq c\}$ is non-empty where s_l, s_u are the minimal and maximal parameter values.*

Slater's condition (Assumption 12) is a common property that is expected to hold for most optimization problems in practice. Dür et al. (2016) states that Slater's condition is a generic property for linear conic programs by showing that it holds for all problem data except in a null set of Lebesgue measure. Assumption 13 also trivially holds in many settings, e.g., when $\mathcal{X} \cap \hat{\mathcal{F}}(s_l)$ is compact or $f(x)$ is coercive. Under these two assumptions, stability results from parametric optimization (Proposition 4.4 in Bonnans and Shapiro (2013)) ensure that the solution path $x^*(s)$ is continuous when the optimal solution is unique for each $OPT(s)$, or piecewise continuous when uniqueness fails at only a finite number of parameter values, leading to Assumption 7. Since other assumptions from Section 4.1 regarding $OPT(s)$ can be readily verified to hold in general, for each considered formulation below we focus on identifying the conditions that guarantee the validity of Assumption 7 in order to ensure the asymptotic feasibility and optimality guarantees. The proofs of all results in this section are presented in Appendix EC.4.

We introduce a condition that will appear in the following discussion. Consider the linear objective $f(x) = c'x$ for some deterministic $c \in \mathbb{R}^d$. We say a finite collection of vectors $\{v_1, \dots, v_k\} \subset \mathbb{R}^d$ with $k \leq d - 1$ satisfies the strict cone inclusion (SCI) condition if

$$\text{SCI: } v_1, \dots, v_k \text{ are linearly independent, and there exist } \lambda_1, \dots, \lambda_k > 0 \text{ such that } \sum_{i=1}^k \lambda_i v_i = -c.$$

SAA: First consider the SAA reformulation for the general stochastic constraint in (1) in the form

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to} \quad \frac{1}{n} \sum_{i=1}^n h(x, \xi_i) \geq \gamma + s \quad (24)$$

where $s > 0$ is the margin to be tuned (and for convenience, in this section only, we use n to represent a generic sample size; in applications this typically refers to the Phase one data size). We have the following result concerning the applicability of Algorithm 4:

THEOREM 9 (Applying univariate Gaussian validator to SAA). *Consider $OPT(s)$ using (24). Suppose Assumptions 11-13 hold. In either of the following two cases:*

- i. $f(x)$ is continuous and strictly convex, $h(x, \xi)$ is continuous and concave in x for every ξ .
- ii. $f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, the functions $f_r, r = 1, \dots, R$ in Assumption 11 are strictly convex, any $k \leq d - 1$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W do not satisfy the SCI condition, and $h(x, \xi)$ either is continuous and strictly concave in x for every ξ or has the form $h(x, \xi) = A(\xi)'x + b(\xi)$ where $A(\xi) \in \mathbb{R}^d$ has a density on $\mathbb{R}^d$.

Assumption 7 holds with $M = 1$ almost surely in the data $\{\xi_1, \dots, \xi_n\}$.

The proof of Theorem 9 (and theorems for other formulations below) mainly consists of establishing the joint continuity of the data-driven constraint (24) in x and s , and the uniqueness of $x^*(s)$, two main ingredients that enable us to apply the stability theory from Bonnans and Shapiro (2013) to conclude the continuity of $x^*(s)$. The former is shown by direct verification, whereas the latter is established from either strict convexity or the SCI condition when the formulation has linear objectives and constraints.

In the case of chance constraint (12), the SAA formulation has the form

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to} \quad \frac{1}{n} \sum_{i=1}^n \mathbf{1}((x, \xi_i) \in A) \geq 1 - \alpha + s. \quad (25)$$

Note that the left hand side can only take values $\frac{j}{n}, j = 0, 1, \dots, n$, therefore all s such that $1 - \alpha + s \in (\frac{j-1}{n}, \frac{j}{n}]$ lead to the same feasible region and hence the same solution $x^*(s)$. As a result, the solution path $\{x^*(s) : s \in S\}$ consists of at most n constant pieces and Assumption 7 holds automatically. Thus we have:

THEOREM 10 (Applying univariate Gaussian validator to SAA under chance constraint).

Consider $OPT(s)$ using (25). Assumption 7 holds for some $M \leq n$.

ϕ -divergence DRO: Given a convex function ϕ on $[0, +\infty)$ such that $\phi(1) = 0$, consider the ϕ -divergence DRO formulation for (1) in the form

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to} \quad \inf \left\{ \sum_{i=1}^n w_i h(x, \xi_i) : \sum_{i=1}^n \frac{1}{n} \phi(nw_i) \leq s, \sum_{i=1}^n w_i = 1, w_i \geq 0 \text{ for all } i \right\} \geq \gamma. \quad (26)$$

We have the following result:

THEOREM 11 (Applying univariate Gaussian validator to ϕ -divergence DRO). Consider $OPT(s)$ using (26). Suppose Assumptions 11-13 hold, and ϕ is continuous and convex on $(0, +\infty)$ with $\phi(1) = 0$. In either of the following three cases:

- i. $f(x)$ is continuous and strictly convex, $h(x, \xi)$ is continuous and concave in x for every ξ .
- ii. $f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, the functions $f_r, r = 1, \dots, R$ in Assumption 11 are strictly convex, any $k \leq d - 1$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W do not satisfy the SCI condition, and $h(x, \xi)$ is continuous and strictly concave in x for every ξ .
- iii. Assume the same conditions as in (ii) except that $h(x, \xi)$ is only concave (instead of strictly concave) in x for every ξ . In addition, ϕ is differentiable and strictly convex on $(0, +\infty)$ with $\lim_{x \rightarrow 0+} \phi(x) = +\infty$. For any $x_1, x_2 \in \mathcal{X}$ let $\widehat{\text{Corr}}(x_1, x_2) = \widehat{\text{Cov}}(h(x_1, \xi), h(x_2, \xi)) / (\hat{\sigma}(x_1)\hat{\sigma}(x_2))$ be the empirical correlation coefficient between $h(x_1, \xi)$ and $h(x_2, \xi)$ based on data $\{\xi_1, \dots, \xi_n\}$. $\hat{\sigma}^2(x) > 0$ for all $x \in \mathcal{X}$, and there exist no distinct x_1, x_2 such that $\widehat{\text{Corr}}(\lambda x_1 + (1 - \lambda)x_2, x_1) = 1$ for all $\lambda \in [0, 1]$.

Assumption 7 holds with $M = 1$ for the ϕ -divergence DRO conditioned on the data $\{\xi_1, \dots, \xi_n\}$.

Wasserstein DRO: Consider the Wasserstein DRO reformulation for the constraint in (1) given by

$$\min_{x \in \mathcal{X}} f(x) \quad \text{subject to} \quad \inf \left\{ \mathbb{E}_G[h(x, \xi)] : d_p(G, F_n) \leq s \right\} \geq \gamma \quad (27)$$

where $F_n = \frac{1}{n} \sum_{i=1}^n \delta_{\xi_i}$ is the empirical distribution and $d_p(G, F_n)$ is the Wasserstein distance between an arbitrary probability measure G and F_n which is defined as

$$d_p^p(G, F_n) = \inf \left\{ \mathbb{E}_\pi[\|\xi - \xi'\|^p] : \pi \text{ is a probability measure on } \Xi^2 \text{ with marginals } G \text{ and } F_n \right\}$$

where Ξ is the known domain of ξ and $\|\cdot\|$ is an arbitrary norm.

The following theorem gives conditions under which Wasserstein DRO satisfies Assumption 7:

THEOREM 12 (Applying univariate Gaussian validator to Wasserstein DRO). *Consider $OPT(s)$ using (27). Suppose Assumptions 11-13 hold, the domain Ξ of ξ is compact, and $1 \leq p < \infty$. In either of the following two cases:*

- i. *$f(x)$ is continuous and strictly convex, $h(x, \xi)$ is jointly continuous in x, ξ and concave in x for every ξ .*
- ii. *$f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, the functions $f_r, r = 1, \dots, R$ in Assumption 11 are strictly convex, any $k \leq d - 1$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W do not satisfy the SCI condition, and $h(x, \xi)$ is jointly continuous in x, ξ and strictly concave in x for every ξ .*

Assumption 7 holds with $M = 1$.

Proving Theorem 12 requires utilizing the recently developed strong duality theory for Wasserstein DRO to show the joint continuity of the constraint (27) and the existence of a worst-case distribution (e.g., Blanchet and Murthy (2019); Gao and Kleywegt (2016)) to establish its strict convexity.

Moment-based DRO: We restrict our discussion in this case to individual linear chance constraints

$$\mathbb{P}_F(a'_i x \leq b_i) \geq 1 - \alpha_i, \quad \text{for } i = 1, \dots, K \quad (28)$$

where each a_i is random and b_i is a deterministic constant, and α_i is an individual tolerance level. This setup also applies to the case of joint linear chance constraint, say, $\mathbb{P}_F(a'_i x \leq b_i \text{ for } i = 1, \dots, K) \geq 1 - \alpha_i$, where one uses the Bonferroni correction to safely approximate with K single chance constraints $\mathbb{P}_F(a'_i x \leq b_i) \geq 1 - \frac{\alpha}{K}, i = 1, \dots, K$. We restrict our discussion to (28) as it enables the tractable use of moment-based DRO; other settings are possible, but would lead to much more elaborate technicality that we do not pursue here.

We consider for each single constraint the following moment-based distributionally robust counterpart

$$\inf_{a_i \sim Q \text{ s.t. } (\mathbb{E}_Q[a_i], \text{Cov}_Q[a_i]) \in \mathcal{U}_i(s)} \mathbb{P}_Q(a'_i x \leq b_i) \geq 1 - \alpha_i \quad \text{for } i = 1, \dots, K$$

where each $\mathcal{U}_i(s)$ is a joint uncertainty set for the mean and covariance of the uncertain quantity a_i , all parametrized by the same s . For a fixed mean μ and covariance Σ , the robust constraint $\inf_{a_i \sim Q \text{ s.t. } \mathbb{E}_Q[a_i] = \mu, \text{Cov}_Q(a_i) = \Sigma} \mathbb{P}_Q(a'_i x \leq b_i) \geq 1 - \alpha_i$ has an analytic expression $\mu'x + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x' \Sigma x} \leq b_i$ (Ghaoui et al. (2003)), therefore this moment-based DRO takes the form

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{subject to} \quad & \sup_{(\mu, \Sigma) \in \mathcal{U}_i(s)} \mu'x + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x' \Sigma x} \leq b_i \text{ for } i = 1, \dots, K. \end{aligned} \quad (29)$$

THEOREM 13 (Applying univariate Gaussian validator to moment-based DRO). *Consider $OPT(s)$ given by (29). Suppose Assumptions 11-13 hold, and that for each i the uncertainty set $\mathcal{U}_i(s)$ satisfies either (1)(2)(3) or (1)(2)(4) among: (1) $\mathcal{U}_i(s)$ is compact for all s ; (ii) $\mathcal{U}_i(s) \subseteq \mathcal{U}_i(s')$ whenever $s < s'$ and $\overline{\cup_{s' < s} \mathcal{U}_i(s')} = \cap_{s' > s} \mathcal{U}_i(s') = \mathcal{U}_i(s)$ for all s ; (3) for every s and every $(\mu, \Sigma) \in \mathcal{U}_i(s)$, Σ is positive definite; (4) $\mathcal{U}_i(s) = \mathcal{U}_i^1(s) \times \mathcal{U}_i^2(s)$, where $\mathcal{U}_i^1(s)$ and $\mathcal{U}_i^2(s)$ are uncertainty sets for the mean and covariance respectively, and there is a positive definite $\Sigma_s \in \mathcal{U}_i^2(s)$ such that $\Sigma \preceq \Sigma_s$ for all $\Sigma \in \mathcal{U}_i^2(s)$ where $\preceq$ is the ordering with respect to the positive semi-definite cone. Then, in either of the following two cases:*

- i. $f(x)$ is continuous and strictly convex.
- ii. $f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, the functions $f_r, r = 1, \dots, R$ in Assumption 11 are strictly convex, any $k \leq d - 1$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W do not satisfy the SCI condition, and each $b_i \neq 0$. Assumption 7 holds with $M = 1$.

Conditions (1) and (2) in Theorem 13 hold for common choices of moment-based uncertainty sets. We discuss some examples where (3) and (4) arise. (3) holds when $\mathcal{U}_i(s)$ is constructed to be a joint confidence region from, e.g., the delta method (Marandi et al. (2019)), for the mean and covariance of a_i whose covariance component converges to the true positive definite covariance as data size grows. (4) happens if the mean and covariance are treated separately and the uncertainty set for covariance takes the form $\mathcal{U}_i^2(s) = \{\Sigma : \Sigma_l(s) \preceq \Sigma \preceq \Sigma_u(s)\}$ (e.g., Delage and Ye (2010)).

RO with polyhedral uncertainty set: Consider the same linear chance constraint (28), and for each i we use the robust counterpart $\sup_{a_i \in \mathcal{U}_i(s)} a'_i x \leq b_i$ where

$$\mathcal{U}_i(s) = \{a_i : \mathcal{W}_i a_i \leq z_i + s e_i\}$$

for some $\mathcal{W}_i \in \mathbb{R}^{l_i \times d}$, $z_i \in \mathbb{R}^{l_i}$ and $e_i \in \mathbb{R}_+^{l_i} := [0, \infty)^{l_i}$. This robust counterpart can be expressed as a set of linear constraints, leading to the following formulation

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{subject to} \quad & (z_i + s e_i)' y_i \leq b_i \\ & W_i' y_i = x \\ & y_i \geq 0 \text{ for all } i = 1, \dots, K \end{aligned} \quad (30)$$

where each $y_i \in \mathbb{R}^{l_i}$ is an auxiliary variable.

THEOREM 14 (Applying univariate Gaussian validator to polyhedral RO). Consider $OPT(s)$ given by (30). Suppose Assumptions 11-13 hold. If $f(x)$ is continuous and strictly convex, then Assumption 7 holds with $M = 1$. Otherwise, if $f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, $R = 0$ in Assumption 11, the uncertainty set $\mathcal{U}_i(s_u)$ of maximal size is bounded for each i , and every $k \leq d - 1$ element in $\{w_1, \dots, w_L\} \cup (\cup_{i=1}^K \{\widetilde{\mathcal{W}}_i^{-1} z_i + s \widetilde{\mathcal{W}}_i^{-1} e_i : \widetilde{\mathcal{W}}_i \in \mathbb{R}^{d \times d} \text{ is an invertible submatrix of } \mathcal{W}_i\})$ satisfies the SCI condition at only finitely many s values, then Assumption 7 holds with some finite M .

The proof of Theorem 14 involves some technical developments to show that $x^*(s)$ has left and right limits at each discontinuity. This consists of transforming (30) into an equivalent parametric linear program whose constraints correspond to the vertices of the uncertainty sets, and then showing that its optimal basis stays constant in a neighborhood of each discontinuity. Lastly, we use the Jordan decomposition of the optimal basis matrix to establish the existence of left and right limits.

RO with ellipsoidal uncertainty set: Consider (28) again, and now for each constraint we consider using $\sup_{a_i \in \mathcal{U}_i(s)} a_i'x \leq b_i$ with

$$\mathcal{U}_i(s) = \{a_i : a_i = \mu_i + \Sigma_i v, \|v\|_2 \leq s\}$$

for some positive definite $\Sigma_i \in \mathbb{R}^{d \times d}$, and $\mu_i \in \mathbb{R}^d$. This robust formulation has the following second-order cone representation

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{subject to} \quad & \mu_i'x + s \|\Sigma_i x\|_2 \leq b_i \text{ for all } i = 1, \dots, K. \end{aligned} \tag{31}$$

THEOREM 15 (Applying univariate Gaussian validator to ellipsoidal RO). Consider $OPT(s)$ given by (31). Suppose Assumptions 11-13 hold, and each Σ_i is positive definite. In either of the following two cases:

- i. $f(x)$ is continuous and strictly convex.
- ii. $f(x) = c'x$ for some non-zero $c \in \mathbb{R}^d$, the functions $f_r, r = 1, \dots, R$ in Assumption 11 are strictly convex, any $k \leq d - 1$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W do not satisfy the SCI condition, and each $b_i \neq 0$.

Assumption 7 holds with $M = 1$.

SO: Consider the CCP (12). Given the data $\{\xi_1, \dots, \xi_n\}$, consider the following sequence $OPT(s)$ of programs

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{subject to} \quad & (x, \xi_i) \in A \text{ for all } i = 1, \dots, s \end{aligned} \tag{32}$$

for $1 \leq s \leq n$, i.e., each $OPT(s)$ uses only the first s sampled constraints. Although s takes integer values only, we can artificially extend the solution path to the continuum $[1, n]$ without introducing new solutions, by letting $x^*(s) = x^*(i)$ for all $s \in [i, i + 1)$. Like the SAA formulation for chance constraints, the solution path $x^*(s)$ can now be viewed as piecewise constant in $s \in [1, n]$ hence Assumption 7 holds. Therefore we have:

THEOREM 16 (Applying univariate Gaussian validator to SO). *Consider $OPT(s)$ given by (32). Assumption 7 holds for some $M \leq n$.*

Lastly, our univariate Gaussian validator also works on a variant of SO called FAST (Carè et al. (2014)), in a sense that we will detail in Section 6.3. FAST differs from the formulations we have discussed so far in that its solution path does not come as solutions of a parametrized optimization problem, but from a line segment connecting two suitably chosen solutions. Nonetheless, the notion of solution-path optimality still applies. In particular, the solution-path optimum is unique if the objective is strictly convex or linear, and all the statistical guarantees in Theorem 5 can be established using the same proof.

6. Numerical Experiments

We present numerical results to demonstrate the performances of our framework in several data-driven reformulations. We consider the following linear CCP

$$\min c'x \text{ subject to } \mathbb{P}_F(\xi'x \leq b) \geq 1 - \alpha \quad (33)$$

where $c \in \mathbb{R}^d$, $b \in \mathbb{R}$ are deterministic, the distribution F of the randomness $\xi \in \mathbb{R}^d$ is multivariate Gaussian with mean μ and covariance Σ , and the tolerance level $1 - \alpha$ is set to 90%.

We consider a range of data-driven reformulations, including RO (or relatedly SCA), DRO (moment-based), and SO (including its variant FAST). In our experiments, we generate i.i.d. data $\xi_1, \dots, \xi_n$ from the underlying true distribution F . Then, using a chosen reformulation, we compute a solution $\hat{x}$ of (33) that attempts to satisfy the chance constraint with a 95% confidence level, while attain an objective value $c'\hat{x}$ as low as possible. For each reformulation, we compare the performance of an existing benchmark with unnormalized and normalized Gaussian supremum validators (Algorithms 2 and 3) and univariate Gaussian validator (Algorithm 4), in terms of both feasibility and optimality. Moreover, we also test a naive validator that directly compares the sample mean to γ when checking feasibility, i.e., without the Gaussian margin $\frac{z_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}}$ in (15), in addition to the three proposed validators, which serves to demonstrate the necessity of the proposed Gaussian margins in the validation procedure. The “plain average” column of each table displays results of this extra validator. “unnorm. GS” denotes the unnormalized Gaussian supremum validator (Algorithm 2), “norm. GS” denotes the normalized Gaussian supremum validator (Algorithm 3), and “uni. Gaussian” denotes the univariate Gaussian validator (Algorithm 4). When applying these validators in all experiments, we use the simple allocation rule of dividing the overall data size into Phases 1 and 2 equally, except only in the case of basic SO where a too small Phase 1 data size is provably subpar in guaranteeing feasibility.

To collect statistically meaningful estimates, for each formulation we repeat the experiments 1000 times each with an independently generated data set and a data-driven solution output. We take down the average objective value achieved by these solutions (the “mean obj. val.” row of each table) and the proportion of

feasible solutions as the empirical feasibility coverage (the “feasibility level” row of each table). Therefore, the smaller the “mean obj. val.” is, the better is the solution in terms of optimality, and “feasibility level” $\geq 95\%$ indicates that the desired feasibility confidence level is achieved and otherwise not.

6.1. RO and SCA

We first test the proposed framework on RO. We use the ellipsoid uncertainty set that leads to a robust counterpart in the form described in Example 2, i.e., $\hat{\mu}'x + \sqrt{s}\|\hat{\Sigma}^{1/2}x\|_2 \leq b$ where $\hat{\mu}$ and $\hat{\Sigma}$ are the sample mean and covariance for ξ computed from Phase one data. The benchmark (“SCA” in the tables) is set to an SCA (equation 2.4.11 of Ben-Tal et al. (2009)) for unbounded ξ , which in our case can be expressed as

$$\mu'x + \sqrt{2\log \frac{1}{\alpha}} \sqrt{\sum_{k=1}^d (z^k'x)^2} = \mu'x + \sqrt{2\log \frac{1}{\alpha}} \|\Sigma^{1/2}x\|_2 \leq b \quad (34)$$

where μ is the true mean, and z^k is the k -th column of the square root $\Sigma^{1/2}$ of the true covariance matrix Σ . Note that (34) is equivalent to the RO formulation with true mean and covariance and parameter value $s = 2\log \frac{1}{\alpha}$. Here, we give this SCA or RO the advantage of knowing the true mean and covariance of the randomness.

To implement our validator, we need to provide a set of parameter values $\{s_1, \dots, s_p\}$ at which the RO is solved. We take the $(1 - \alpha)n_1$ -th order statistic $\hat{s}_{1-\alpha}$ of $\{(\xi_{n_2+i} - \hat{\mu})'\hat{\Sigma}^{-1}(\xi_{n_2+i} - \hat{\mu}) : i = 1, \dots, n_1\}$, where $\xi_{n_2+i}, i = 1, \dots, n_1$ are the Phase one data, so that $\{\xi : (\xi - \hat{\mu})'\hat{\Sigma}^{-1}(\xi - \hat{\mu}) \leq \hat{s}_{1-\alpha}\}$ is roughly a $(1 - \alpha)$ -content set for ξ (such type of quantile-based selection has been used in Hong et al. (2017)). We then set the values $s_j = (\hat{s}_{1-\alpha} + 20) \frac{j}{50}$ for $j = 1, \dots, 50$ ($p = 50$). Table 6.1 shows the range of feasible hyperparameters and Figure 6.1 summarize the results under different problem dimensions and data sizes.

Table 1 Feasible RO hyperparameter values. Data size ranges from $n = 100$ to 2000.

n	uni. Gaussian	Sectioning	Cross Validation	Bootstrapping
100	20-60%	20-30%	10, 5, 3	10, 5, 3
200	10-80%	10-70%	10, 5, 3	10, 5, 3
300	10-80%	10-80%	10, 5, 3	10, 5, 3
400	10-90%	10-80%	10, 5, 3	10, 5, 3
500	10-90%	10-80%	10, 5, 3	10, 5, 3
≥ 600	10-90%	10-90%	10, 5, 3	10, 5, 3

We highlight a few observations. First, our framework with the three proposed validators outperforms the SCA benchmark. In terms of the objective performance, both our unnormalized and normalized Gaussian supremum validators, and univariate Gaussian validators, achieve lower objective value than SCA (with a difference ≥ 0.6), while at the same time retain the feasibility confidence to above 95% in all the three tables. In particular, as the dimension grows from 10 (Tables ?? and ??) to 50 (Table ??), the feasibility confidence level remains above 95%, consistent with the dimension-free feasibility guarantee of our methods.

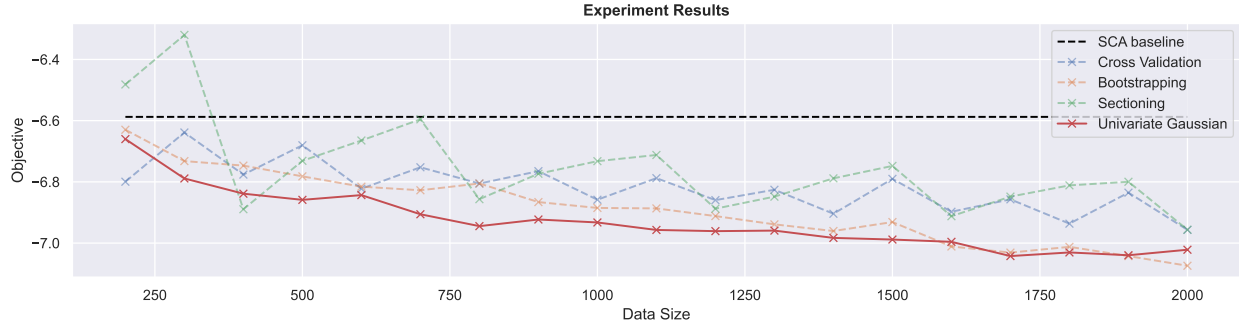


Figure 1 Performance of univariate gaussian validator against other validation methods on RO.

Second, among the three proposed validators, the univariate Gaussian validator appears less conservative than the Gaussian supremum counterparts in achieving better objective values, and relatedly tighter feasibility confidence levels (i.e., closer to 95%). Specifically, the univariate Gaussian validator gives a feasibility confidence level around 98% in all the three tables, whereas the Gaussian supremum validators give a level between 99%-100% (and also 0.1-0.4 higher mean objective values). Finally, we comment that the “plain average” scheme does not have the desired feasibility confidence level even when the data size is as large as 500 (Table ??), which shows that margin adjustments to the naive sample average in the validators is necessary to ensure feasibility.

6.2. Moment-based DRO

The second formulation we consider is a moment-based DRO. We use the formulation

$$\inf_{\xi \sim Q \text{ s.t. } (\mathbb{E}_Q[\xi], \text{Cov}_Q(\xi)) \in \mathcal{U}_s} \mathbb{P}_Q(\xi'x \leq b) \geq 1 - \alpha \quad (35)$$

where $\mathcal{U}_s$ is a confidence region for the true mean and covariance of ξ obtained via the delta method described in Example 3 (see Section 6 of Marandi et al. (2019) for details). According to (29), (35) can be expressed as $\sup_{(\mu, \Sigma) \in \mathcal{U}_s} \mu'x + \sqrt{\frac{1-\alpha}{\alpha}} \|\Sigma^{1/2}x\|_2 \leq b$, which can be further reformulated as a conic constraint (see Theorem 1 of Marandi et al. (2019)). In the benchmark case “DRO (χ^2 quantile)” we choose s to be the 95% quantile of the limiting χ^2 distribution as suggested in Marandi et al. (2019) so that $\mathcal{U}_s$ is a valid 95% confidence region. In our framework, we solve the DRO formulation at parameter values $s_j = 1.5\hat{s}_{0.95} \frac{j}{50}$ for $j = 1, \dots, 50$ where $\hat{s}_{0.95}$ is the χ^2 quantile used in the benchmark. Tables 2 and 3 show the experimental results under different data sizes.

Table 2 Moment-based DRO. $d = 10, n = 200$. Data are split to $n_1 = 100, n_2 = 100$.

	DRO (χ^2 quantile)	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-1.83	-2.73	-2.73	-2.73	-2.73
feasibility level	100%	100%	100%	100%	100%

Table 3 Moment-based DRO. $d = 10, n = 500$. Data are split to $n_1 = 250, n_2 = 250$.

	DRO (χ^2 quantile)	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-2.00	-2.62	-2.62	-2.62	-2.62
feasibility level	100%	100%	100%	100%	100%

The comparisons between the benchmark and our framework here share similarities with the RO setting. The solutions output from our validators possess superior objective performance (with a difference of 0.6-0.9) than simply setting s to be the 95%-level χ^2 quantile, while still attain the desired feasibility confidence level. Note that all validators (including the “plain average”) give the same objective value (-2.73 in Table 2 and -2.62 in Table 3), and have a 100% feasibility confidence. This is because the chosen parameter s turns out to be 0 for all of them. In other words, setting the moment constraints as equalities (to the estimated moments from Phase one data) is statistically feasible and achieves the best objective value, and any relaxation from this would lead to a deterioration of solution quality. This hints that the conventional choices of moment set size suggested in the literature could be very conservative.

6.3. SO

Given the Phase one data $\{\xi_{n_2+1}, \dots, \xi_n\}$, we consider the data-driven feasible region specified by the first s sampled constraints, $\xi'_{n_2+i}x \leq b$ for $i = 1, \dots, s$, and tune the number of satisfied constraints $s \in \{1, 2, \dots, n_1\}$. The benchmark “SO” in this case is to impose all the constraints given by the whole data set $\{\xi_1, \dots, \xi_n\}$. Tables 4 and 5 summarize the results for data size $n = 200, 500$ respectively.

Table 4 SO. $d = 10, n = 200$. Data are split to $n_1 = 150, n_2 = 50$.

	SO	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-3.90	-4.24	-4.31	-4.46	-4.91
feasibility level	99.7%	95.2%	94.0%	85.1%	44.7%

Table 5 SO. $d = 10, n = 500$. Data are split to $n_1 = 250, n_2 = 250$.

	SO	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-3.28	-3.86	-4.10	-4.30	-4.69
feasibility level	100%	99.7%	98.7%	95.6%	62.0%

We observe the gain in objective performance of our validators compared to SO (a difference of 0.3-0.6 in Table 4 and 0.6-1.0 in Table 5). We also note the drastic failure of “plain average” in rendering the desired 95% feasibility confidence, thus showing that a margin adjustment to the validators is necessary. Our validators maintain feasibility in all cases, except the univariate Gaussian validator for $n = 200$. This deficiency is attributed to two potential reasons. First is that with $n_1 = 150$ there is a non-negligible chance that none of the n_1 solutions $x^*(s), s = 1, \dots, n_1$, produced in Phase one is feasible, thus violating Assumption 8. In

fact, the infeasibility probability of the solution derived by an SO using all the n_1 constraints can be computed to be 6% (Campi and Garatti (2008)), leaving the actual confidence of obtaining a feasible solution at most 94%. The second possible cause is the finite-sample coverage error of the univariate Gaussian validator, seeing that the validation data size $n_2 = 50$ is relatively small. When both n_1 and n_2 increase to 250 in Table 5, the desired feasibility confidence level is recovered for the univariate Gaussian validator as the chance of all solution candidates being infeasible decreases to $< 0.2\%$ and the finite-sample error is reduced due to a larger validation data size. Finally, although we do not pursue here, we should mention that the performances of the basic SO considered in the tables can plausibly be boosted by using techniques such as sampling-and-discarding (Campi and Garatti (2011)) and wait-and-judge (Campi and Garatti (2018)). Comprehensive comparisons with these enhanced techniques would be left as important future work.

Lastly, we consider a variant of SO called FAST (Carè et al. (2014)), designed originally to tone down the sample size requirement in basic SO. Our comparison with FAST here is motivated by its similarity with our framework in that it also splits the data into two portions and uses a validation-based idea. With the first portion of data $\{\xi_{n_2+i}, i = 1, \dots, n_1\}$, FAST computes a solution $\hat{x}$ by imposing all the n_1 constraints $\xi'_{n_2+i}x \leq b$ as in the basic SO, and then uses the second portion to obtain the final solution $\hat{x}^*$ by solving the following program

$$\min c'((1-s)x_o + s\hat{x}) \text{ subject to } \xi'_i((1-s)x_o + s\hat{x}) \leq b \text{ for all } i = 1, \dots, n_2 \text{ and } 0 \leq s \leq 1$$

where x_o is a feasible solution of (33) with $\mathbb{P}_F(\xi'x_o \leq b) = 1$. One particular choice of x_o for problem (33) is the vector of all zeros and is used in the experiment. When applying our framework to FAST, we search for the best feasible solution along the line segment $\{x^*(s) = (1-s)x_o + s\hat{x} : s \in [0, 1]\}$ by validating solutions $x^*(s_j)$ at parameter values $s_j = \frac{j-1}{10}$ for $j = 1, \dots, 11$ ($p = 11$). Tables 6 and 7 show the results under different dimensions and data sizes.

Table 6 FAST. $d = 10, n = 200$. Data are split to $n_1 = 100, n_2 = 100$.

	FAST	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-2.54	-3.55	-3.68	-3.87	-4.44
feasibility level	100%	98.9%	98.9%	97.3%	79.6%

Table 7 FAST. $d = 50, n = 500$. Data are split to $n_1 = 250, n_2 = 250$.

	FAST	unnorm. GS	norm. GS	uni. Gaussian	plain average
mean obj. val.	-8.92	-14.11	-15.06	-15.80	-18.14
feasibility level	100%	99.8%	99.3%	98.0%	76.7%

Similar phenomena persist from our previous settings. Our three validators give tighter feasibility confidence levels and better objective performances (with a difference of ≥ 1 in Table 6 and ≥ 5 in Table

7) compared to FAST. Among them, univariate Gaussian validator gives the tightest feasibility confidence level and best objective value. The naive “plain average” validator fails in attaining the desired feasibility confidence. Here we have used a rather coarse mesh with only 11 parameter values, and expect a sharper improvement should a finer mesh be used.

6.4. Comparison with Existing Validation Schemes

For the rest of this section we compare the performance of the normalized Gaussian supremum validator and univariate Gaussian validator against K -fold cross validation and bootstrapping, two commonly used validation schemes, as well as sectioning, a validation scheme introduced by Xie (2018) (see Appendix A for details to these schemes). We consider the same SO setup as in the previous section, as well as an SAA formulation and a Wasserstein DRO formulation. The SAA formulation takes the form $\frac{1}{n_1} \sum_{i=1}^{n_1} \mathbf{1}(\xi'_{n_2+i} x \leq b) \geq 1 - \alpha + s$, which we solved at parameter values $s_j = 0.002j$ for $j = 1, \dots, 25$. The Wasserstein DRO takes the form $\inf_{G \in \mathcal{B}(\hat{F}, s)} P_G(\xi' x \leq b) \geq 1 - \alpha$ where $\hat{F} = \frac{1}{n_1} \sum_{i=1}^{n_1} \delta_{\xi_{n_2+i}}$ is the empirical distribution with δ_ξ denoting the Dirac measure at ξ and $\mathcal{B}(\hat{F}, s)$ is the Wasserstein ball of radius $s > 0$ surrounding $\hat{F}$, that is, $B(\hat{F}, s) := \{G : G \text{ is a probability distribution on } \mathbb{R}^d \text{ s.t. } d_w(G, \hat{F}) \leq s\}$ with

$$d_w(G_1, G_2) := \inf \left\{ E_P[\|\xi - \xi'\|] : \begin{array}{l} P \text{ is a joint probability distribution of } (\xi, \xi') \in \mathbb{R}^d \times \mathbb{R}^d \\ \text{with marginals } G_1 \text{ and } G_2 \text{ for } \xi \text{ and } \xi' \text{ respectively} \end{array} \right\}.$$

If $b \geq 0$, this distributionally robust chance constraint $Z := \{x \in \mathbb{R}^d : \inf_{G \in \mathcal{B}(\hat{F}, s)} P_G(\xi' x \leq b) \geq 1 - \alpha\}$ has the following equivalent reformulation (see Theorem 2 of Xie (2018))

$$Z = \left\{ x \in \mathbb{R}^d : \begin{array}{l} sv + \frac{1}{n_1} \sum_{i=1}^{n_1} z_i \leq \alpha r \\ r \leq z_i + s_i \text{ for } i = 1, \dots, n_1 \\ s_i \leq b - \xi'_{n_2+i} x + M(1 - y_i) \text{ for } i = 1, \dots, n_1 \\ s_i \leq M y_i \text{ for } i = 1, \dots, n_1 \\ \|x\|_* \leq v \\ v \geq 0, r \geq 0, s_i \geq 0, z_i \geq 0, y_i \in \{0, 1\} \text{ for } i = 1, \dots, n_1 \end{array} \right\}$$

where $\|\cdot\|_*$ is the dual norm of $\|\cdot\|$, and M can be any number such that

$$M \geq \max_{i=1, \dots, n_1} \max_{x: 0 \leq x_l \leq 1 \text{ for each } l} |b - \xi'_{n_2+i} x|.$$

We set $M = |b| + \max_{i=1, \dots, n} \|\xi_i\|_1$, with $\|\cdot\|_1$ denoting the l_1 -norm, and solved the DRO formulation (as a mixed-integer program) at parameter values $s \in \{\frac{3}{2^j} : j = 0, \dots, 9\} \cup \{\frac{3/512}{1.2^j} : j = 1, \dots, 15\}$. These experiments were performed with a dimension size d of 10, a tolerance level $1 - \alpha$ of 95%, and a confidence level $1 - \beta$ of 90%. Cross validation (bootstrapping) was implemented with 3, 5, and 10 folds (bootstrap resamples). Sectioning, with 10 folds, as well as our proposed approaches were implemented at n_1 (training set) values of 10% to 90% of the data size n , in intervals of 10%, with the rest of the data allocated to n_2 (validation set).

For the SO formulation, we see that for sufficiently large data sizes and a sensible percentage of data allocated to Phase one, our proposed methods perform well (see Figure 2). If the percentage of data allocated

to n_1 is too small, however, the solutions are far from meeting the desired confidence level. For smaller data sizes, cross validation and bootstrapping seem to provide closer confidence levels to that desired. For larger data sizes where all the validators begin meeting the confidence level, the existing validation methods do not appear to provide better objective values compared to our proposed validators. Finally, all methods seem to outperform the “plain average.” Similar observations for the SAA and Wasserstein DRO formulations can be seen for the existing methods and our proposed validators (see Appendix B).

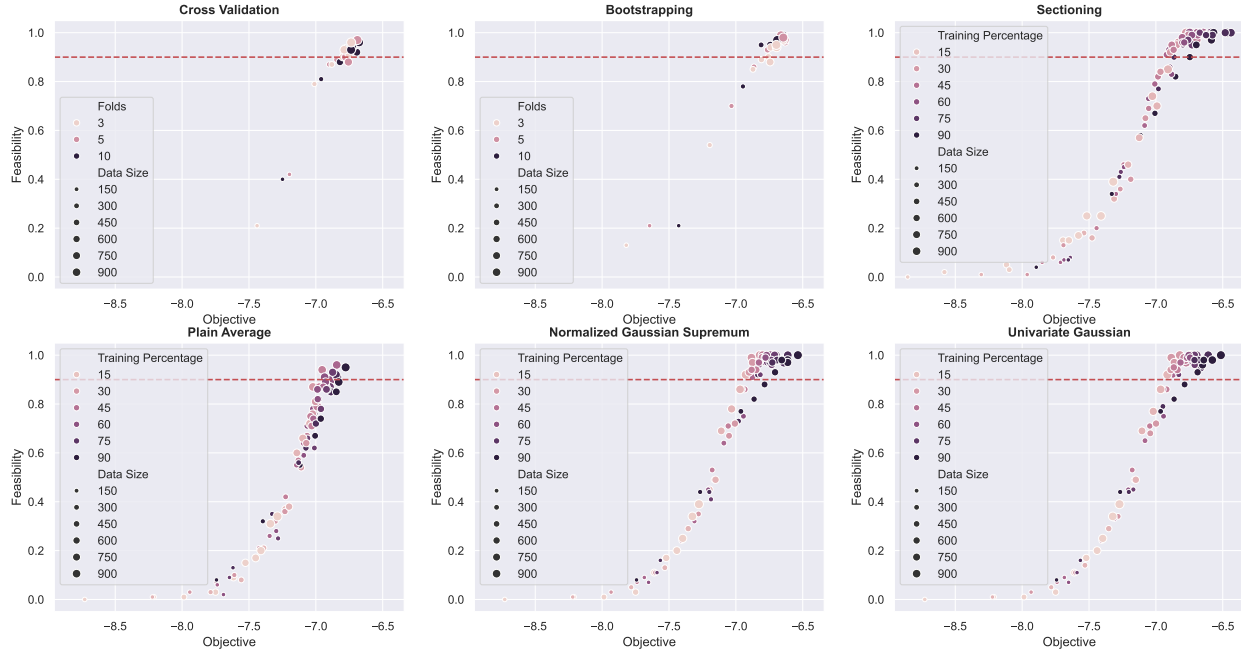


Figure 2 Performance of the validation methods considered at different parameter values. The red dashed line indicates the 90% feasibility confidence level.

Tables ??, ??, and ?? show the hyperparameter values that return solutions which meet the confidence level for their respective reformulation. For our proposed approaches and sectioning, these values are the percentage of data allocated to n_1 (e.g., 70% means that $n_1 = 0.7n$ and $n_2 = 0.3n$). For cross validation and bootstrapping, these values are the number of folds and bootstrap resamples, respectively. “-” indicates that no tested hyperparameter values return a solution meeting the confidence level. Data sizes smaller than those tested also did not return a solution meeting the confidence level. For data sizes greater than those tested, using cross validation and bootstrapping started to become computationally intractable.

We observe that, for small data sizes, no relative budgeting of samples between the two phases, that is, values of n_1 and n_2 , lead to a solution that meets the confidence level. Additionally, for small data sizes, the existing methods do not lead to a solution that meet the confidence level as well. However, we note that for the SAA formulation, cross validation and bootstrapping begin to return solutions that meet the confidence level at a smaller data size of $n = 200$ compared to our proposed validators. This aligns with

our earlier observation, where cross validation and bootstrapping return solutions with higher confidence levels at lower data sizes and hence may be preferred. As the data size increases, solutions that meet the confidence level are found and the relative budgeting of the samples between the two phases can be done more flexibly. For both our proposed approaches, an allocation of 70% of the data to n_1 seems to work best in terms of meeting the confidence level, as it does so for all data sizes sufficiently large. These empirical results provide insight in the relative budgeting between n_1 and n_2 , particularly showing the importance of n_1 . A rigorous methodology for selecting n_1 and n_2 would definitely be worthwhile exploring in future work, especially since most of our assumptions and theorems are based on n_2 and do not carry explicitly the notions of n_1 . Finally, we see that for cross validation and bootstrapping, 10 folds (respectively, bootstrap resamples) seem to work best in meeting the confidence level across the different data sizes.

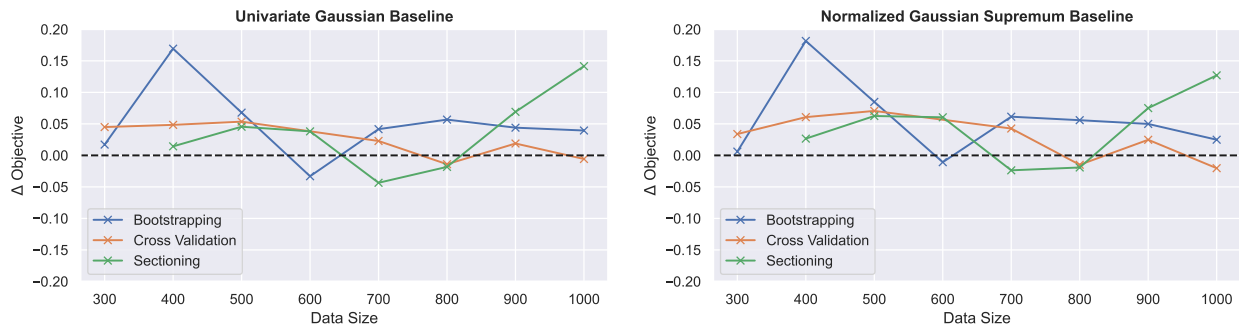


Figure 3 Increase in objective value of the existing validation methods compared to the baseline Univariate Gaussian (left) and Normalized Gaussian Supremum (right) validators for the SO formulation.

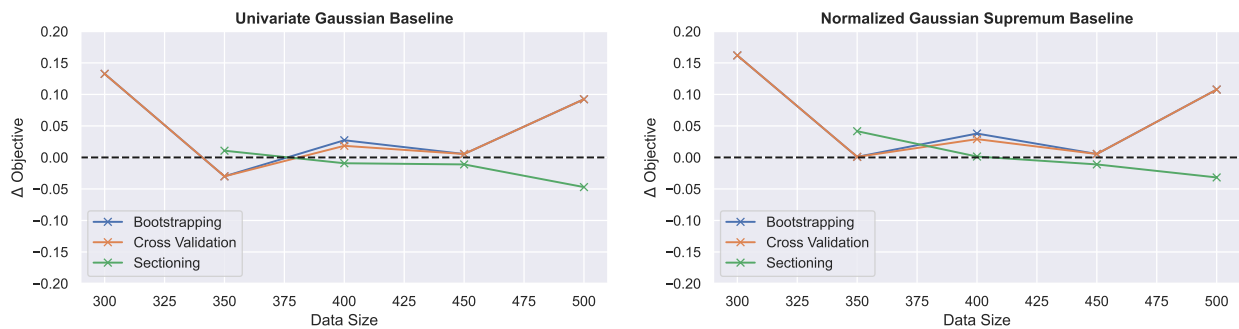


Figure 4 Increase in objective value of the existing validation methods compared to the baseline Univariate Gaussian (left) and Normalized Gaussian Supremum (right) validators for the SAA formulation.

(TO EDIT) If we assign 70% of the data to Phase one for our proposed validators and set the number of folds/bootstrap resamples to 10 for the existing methods, we see that... Since CV and SEC both require significantly greater computational cost (at least a factor of K times greater), prefer to use proposed validators when the data size is large enough. For smaller data sizes which are faster to solve, CV or BS may

be preferred as they seem to be more likely to lead to solutions that either meet the confidence level or are closer to meeting the confidence level compared to our proposed approaches.

7. Conclusion

We have studied a validation-based framework to combat the conservativeness in data-driven optimization with uncertain constraints. We have demonstrated how the conventional approaches in several optimization paradigms, including SAA, RO and DRO, implicitly estimate the whole feasible region. This in turn leads to over-conservativeness caused by the need to control huge simultaneous estimation errors, especially for high-dimensional problems. On the other hand, we have also demonstrated that the solution output from these reformulation classes can often be represented in a low-dimensional manifold parametrized by key conservativeness parameters. Our framework leverages this low dimensionality by extracting the parametrized solution path and selecting the best parameter value. We have proposed two types of validators for this parameter selection, one utilizing a multivariate Gaussian supremum (unnormalized or normalized) and another utilizing a univariate Gaussian, to set the margin in a sample average constraint when optimizing over the solution path. We have shown that the obtained solutions enjoy asymptotic and finite-sample performance guarantees on feasibility that scale lightly with the problem dimension, and asymptotic optimality within the reformulation class. The Gaussian supremum validator requires less regularity conditions and is applicable more generally, whereas the univariate Gaussian validator provides tighter guarantees when applicable. Our numerical results support these findings and show that our framework and validators consistently provide better solutions compared to several benchmarks in terms of better objective values and tighter feasibility confidence. Our study provides a first rigorous validation-based framework to tackle over-conservativeness in data-driven constrained optimization, and is foreseen to open up follow-up investigations on more powerful validation strategies and refined statistical guarantees regarding joint feasibility and optimality.

References

- Atlason J, Epelman MA, Henderson SG (2004) Call center staffing with simulation and cutting plane methods. *Annals of Operations Research* 127(1-4):333–358.
- Bayraksan G, Love DK (2015) Data-driven stochastic programming using phi-divergences. *INFORMS TutORials in Operations Research* 1–19.
- Ben-Tal A, Den Hertog D, De Waegenaere A, Melenberg B, Rennen G (2013) Robust solutions of optimization problems affected by uncertain probabilities. *Management Science* 59(2):341–357.
- Ben-Tal A, El Ghaoui L, Nemirovski A (2009) *Robust optimization* (Princeton, NJ: Princeton University Press).
- Ben-Tal A, Nemirovski A (2000) Robust solutions of linear programming problems contaminated with uncertain data. *Mathematical Programming* 88(3):411–424.

- Bertsimas D, Brown DB, Caramanis C (2011) Theory and applications of robust optimization. *SIAM Review* 53(3):464–501.
- Bertsimas D, Gupta V, Kallus N (2018) Data-driven robust optimization. *Mathematical Programming* 167(2):235–292.
- Blanchet J, Kang Y (2016) Sample out-of-sample inference based on Wasserstein distance. *arXiv preprint arXiv:1605.01340*.
- Blanchet J, Kang Y, Murthy K (2016) Robust Wasserstein profile inference and applications to machine learning. *arXiv preprint arXiv:1610.05627*.
- Blanchet J, Murthy K (2019) Quantifying distributional model risk via optimal transport. *Mathematics of Operations Research* 44(2):565–600.
- Bonnans JF, Shapiro A (2013) *Perturbation analysis of optimization problems* (Springer Science & Business Media).
- Boucheron S, Lugosi G, Massart P (2013) *Concentration inequalities: A nonasymptotic theory of independence* (Oxford University Press).
- Calafiore G, Campi MC (2005) Uncertain convex programs: randomized solutions and confidence levels. *Mathematical Programming* 102(1):25–46.
- Calafiore GC (2017) Repetitive scenario design. *IEEE Transactions on Automatic Control* 62(3):1125–1137.
- Campi MC, Carè A (2013) Random convex programs with L₁-regularization: sparsity and generalization. *SIAM Journal on Control and Optimization* 51(5):3532–3557.
- Campi MC, Garatti S (2008) The exact feasibility of randomized solutions of uncertain convex programs. *SIAM Journal on Optimization* 19(3):1211–1230.
- Campi MC, Garatti S (2011) A sampling-and-discarding approach to chance-constrained optimization: feasibility and optimality. *Journal of Optimization Theory and Applications* 148(2):257–280.
- Campi MC, Garatti S (2018) Wait-and-judge scenario optimization. *Mathematical Programming* 167(1):155–189.
- Carè A, Garatti S, Campi MC (2014) Fast—fast algorithm for the scenario technique. *Operations Research* 62(3):662–671.
- Chernozhukov V, Chetverikov D, Kato K, et al. (2017) Central limit theorems and bootstrap in high dimensions. *The Annals of Probability* 45(4):2309–2352.
- Delage E, Ye Y (2010) Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Operations Research* 58(3):595–612.
- Doan XV, Li X, Natarajan K (2015) Robustness to dependency in portfolio optimization using overlapping marginals. *Operations Research* 63(6):1468–1488.
- Duchi J, Glynn P, Namkoong H (2016) Statistics of robust optimization: A generalized empirical likelihood approach. *arXiv preprint arXiv:1610.03425*.

- Dupuis P, Katsoulakis MA, Pantazis Y, Plecháč P (2016) Path-space information bounds for uncertainty quantification and sensitivity analysis of stochastic dynamics. *SIAM/ASA Journal on Uncertainty Quantification* 4(1):80–111.
- Dür M, Jargalsaikhan B, Still G (2016) Genericity results in linear conic programming—a tour d’horizon. *Mathematics of Operations Research* 42(1):77–94.
- Esfahani PM, Kuhn D (2018) Data-driven distributionally robust optimization using the wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming* 171(1-2):115–166.
- Fournier N, Guillin A (2015) On the rate of convergence in wasserstein distance of the empirical measure. *Probability Theory and Related Fields* 162(3-4):707–738.
- Freund RM (1985) Postoptimal analysis of a linear program under simultaneous changes in matrix coefficients. *Mathematical Programming Essays in Honor of George B. Dantzig Part I*, 1–13 (Springer).
- Gao R, Kleywegt AJ (2016) Distributionally robust stochastic optimization with wasserstein distance. *arXiv preprint arXiv:1604.02199*.
- Ghaoui LE, Oks M, Oustry F (2003) Worst-case value-at-risk and robust portfolio optimization: A conic programming approach. *Operations research* 51(4):543–556.
- Glasserman P, Xu X (2014) Robust risk measurement and model risk. *Quantitative Finance* 14(1):29–58.
- Goh J, Sim M (2010) Distributionally robust optimization and its tractable approximations. *Operations Research* 58(4-part-1):902–917.
- Goldfarb D, Iyengar G (2003) Robust portfolio selection problems. *Mathematics of Operations Research* 28(1):1–38.
- Gotoh Jy, Kim MJ, Lim AE (2018) Robust empirical optimization is almost the same as mean–variance optimization. *Operations research letters* 46(4):448–452.
- Gupta V (2019) Near-optimal bayesian ambiguity sets for distributionally robust optimization. *Management Science, Articles in Advance* 1–19.
- Hanasusanto GA, Roitch V, Kuhn D, Wiesemann W (2015) A distributionally robust perspective on uncertainty quantification and chance constrained programming. *Mathematical Programming* 151(1):35–62.
- Hoeffding W (1963) Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association* 58(301):13–30.
- Hong LJ, Huang Z, Lam H (2017) Learning-based robust optimization: Procedures and statistical guarantees. *arXiv preprint arXiv:1704.04342*.
- Hu Z, Hong LJ (2013) Kullback-Leibler divergence constrained distributionally robust optimization. *Available at Optimization Online*.
- Jiang R, Guan Y (2016) Data-driven chance constrained stochastic program. *Mathematical Programming* 158(1-2):291–327.
- Kosorok MR (2008) *Introduction to empirical processes and semiparametric inference*. (Springer).

- Krokhmal P, Palmquist J, Uryasev S (2002) Portfolio optimization with conditional value-at-risk objective and constraints. *Journal of Risk* 4:43–68.
- Lam H (2016) Robust sensitivity analysis for stochastic systems. *Mathematics of Operations Research* 41(4):1248–1275.
- Lam H (2018) Sensitivity to serial dependency of input processes: A robust approach. *Management Science* 64(3):1311–1327.
- Lam H (2019) Recovering best statistical guarantees via the empirical divergence-based distributionally robust optimization. *Operations Research* 67(4):1090–1105.
- Lam H, Zhou E (2017) The empirical likelihood approach to quantifying uncertainty in sample average approximation. *Operations Research Letters* 45(4):301–307.
- Lifshits MA (2013) *Gaussian random functions*, volume 322 (Springer Science & Business Media).
- Luedtke J, Ahmed S (2008) A sample approximation approach for optimization with probabilistic constraints. *SIAM Journal on Optimization* 19(2):674–699.
- Marandi A, Ben-Tal A, Hertog Dd, Melenberg B (2019) Extending the scope of robust quadratic optimization. *arXiv preprint arXiv:1909.01762* .
- Maurer A, Pontil M (2009) Empirical bernstein bounds and sample variance penalization. *arXiv preprint arXiv:0907.3740* .
- Natarajan K, Pachamanova D, Sim M (2008) Incorporating asymmetric distributional information in robust value-at-risk optimization. *Management Science* 54(3):573–585.
- Nemirovski A (2003) On tractable approximations of randomly perturbed convex constraints. *42nd IEEE International Conference on Decision and Control*, volume 3, 2419–2422 (IEEE).
- Nemirovski A, Shapiro A (2006) Convex approximations of chance constrained programs. *SIAM Journal on Optimization* 17(4):969–996.
- Peel T, Anthoine S, Ralaivola L (2010) Empirical bernstein inequalities for U-statistics. *Advances in Neural Information Processing Systems*, 1903–1911.
- Petersen IR, James MR, Dupuis P (2000) Minimax optimal control of stochastic uncertain systems with relative entropy constraints. *IEEE Transactions on Automatic Control* 45(3):398–412.
- Prékopa A (2003) Probabilistic programming. *Handbooks in operations research and management science* 10:267–351.
- Schildbach G, Fagiano L, Morari M (2013) Randomized solutions to convex programs with multiple chance constraints. *SIAM Journal on Optimization* 23(4):2479–2501.
- Shapiro A, Dentcheva D, Ruszczyński A (2009) *Lectures on Stochastic Programming: Modeling and Theory* (SIAM).
- Tulabandhula T, Rudin C (2014) Robust optimization using machine learning for uncertainty sets. *arXiv preprint arXiv:1407.1097* .

- Van der Vaart AW, Wellner JA (1996) *Weak Convergence and Empirical Processes with Applications to Statistics* (Springer).
- Van Parys BP, Goulart PJ, Kuhn D (2016) Generalized gauss inequalities via semidefinite programming. *Mathematical Programming* 156(1-2):271–302.
- Vershynin R (2010) Introduction to the non-asymptotic analysis of random matrices. *arXiv preprint arXiv:1011.3027*.
- Wand MP, Jones MC (1994) *Kernel smoothing* (Chapman and Hall/CRC).
- Wang W, Ahmed S (2008) Sample average approximation of expected value constrained stochastic programs. *Operations Research Letters* 36(5):515–519.
- Wiesemann W, Kuhn D, Sim M (2014) Distributionally robust convex optimization. *Operations Research* 62(6):1358–1376.
- Xie W (2018) On distributionally robust chance constrained programs with Wasserstein distance. *arXiv preprint arXiv:1806.07418*.
- Xu H, Caramanis C, Mannor S (2012) Optimization under probabilistic envelope constraints. *Operations Research* 60(3):682–699.

Appendix A: Validation Schemes

Algorithm 5 K-Fold Cross Validation

Input: $\Xi = \{\xi_1, \dots, \xi_n\}$, $\{s_1, \dots, s_p\}$, $1 - \beta$, the number of folds K

1. Randomly partition the data into K subsets of equal size, $\Xi_k, k = 1, \dots, K$ such that each Ξ_k contains n/K data and $\Xi = \cup_{k=1}^K \Xi_k$.
2. For each $k = 1, \dots, K$ and each $j = 1, \dots, p$, use all the data except the k -th group to solve the given reformulation with $s = s_j$ and $\hat{F} = \frac{K}{(K-1)n} \sum_{\xi \in \Xi \setminus \Xi_k} s_\xi$ to obtain its optimal solution $x_k^*(s_j)$, and then estimate the chance constraint $\hat{P}_k(s_j) = \frac{K}{n} \sum_{\xi \in \Xi_k} \mathbf{1}\{\xi^T x_k^*(s_j) \leq b\}$.

3. Compute

$$s^* = \arg \min_{s_j, 1 \leq j \leq p} \left\{ \frac{1}{K} \sum_{k=1}^K c^T x_k^*(s_j) : \frac{\sum_{k=1}^K \mathbf{1}\{\hat{P}_k(s_j) \geq 1 - \alpha\}}{K} \geq 1 - \beta, j = 1, \dots, p \right\}$$

4. Solve the reformulation with $s = s^*$, $\hat{F} = \frac{1}{n} \sum_{i=1}^n s_{\xi_i}$, and all the data Ξ to obtain $x^*(s^*)$.

Output: $s^*, x^*(s^*)$.

Algorithm 6 Bootstrapping (Section 7.2.3 in Esfahani and Kuhn (2018))

Input: $\Xi = \{\xi_1, \dots, \xi_n\}, \{s_1, \dots, s_p\}, 1 - \beta$, the number of bootstrap resamples B

1. For each $b = 1, \dots, B$, uniformly draw n data with replacement from the data set Ξ to get the resampled data $\xi_1^b, \dots, \xi_n^b$, and let $\Xi_b^c = \Xi \setminus \{\xi_1^b, \dots, \xi_n^b\}$ be the remaining data.

2. For each $b = 1, \dots, B$ and each $j = 1, \dots, p$, use the b -th resample $\xi_1^b, \dots, \xi_n^b$ to solve the reformulation with $s = s_j$ and $\hat{F} = \frac{1}{n} \sum_{i=1}^n s_{\xi_i^b}$ to obtain its optimal solution $x_b^*(s_j)$, and then estimate the chance constraint $\hat{P}_b(s_j) = \frac{1}{\text{card}(\Xi_b^c)} \sum_{\xi \in \Xi_b^c} \mathbf{1}\{\xi^T x_b^*(s_j) \leq b\}$, where $\text{card}(\cdot)$ denotes the cardinality of a set.

3. Compute

$$s^* = \arg \min_{s_j, 1 \leq j \leq p} \left\{ \frac{1}{B} \sum_{k=1}^B c^T x_k^*(s_j) : \frac{\sum_{b=1}^B \mathbf{1}\{\hat{P}_b(s_j) \geq 1 - \alpha\}}{B} \geq 1 - \beta, j = 1, \dots, p \right\}$$

4. Solve the reformulation with $s = s^*$, $\hat{F} = \frac{1}{n} \sum_{i=1}^n s_{\xi_i}$, and all the data Ξ to obtain $x^*(s^*)$.

Output: $s^*, x^*(s^*)$.

Algorithm 7 Sectioning (Section 5.2 in Xie (2018))

Input: $\Xi = \{\xi_1, \dots, \xi_n\}, \{s_1, \dots, s_p\}, 1 - \beta, n_1, n_2$, the number of sections K

1. Randomly split Ξ into Ξ_t and Ξ_v such that $\text{card}(\Xi_t) = n_1$ and $\text{card}(\Xi_v) = n_2$, and further split Ξ_v evenly into $\Xi_{v,k}, k = 1, \dots, K$ such that $\text{card}(\Xi_{v,k}) = n_2/K$ for each k .

2. For each $j = 1, \dots, p$, use the training set Ξ_t to solve the reformulation with $s = s_j$ and $\hat{F} = \frac{1}{n_1} \sum_{\xi \in \Xi_t} s_{\xi}$ to obtain its optimal solution $x^*(s_j)$, and then for each $k = 1, \dots, K$ estimate the chance constraint $\hat{P}_k(s_j) = \frac{1}{\text{card}(\Xi_{v,k})} \sum_{\xi \in \Xi_{v,k}} \mathbf{1}\{\xi^T x^*(s_j) \leq b\}$.

3. Compute

$$s^* = \arg \min_{s_j, 1 \leq j \leq p} \left\{ c^T x^*(s_j) : \frac{\sum_{k=1}^K \mathbf{1}\{\hat{P}_k(s_j) \geq 1 - \alpha\}}{K} \geq 1 - \beta, j = 1, \dots, p \right\}$$

Output: $s^*, x^*(s^*)$.

Appendix B: Additional Figures

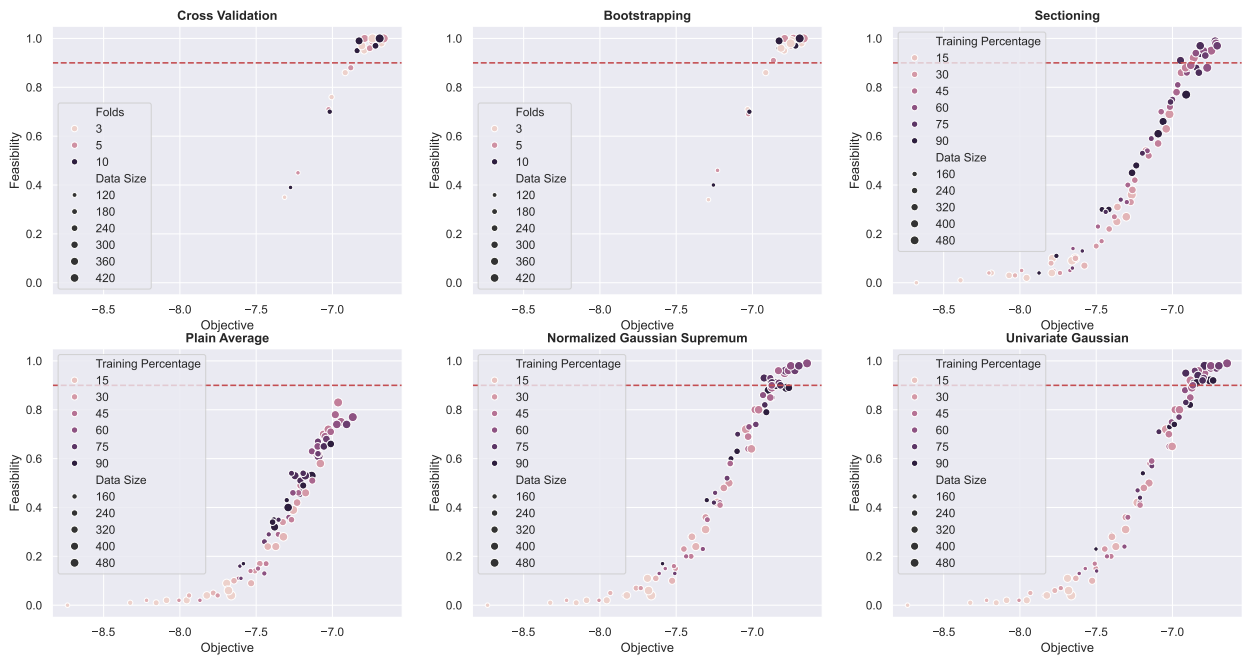


Figure 5 Caption

Proofs of Statements

In all the proofs, for universal constants which are usually denoted C or c , we abuse notation slightly to allow C or c to take a different value in each appearance. For example, consider three quantities x, y, z such that $x \leq Cy$ and $z \leq 2x$. This implies $z \leq 2Cy$, but we would write as $z \leq Cy$ to simplify the notation.

Appendix EC.1: Existing Central Limit Theorems in High Dimensions

This section reviews some results on high-dimensional central limit theorems that are needed subsequently in our proofs. We start with some notations. Let $\mathbf{X}_i := (X_{i,1}, \dots, X_{i,p})$, $i = 1, \dots, n$ be n i.i.d. copies of the random vector $\mathbf{X} := (X_1, \dots, X_p) \in \mathbb{R}^p$, and $\mu_j := E[X_j]$ for $j = 1, \dots, p$. Let $\bar{X}_j = \sum_{i=1}^n X_{i,j}/n$ be the sample mean of the j -th component. We denote by $\mathbf{Z} := (Z_1, \dots, Z_p)$ a p -dimensional Gaussian random vector with $E[Z_j] = 0$ and covariance structure $Cov(Z_j, Z_{j'}) = \Sigma(j, j') := Cov(X_j, X_{j'})$ for $j, j' = 1, \dots, p$, and by $\hat{\mathbf{Z}} := (\hat{Z}_1, \dots, \hat{Z}_p)$ a p -dimensional centered Gaussian random vector with covariance $\hat{\Sigma}$, where

$$\hat{\Sigma}(j, j') = \frac{1}{n} \sum_{i=1}^n X_{i,j} X_{i,j'} - \bar{X}_j \bar{X}_{j'}$$

is the sample covariance of all $\mathbf{X}_i$'s. We also denote $\sigma_j^2 = \Sigma(j, j)$ and $\hat{\sigma}_j^2 = \hat{\Sigma}(j, j)$.

We make the following assumption:

ASSUMPTION EC.1. *There exist constants $b > 0$ and $B \geq 1$ such that*

$$\begin{aligned} \text{Var}[X_j] &\geq b \text{ and } E[\exp(|X_j - \mu_j|^2 / B^2)] \leq 2 \text{ for all } j = 1, \dots, p \\ E[|X_j - \mu_j|^{2+k}] &\leq B^k \text{ for all } j = 1, \dots, p \text{ and } k = 1, 2. \end{aligned}$$

Note that, since the sub-exponential norm of a random variable is always bounded above by its sub-Gaussian norm up to some universal constant C , the exponential condition in Assumption EC.1 implies $E[\exp(|X_j - \mu_j| / (CB))] \leq 2$. Chernozhukov et al. (2017) proved the following CLT:

THEOREM EC.1 (First half of Proposition 2.1 in Chernozhukov et al. (2017)). *If Assumption EC.1 holds, then*

$$\sup_{a_j \leq b_j, j=1, \dots, p} |P(a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq b_j \text{ for all } j) - P(a_j \leq Z_j \leq b_j \text{ for all } j)| \leq C_1 \left(\frac{B^2 \log^7(pn)}{n} \right)^{\frac{1}{6}}$$

where the constant C_1 depends only on b .

To derive confidence bounds based on the CLT, one needs to properly estimate the quantile of the limit Gaussian vector $\mathbf{Z} \sim N_p(0, \Sigma)$. One common approach is to use the Gaussian vector $\hat{\mathbf{Z}} \sim N_p(0, \hat{\Sigma})$, where $\hat{\Sigma}$ is the sample covariance matrix, to approximate $\mathbf{Z}$. This approach is also called the multiplier bootstrap. Chernozhukov et al. (2017) gave the following result concerning the statistical accuracy of the multiplier bootstrap:

THEOREM EC.2 (First half of Corollary 4.2 in Chernozhukov et al. (2017)). *If Assumption EC.1 holds, then for any constant $0 < \alpha < \frac{1}{e}$ we have*

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \hat{Z}_j \leq b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq Z_j \leq b_j \text{ for all } j) \right| \leq C_2 \left(\frac{B^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{\frac{1}{6}}$$

with probability at least $1 - \alpha$, where the constant C_2 depends only on b .

Appendix EC.2: Proofs of Results in Section 3

This section proves the performance guarantees of our Gaussian supremum validators. Section EC.2.1 adapts the high-dimensional CLTs in Appendix EC.1 to handle small-variance situations that potentially arise in our optimization context. Section EC.2.2 extends them to the case where the sample means are normalized by their standard deviations, a key step to justify our normalized Gaussian supremum validator. Section EC.2.3 presents results on the consistency of the multiplier bootstrap to approximate the limiting Gaussian distributions. Section EC.2.4 puts together all these results to synthesize the main proofs for Section 3.

EC.2.1. A CLT for Random Vectors with Potentially Small Variances

Note that in both Theorems EC.1 and EC.2, the constants C_1, C_2 depend on the minimal variance b . By tracing the proof in Chernozhukov et al. (2017), the constant C_1 is of the form $c_1(b^{-1} \vee c_2)$ where c_1, c_2 are two universal constants. Due to such a dependence on the minimal variance, the bound can deteriorate when the noise levels across different components of $\mathbf{X}$ are not of the same scale, e.g., in the case of CCPs. To resolve this issue, we derive an alternate CLT that applies to normalized random vectors. We assume:

ASSUMPTION EC.2. *$\text{Var}[X_j] > 0$ for all $j = 1, \dots, p$ and there exists some constant $D_1 \geq 1$ such that*

$$E \left[\exp \left(\frac{|X_j - \mu_j|^2}{D_1^2 \text{Var}[X_j]} \right) \right] \leq 2 \text{ for all } j = 1, \dots, p \quad (\text{EC.1})$$

$$E \left[\left(\frac{|X_j - \mu_j|}{\sqrt{\text{Var}[X_j]}} \right)^{2+k} \right] \leq D_1^k \text{ for all } j = 1, \dots, p \text{ and } k = 1, 2. \quad (\text{EC.2})$$

Note that rectangles in $\mathbb{R}^p$ are invariant with respect to component-wise rescaling, i.e., for any rectangle $R = \{(x_1, \dots, x_p) : a_j \leq x_j \leq b_j, j = 1, \dots, p\}$, the rescaled set $R' := \{(\lambda_1 x_1, \dots, \lambda_p x_p) : (x_1, \dots, x_p) \in R\}$ with each $\lambda_j > 0$ is still a rectangle which can be represented as $R' = \{(x_1, \dots, x_p) : \lambda_j a_j \leq x_j \leq \lambda_j b_j, j = 1, \dots, p\}$. Hence one can show the following CLT by applying Theorem EC.1 to the rescaled data:

THEOREM EC.3. *If Assumption EC.2 holds, then*

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq b_j \text{ for all } j) - P(a_j \leq Z_j \leq b_j \text{ for all } j) \right| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}}$$

where C is a universal constant.

Proof of Theorem EC.3. Consider the rescaled data $Y_{i,j} = (X_{i,j} - \mu_j) / \sqrt{\text{Var}[X_j]}$. Due to Assumption EC.2, $Y_{i,j}$'s satisfy Assumption EC.1 with $b = 1$ and $B = D_1$, and has covariance structure $\Sigma_Y(j, j') = \Sigma(j, j') / \sqrt{\Sigma(j, j) \Sigma(j', j')}$. Let $\bar{Y}_j = \sum_{i=1}^n Y_{i,j} / n$. By Theorem EC.1 we have

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \sqrt{n} \bar{Y}_j \leq b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sqrt{\text{Var}[X_j]}} \leq b_j \text{ for all } j) \right| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}}.$$

The theorem follows from

$$\begin{aligned} & \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq b_j \text{ for all } j) - P(a_j \leq Z_j \leq b_j \text{ for all } j) \right| \\ &= \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\sqrt{\text{Var}[X_j]} a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \sqrt{\text{Var}[X_j]} b_j \text{ for all } j) \right. \\ & \quad \left. - P(\sqrt{\text{Var}[X_j]} a_j \leq Z_j \leq \sqrt{\text{Var}[X_j]} b_j \text{ for all } j) \right| \\ &= \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \sqrt{n} \bar{Y}_j \leq b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sqrt{\text{Var}[X_j]}} \leq b_j \text{ for all } j) \right|. \end{aligned}$$

□

Similarly, we have the following result regarding the multiplier bootstrap:

THEOREM EC.4. *If Assumption EC.2 holds, then for any constant $0 < \alpha < \frac{1}{e}$ we have*

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \hat{Z}_j \leq b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq Z_j \leq b_j \text{ for all } j) \right| \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{\frac{1}{6}}$$

with probability at least $1 - \alpha$, where C is a universal constant.

Proof of Theorem EC.4. Again, consider the rescaled data $Y_{i,j} = (X_{i,j} - \mu_j) / \sqrt{\text{Var}[X_j]}$. Note that the sample covariance of $Y_{i,j}$ is the same as the covariance structure of $(\hat{Z}_1 / \sqrt{\text{Var}[X_1]}, \dots, \hat{Z}_m / \sqrt{\text{Var}[X_p]})$.

Theorem EC.4 entails that

$$\begin{aligned} & \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \frac{\hat{Z}_j}{\sqrt{\text{Var}[X_j]}} \leq b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq \frac{Z_j}{\sqrt{\text{Var}[X_j]}} \leq b_j \text{ for all } j) \right| \\ & \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{\frac{1}{6}} \end{aligned}$$

with probability at least $1 - \alpha$. The desired conclusion then follows by invariance of the class of rectangles under component-wise rescaling. □

Theorems EC.3 and EC.4 rely on conditions more pertinent to our context than those in Theorems EC.1 and EC.2. The first condition (EC.1) of Assumption EC.2 measures the ratio of the sub-exponential norm to the L_2 norm of each component of the vector, whereas the second condition (EC.2) concerns the kurtosis of each component. Therefore, to guarantee a valid CLT, we need the vector to be sufficiently light-tailed after being normalized to have unit variance.

EC.2.2. CLTs for Sample Means Normalized by Standard Deviations

We establish CLTs for sample means normalized by sample standard deviations, needed to prove results regarding our normalized supremum validator. Note that when the dimension p is fixed, such CLTs can be established by Slutsky's theorem, but when p is huge or grows with the data size n this is no longer applicable. Instead, we need to develop concentration inequalities for sample variances, which we state below.

LEMMA EC.1 (Concentration inequalities for sample variances). *Let $\xi_1, \dots, \xi_n$ be n i.i.d. copies of the random variable $\xi \in \mathbb{R}$, $\sigma^2 := \text{Var}[\xi]$ be the true variance, and $\hat{\sigma}^2 = \sum_{i=1}^n (\xi_i - \bar{\xi})^2/n$, where $\bar{\xi} = \sum_{i=1}^n \xi_i/n$ is the sample mean, be the sample variance. We have the following concentration inequalities:*

1. *if ξ is $[0, 1]$ -valued, then there exists a universal constant C such that for any $t > 0$*

$$P(|\hat{\sigma}^2 - \sigma^2| > t) \leq 2 \exp\left(-\frac{Cnt^2}{\sigma^2 + t}\right). \quad (\text{EC.3})$$

2. *if $\xi - E[\xi]$ has a sub-Gaussian norm at most K , i.e. $E[\exp((\xi - E[\xi])^2/K^2)] \leq 2$, then there exists a universal constant C such that for any $t > 0$*

$$P(|\hat{\sigma}^2 - \sigma^2| > t) \leq 4 \exp\left(-\frac{Cnt^2}{K^4 + K^2 t}\right). \quad (\text{EC.4})$$

Proof of Lemma EC.1. Case 1: Since the unbiased sample variance, $(n/(n-1))\hat{\sigma}^2$, is a U-statistic of degree 2, Hoeffding's inequality for U-statistics (see, e.g., Hoeffding (1963), Peel et al. (2010)) entails that with probability at least $1 - \alpha$

$$\left|\frac{n}{n-1}\hat{\sigma}^2 - \sigma^2\right| \leq \sqrt{\frac{4\text{Var}[(\xi - \xi')^2]}{n} \log \frac{2}{\alpha}} + \frac{4}{3n} \log \frac{2}{\alpha}$$

where ξ, ξ' are i.i.d. copies. Note that $\text{Var}[(\xi - \xi')^2] \leq E[(\xi - \xi')^4] \leq E[(\xi - \xi')^2] = 2\sigma^2$ because $|\xi - \xi'| \leq 1$, and $\hat{\sigma}^2 \leq 1$ for the same reason. Therefore with probability at least $1 - \alpha$

$$\begin{aligned} |\hat{\sigma}^2 - \sigma^2| &\leq \sqrt{\frac{8\sigma^2}{n} \log \frac{2}{\alpha}} + \frac{4}{3n} \log \frac{2}{\alpha} + \frac{\hat{\sigma}^2}{n-1} \\ &\leq \sqrt{\frac{8\sigma^2}{n} \log \frac{2}{\alpha}} + \frac{5}{n} \log \frac{2}{\alpha} \end{aligned}$$

and the conclusion easily follows by fixing the right hand side and solving for α .

Case 2: Since the sub-Gaussian norm of $\xi - E[\xi]$ is at most K , $(\xi - E[\xi])^2$ has a sub-exponential norm of at most K^2 by definition. Centering a variable can only inflate its sub-exponential norm by a constant factor (Remark 5.18 in Vershynin (2010)), that is, $(\xi - E[\xi])^2 - \sigma^2$ must have a sub-exponential norm of at most CK^2 where C is a universal constant. By Hoeffding's inequality and Bernstein's inequality for

sums of independent variables (Propositions 5.10 and 5.16 in Vershynin (2010)) we have for some universal constant C and any $t > 0$

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n (\xi_i - E[\xi])^2 - \sigma^2\right| > t\right) \leq 2 \exp\left(-\frac{Cnt^2}{K^4 + K^2 t}\right)$$

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i - E[\xi]\right| > t\right) \leq 2 \exp\left(-\frac{Cnt^2}{K^2}\right).$$

Note that the sample variance can be expressed as

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\xi_i - E[\xi])^2 - \left(\frac{1}{n} \sum_{i=1}^n \xi_i - E[\xi]\right)^2.$$

Hence by a union bound

$$\begin{aligned} P(|\hat{\sigma}^2 - \sigma^2| > t) &\leq P\left(\left|\frac{1}{n} \sum_{i=1}^n (\xi_i - E[\xi])^2 - \sigma^2\right| > t/2\right) + P\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i - E[\xi]\right| > \sqrt{t/2}\right) \\ &\leq 2 \exp\left(-\frac{Cnt^2}{K^4 + K^2 t}\right) + 2 \exp\left(-\frac{Cnt}{K^2}\right) \\ &\leq 4 \exp\left(-\frac{Cnt^2}{K^4 + K^2 t}\right). \end{aligned}$$

This completes the proof. $\square$

Note that inequality (EC.3) cannot be deduced from inequality (EC.4) as a special case because of the appearance of σ^2 in the bound. In fact (EC.3) is a sharper bound than (EC.4) when the variable ξ is Bernoulli, e.g., in the case of chance constrained optimization, because the sub-Gaussian norm of $\xi - E[\xi]$ is of order $K^2 = \Theta(\log^{-1}(1/\epsilon)) \gg \Theta(\epsilon) = \sigma^2$ when the success probability ϵ is small.

We also need the following anti-concentration inequality for Gaussian distribution:

LEMMA EC.2 (Nazarov's inequality). *Let $(Y_1, \dots, Y_p)$ be an p -dimensional centered Gaussian random vector such that $\text{Var}[Y_j] \geq b$ for all $j = 1, \dots, p$ and some constant $b > 0$. Then for every $-\infty \leq a_j \leq b_j \leq +\infty, j = 1, \dots, p$ and every $\delta > 0$ it holds*

$$P(a_j - \delta \leq Y_j \leq b_j + \delta \text{ for all } j) - P(a_j \leq Y_j \leq b_j \text{ for all } j) \leq C_3 \delta \sqrt{\log p}$$

where C_3 is a constant that depends only on b .

A special case of this inequality where $a_j = -\infty$ for all $j = 1, \dots, p$ has appeared in Chernozhukov et al. (2017). Establishing a similar inequality for the case of possibly finite a_j 's involves a routine application of union bound. For completeness we provide a proof here.

Proof of Lemma EC.2. Lemma A.1 in Chernozhukov et al. (2017) states that for every $\delta > 0$ and every $b_1, \dots, b_p$ the following bound holds

$$P(Y_j \leq b_j + \delta \text{ for all } j) - P(Y_j \leq b_j \text{ for all } j) \leq C \delta \sqrt{\log p}$$

where C depends on b only. Applying the same bound to $-Y_j, j = 1, \dots, p$ and $-a_j, j = 1, \dots, p$ gives

$$P(a_j - \delta \leq Y_j \text{ for all } j) - P(a_j \leq Y_j \text{ for all } j) \leq C\delta\sqrt{\log p}.$$

Therefore

$$\begin{aligned} & P(a_j - \delta \leq Y_j \leq b_j + \delta \text{ for all } j) - P(a_j \leq Y_j \leq b_j \text{ for all } j) \\ &= P(a_j - \delta \leq Y_j \leq b_j + \delta \text{ for all } j) - P(a_j - \delta \leq Y_j \leq b_j \text{ for all } j) \\ &\quad + P(a_j - \delta \leq Y_j \leq b_j \text{ for all } j) - P(a_j \leq Y_j \leq b_j \text{ for all } j) \\ &\leq P(Y_j \leq b_j + \delta \text{ for all } j) - P(Y_j \leq b_j \text{ for all } j) \\ &\quad + P(a_j - \delta \leq Y_j \text{ for all } j) - P(a_j \leq Y_j \text{ for all } j) \\ &\leq 2C\delta\sqrt{\log p}. \end{aligned}$$

This completes the proof. □

We have the following CLT with componentwise normalization. Recall that $\hat{\sigma}_j^2$ is the sample variance of X_j computed from the data $\{X_{1,j}, \dots, X_{n,j}\}$.

THEOREM EC.5. *Under Assumption EC.2 we have*

$$\begin{aligned} & \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(\sigma_j a_j \leq Z_j \leq \sigma_j b_j \text{ for all } j) \right| \\ & \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp \left(- \frac{cn^{2/3}}{D_1^{10/3}} \right) \end{aligned}$$

for some universal constants C, c .

If each component of the random vector is $[0, 1]$ -valued, we assume:

ASSUMPTION EC.3. *Each X_j is $[0, 1]$ -valued and $\sigma_j^2 := \text{Var}[X_j] \geq \delta$ for all $j = 1, \dots, p$ and some constant $\delta > 0$.*

Then we have an alternate CLT:

THEOREM EC.6. *Under Assumptions EC.2 and EC.3 we have*

$$\begin{aligned} & \sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(\sigma_j a_j \leq Z_j \leq \sigma_j b_j \text{ for all } j) \right| \\ & \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp \left(- c\delta D_1^{2/3} n^{2/3} \right) \end{aligned}$$

for some universal constants C, c .

Proof of Theorems EC.5 and EC.6. For any $a_j \leq b_j, j = 1, \dots, p$ and $0 < \epsilon < 1/2$

$$\begin{aligned}
& P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) \\
&= P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\sqrt{n}(\bar{X}_j - \mu_j)}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j \text{ for all } j\right) \\
&= P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\sqrt{n}(\bar{X}_j - \mu_j)}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j, \left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| \leq \epsilon \text{ for all } j\right) \\
&\quad + P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\sqrt{n}(\bar{X}_j - \mu_j)}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j \text{ for all } j, \left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| > \epsilon \text{ for some } j\right) \\
&\leq P(a_j - \epsilon |a_j| \leq \frac{\sqrt{n}(\bar{X}_j - \mu_j)}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j) + P\left(\left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| > \epsilon \text{ for some } j\right) \\
&\leq P(a_j - \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j) + C \left(\frac{D_1^2 \log^7(pn)}{n}\right)^{\frac{1}{6}} + \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j)
\end{aligned}$$

where in the last inequality we use Theorem EC.3 for the first probability and apply the union bound to the second probability. Note that $\epsilon < 1/2$ hence for any constant $M > 0$ if we denote by $f_M(x) = -\infty \cdot \mathbf{1}(x < -M) + \infty \cdot \mathbf{1}(x > M)$ then we have

$$\begin{aligned}
& P(a_j - \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j) \\
&\leq P(f_M(a_j) + a_j - \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq f_M(b_j) + b_j + \epsilon |b_j| \text{ for all } j) + 2p \exp(-cM^2) \\
&\leq P(f_M(a_j) + a_j \leq \frac{Z_j}{\sigma_j} \leq f_M(b_j) + b_j \text{ for all } j) + C\epsilon M \sqrt{\log p} + 2p \exp(-cM^2) \\
&\leq P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j) + 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p}
\end{aligned}$$

where in the second inequality we use Lemma EC.2 (note that $f_M(a_j) + a_j - \epsilon |a_j|$ is either ∞ or its absolute value $\leq \frac{3}{2}M$, so is $f_M(b_j) + b_j + \epsilon |b_j|$), the term $\exp(-cM^2)$ is the tail bound of the univariate standard normal distribution, and C, c are universal constants. Therefore we have derived the following upper bound

$$\begin{aligned}
& P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) \\
&\leq P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j) + 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p} + C \left(\frac{D_1^2 \log^7(pn)}{n}\right)^{\frac{1}{6}} + \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j)
\end{aligned}$$

which holds true for $\epsilon < 1/2$ and $M > 0$. Similarly, one can show

$$\begin{aligned}
& P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) \\
&\geq P(a_j + \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j - \epsilon |b_j| \text{ for all } j) - C \left(\frac{D_1^2 \log^7(pn)}{n}\right)^{\frac{1}{6}} - \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j)
\end{aligned}$$

along with

$$P(a_j + \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j - \epsilon |b_j| \text{ for all } j)$$

$$\begin{aligned}
&\geq P(f_M(a_j) + a_j + \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq f_M(b_j) + b_j - \epsilon |b_j| \text{ for all } j) - 2p \exp(-cM^2) \\
&\geq P(f_M(a_j) + a_j \leq \frac{Z_j}{\sigma_j} \leq f_M(b_j) + b_j \text{ for all } j) - C\epsilon M \sqrt{\log p} - 2p \exp(-cM^2) \\
&\geq P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j) - C\epsilon M \sqrt{\log p} - 4p \exp(-cM^2).
\end{aligned}$$

This leads to

$$\begin{aligned}
&|P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
&\leq 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p} + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j) \\
&\leq 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p} + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + \sum_{j=1}^p P(|\hat{\sigma}_j^2 - \sigma_j^2| > \epsilon \sigma_j^2) \\
&\leq 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p} + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + 4p \exp\left(-\frac{cn\epsilon^2}{D_1^4 + D_1^2 \epsilon}\right) \quad (\text{EC.5})
\end{aligned}$$

where the last inequality holds because Assumption EC.2 guarantees that the sub-Gaussian norm of $X_j - \mu_j$ is at most $D_1 \sigma_j$ and one then applies Lemma EC.1. Now set

$$M = \sqrt{\frac{1}{c} \log(pn)}, \quad \epsilon = \left(\frac{D_1^2}{n} \right)^{\frac{1}{6}}$$

and note that $\epsilon < 1/2$ can be assumed since otherwise the first term in the desired bound is already greater than 1 (by enlarging the universal constant if necessary) and the bound is trivial. We get the uniform bound

$$\begin{aligned}
&|P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
&\leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp\left(-\frac{cD_1^{2/3} n^{2/3}}{D_1^4 + D_1^2 \epsilon}\right) \\
&\leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp\left(-\frac{cn^{2/3}}{D_1^{10/3}}\right)
\end{aligned}$$

where the second inequality holds because $D_2 \geq 1$ and $\epsilon < 1/2$. In particular, if X_j 's are $[0, 1]$ -valued, we use the concentration inequality (EC.3) instead of (EC.4) to refine the bound (EC.5) to be

$$\begin{aligned}
&|P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
&\leq 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p} + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + 4p \exp\left(-\frac{cn\delta\epsilon^2}{1 + \epsilon}\right).
\end{aligned}$$

Likewise, letting M and ϵ take the same values as before, we obtain

$$\begin{aligned}
&|P(\hat{\sigma}_j a_j \leq \sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j b_j \text{ for all } j) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
&\leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp\left(-c\delta D_1^{2/3} n^{2/3}\right).
\end{aligned}$$

This completes the proof of Theorems EC.5 and EC.6. □

The following are corresponding results for the multiplier bootstrap:

THEOREM EC.7. *If Assumption EC.2 holds, for any constant $0 < \alpha < \frac{1}{e}$ we have*

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(\sigma_j a_j \leq Z_j \leq \sigma_j b_j \text{ for all } j) \right| \\ \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{\frac{1}{6}} + \frac{CD_1^2 \log(pn) \log(p/\alpha)}{\sqrt{n}}$$

with probability at least $1 - \alpha$, where C is a universal constant.

THEOREM EC.8. *If Assumptions EC.2 and EC.3 hold, for any constant $0 < \alpha < \frac{1}{e}$ we have*

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(\sigma_j a_j \leq Z_j \leq \sigma_j b_j \text{ for all } j) \right| \\ \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{\frac{1}{6}} + \frac{C \log(pn) \log(p/\alpha)}{\sqrt{n\delta}}$$

with probability at least $1 - \alpha$, where C is a universal constant.

Proof of Theorems EC.7 and EC.8. For any $a_j \leq b_j, j = 1, \dots, p$ and $0 < \epsilon < 1/2$

$$\begin{aligned} & P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) \\ &= P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\hat{Z}_j}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n\right) \\ &= P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\hat{Z}_j}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j, \left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| \leq \epsilon \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n\right) \\ &\quad + P\left(\frac{\hat{\sigma}_j}{\sigma_j} a_j \leq \frac{\hat{Z}_j}{\sigma_j} \leq \frac{\hat{\sigma}_j}{\sigma_j} b_j \text{ for all } j, \left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| > \epsilon \text{ for some } j | \{\mathbf{X}_i\}_{i=1}^n\right) \\ &\leq P(a_j - \epsilon |a_j| \leq \frac{\hat{Z}_j}{\sigma_j} \leq b_j + \epsilon |b_j|, \left|\frac{\hat{\sigma}_j}{\sigma_j} - 1\right| \leq \epsilon \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) \\ &\quad \text{with probability at least } 1 - \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j) \\ &\leq P(a_j - \epsilon |a_j| \leq \frac{\hat{Z}_j}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) \\ &\leq P(a_j - \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j) + C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} \\ &\quad \text{with probability at least } 1 - \frac{\alpha}{4}. \end{aligned}$$

In the proof of Theorems EC.5 and EC.6 we show

$$P(a_j - \epsilon |a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j + \epsilon |b_j| \text{ for all } j) \leq P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j) + 4p \exp(-cM^2) + C\epsilon M \sqrt{\log p}.$$

Similarly we can show the other direction

$$\begin{aligned}
& P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) \\
& \geq P(a_j + \epsilon | a_j| \leq \frac{Z_j}{\sigma_j} \leq b_j - \epsilon | b_j| \text{ for all } j) - C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} \\
& \quad \text{with probability at least } 1 - \frac{\alpha}{4} - \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j) \\
& \geq P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j) - C \epsilon M \sqrt{\log p} - 4p \exp(-cM^2) - C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}}.
\end{aligned}$$

Therefore the following uniform bound holds with probability at least $1 - \frac{\alpha}{2} - 2 \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j)$

$$\begin{aligned}
& |P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
& \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} + C \epsilon M \sqrt{\log p} + 4p \exp(-cM^2).
\end{aligned}$$

Note that

$$2 \sum_{j=1}^p P(|\hat{\sigma}_j - \sigma_j| > \epsilon \sigma_j) \leq 8p \exp\left(-\frac{c n \epsilon^2}{D_1^4 + D_1^2 \epsilon}\right).$$

By setting the right hand side of the above inequality to be $\alpha/2$ and $M = \sqrt{\frac{1}{c} \log(pn)}$ we get

$$\begin{aligned}
& |P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
& \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} + C \left(\sqrt{\frac{D_1^4}{n} \log \frac{16p}{\alpha}} + \frac{D_1^2}{n} \log \frac{16p}{\alpha} \right) \log(pn) \\
& \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} + C \frac{D_1^2}{\sqrt{n}} \log \frac{16p}{\alpha} \log(pn)
\end{aligned}$$

with probability at least $1 - \alpha$.

In case of $[0, 1]$ -valued variables, we use (EC.3) instead of (EC.4) to get

$$\begin{aligned}
& |P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq \frac{Z_j}{\sigma_j} \leq b_j \text{ for all } j)| \\
& \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} + C \left(\sqrt{\frac{1}{n\delta} \log \frac{16p}{\alpha}} + \frac{1}{n\delta} \log \frac{16p}{\alpha} \right) \log(pn) \\
& \leq C \left(\frac{D_1^2 \log^5(pn) \log^2(4/\alpha)}{n} \right)^{\frac{1}{6}} + C \left(\frac{1}{\sqrt{n\delta}} + \frac{1}{n\delta} \right) \log \frac{16p}{\alpha} \log(pn)
\end{aligned}$$

with probability at least $1 - \alpha$. We can assume that $n\delta \geq 1$ to get the desired bound since otherwise the second term in the bound is already greater than 1. $\square$

EC.2.3. Coverage Probability through Multiplier Bootstrap

THEOREM EC.9 (Coverage probability for unnormalized supremum). *Under Assumption EC.2, for every $0 < \beta < 1$ we have*

$$|P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}}$$

where $\hat{q}_{1-\beta}$ is such that

$$P(\hat{Z}_j \leq \hat{q}_{1-\beta} \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) = 1 - \beta$$

and C is a universal constant.

Proof of Theorem EC.9. Denote by $\epsilon = C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{1/6}$ and by A_α the event that

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(a_j \leq \hat{Z}_j \leq b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(a_j \leq Z_j \leq b_j \text{ for all } j) \right| \leq \epsilon.$$

Then we can rewrite

$$\begin{aligned} & P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j) \\ &= P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j, \text{ and } A_\alpha) + P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j, \text{ and } A_\alpha^c) \end{aligned}$$

The second term is bounded by α because of Theorem EC.4. To study the first term, denote by $q_{1-\beta}$ the true $(1 - \beta)$ -level quantile of the limit distribution, i.e., $q_{1-\beta}$ is such that $P(Z_j \leq q_{1-\beta} \text{ for all } j) = 1 - \beta$. On event A_α we have $q_{1-\beta-\epsilon} \leq \hat{q}_{1-\beta} \leq q_{1-\beta+\epsilon}$, therefore

$$\begin{aligned} P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta-\epsilon} \text{ for all } j, \text{ and } A_\alpha) &\leq P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j, \text{ and } A_\alpha) \\ &\leq P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta+\epsilon} \text{ for all } j, \text{ and } A_\alpha). \end{aligned}$$

From this two-sided bound we get

$$\begin{aligned} & P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j) \\ &\leq P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta+\epsilon} \text{ for all } j) + \alpha \\ &\leq P(Z_j \leq q_{1-\beta+\epsilon} \text{ for all } j) + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{1/6} + \alpha \\ &= 1 - \beta + \epsilon + C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{1/6} + \alpha. \end{aligned}$$

Similarly the lower bound can be derived as

$$\begin{aligned} & P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j) \\ &\geq P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta-\epsilon} \text{ for all } j, \text{ and } A_\alpha) \\ &= P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta+\epsilon} \text{ for all } j) - P(\sqrt{n}(\bar{X}_j - \mu_j) \leq q_{1-\beta-\epsilon} \text{ for all } j, \text{ and } A_\alpha^c) \\ &\geq P(Z_j \leq q_{1-\beta-\epsilon} \text{ for all } j) - C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{1/6} - \alpha \\ &= 1 - \beta - \epsilon - C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{1/6} - \alpha. \end{aligned}$$

This gives the following bound for any $\alpha < 1/e$

$$|P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + \epsilon + \alpha.$$

Set $\alpha = 1/n$ and note that $1/n$ is less than the leading term, thus we have shown the desired conclusion.

□

THEOREM EC.10 (Coverage probability for normalized supremum). *Under Assumptions EC.2, for every $0 < \beta < 1$ we have*

$$\begin{aligned} & |P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \\ & \leq C \left(\left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + \frac{D_1^2 \log^2(pn)}{\sqrt{n}} + p \exp \left(- \frac{cn^{2/3}}{D_1^{10/3}} \right) \right). \end{aligned}$$

If Assumption EC.3 also holds, then

$$\begin{aligned} & |P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \\ & \leq C \left(\left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + \frac{\log^2(pn)}{\sqrt{n\delta}} + p \exp \left(- c\delta D_1^{2/3} n^{2/3} \right) \right). \end{aligned}$$

Here $\hat{q}_{1-\beta}$ is such that

$$P(\hat{Z}_j \leq \hat{\sigma}_j \hat{q}_{1-\beta} \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) = 1 - \beta$$

and C, c are universal constants.

Proof of Theorem EC.10. Let $\epsilon = C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{1/6} + \frac{CD_1^2 \log(pn) \log(p/\alpha)}{\sqrt{n}}$ and A_α be the event that

$$\sup_{a_j \leq b_j, j=1, \dots, p} \left| P(\hat{\sigma}_j a_j \leq \hat{Z}_j \leq \hat{\sigma}_j b_j \text{ for all } j | \{\mathbf{X}_i\}_{i=1}^n) - P(\sigma_j a_j \leq Z_j \leq \sigma_j b_j \text{ for all } j) \right| \leq \epsilon.$$

We know that $P(A_\alpha) \leq \alpha$ from Theorem EC.7. Following the same line of the proof for Theorem EC.9 and using the CLT in Theorem EC.5 we can derive that

$$|P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp \left(- \frac{cn^{2/3}}{D_1^{10/3}} \right) + \epsilon + \alpha.$$

Again setting $\alpha = \frac{1}{n}$ leads to the first bound.

The second bound can be derived similarly. Define $\epsilon = C \left(\frac{D_1^2 \log^5(pn) \log^2(1/\alpha)}{n} \right)^{1/6} + \frac{C \log(pn) \log(p/\alpha)}{\sqrt{n\delta}}$, and now Theorem EC.8 entails that $P(A_\alpha) \leq \alpha$ again. Using the CLT in Theorem EC.6 gives

$$|P(\sqrt{n}(\bar{X}_j - \mu_j) \leq \hat{\sigma}_j \hat{q}_{1-\beta} \text{ for all } j) - (1 - \beta)| \leq C \left(\frac{D_1^2 \log^7(pn)}{n} \right)^{\frac{1}{6}} + Cp \exp \left(- c\delta D_1^{2/3} n^{2/3} \right) + \epsilon + \alpha.$$

The second bound follows from setting $\alpha = \frac{1}{n}$. □

EC.2.4. Proofs of Main Statistical Guarantees

We now put together all the previous results to prove the statistical guarantees of our validators. For convenience, we suppress the subscript $\xi_{1:n_2}$ in the probability notation.

Proof of Theorem 1. We bound the probability as follows

$$\begin{aligned}
& P(x^*(\hat{s}^*) \text{ is feasible for (1)}) \\
& \geq P(\hat{H}_j \geq \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for some } j = 1, \dots, p \text{ in (8) and } H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
& \geq P(H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) - P(\hat{H}_j < \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
& \geq 1 - \beta - C \left(\frac{D_1^2 \log^7(pn_2)}{n_2} \right)^{\frac{1}{6}} - P(\hat{H}_j < \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \tag{EC.6}
\end{aligned}$$

where we use Theorem EC.9 for the first probability by letting $X_{i,j} = h(x^*(s_j), \xi_i)$. To bound the second probability, we recall that $\hat{\sigma}_j^2$ is the sample variance computed from $\{h(x^*(s_j), \xi_1), \dots, h(x^*(s_j), \xi_{n_2})\}$ and write

$$\begin{aligned}
& P(\hat{H}_j < \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
& \leq P(\hat{H}_j < \gamma + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ and } \hat{\sigma}_j^2 \leq 2\sigma^2(x^*(s_j)) \text{ for all } j = 1, \dots, p) + P(\hat{\sigma}_j^2 > 2\sigma^2(x^*(s_j)) \text{ for some } j = 1, \dots, p) \\
& \leq P(\hat{H}_j < \gamma + \frac{C\bar{\sigma}\sqrt{\log(p/\beta)}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) + P(\hat{\sigma}_j^2 > 2\sigma^2(x^*(s_j)) \text{ for some } j = 1, \dots, p) \\
& \quad \text{because of the fact that } q_{1-\beta} \leq C \max_j \hat{\sigma}_j \sqrt{\log(p/\beta)} \text{ for some universal constant } C \\
& \leq P(\hat{H}_{\bar{j}} - \bar{H} < \gamma + \frac{C\bar{\sigma}\sqrt{\log(p/\beta)}}{\sqrt{n_2}} - \bar{H}) + Cp \exp(-\frac{cn_2}{D_1^4}) \\
& \quad \text{where } \bar{j} \text{ is the index such that } H(x^*(s_{\bar{j}})) = \bar{H} \text{ and the concentration (EC.4) is used} \\
& \leq C \exp(-\frac{cn_2\epsilon^2}{D_1^2\bar{\sigma}^2}) + Cp \exp(-\frac{cn_2}{D_1^4}) \text{ because the sub-Gaussian norm of } h(x^*(s_{\bar{j}}), \xi) \text{ is at most } D_1\bar{\sigma}.
\end{aligned}$$

Substituting this bound into (EC.6) gives the desired conclusion. $\square$

Proof of Theorem 2. Similar to the proof of Theorem 1, we have the bound

$$\begin{aligned}
& P(x^*(\hat{s}^*) \text{ is feasible for (1)}) \\
& \geq P(H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) - P(\hat{H}_j < \gamma + \frac{q_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
& \geq 1 - \beta - C \left(\left(\frac{D_1^2 \log^7(pn_2)}{n_2} \right)^{\frac{1}{6}} + \frac{D_1^2 \log^2(pn_2)}{\sqrt{n_2}} + p \exp(-\frac{cn_2^{2/3}}{D_1^{10/3}}) \right) - P(\hat{H}_j < \gamma + \frac{q_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
& \quad \text{where the first bound is due to Theorem EC.10.}
\end{aligned}$$

For the second probability we write

$$P(\hat{H}_j < \gamma + \frac{q_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p)$$

$$\begin{aligned}
&\leq P(\hat{H}(x^*(\bar{s})) < \gamma + \frac{q_{1-\beta}\hat{\sigma}(x^*(\bar{s}))}{\sqrt{n_2}}) \\
&\leq P(\hat{H}(x^*(\bar{s})) < \gamma + \frac{q_{1-\beta}\sqrt{2}\sigma(x^*(\bar{s}))}{\sqrt{n_2}}) + P(\hat{\sigma}^2(x^*(\bar{s})) > 2\sigma^2(x^*(\bar{s}))) \\
&\leq P(\hat{H}(x^*(\bar{s})) < \gamma + \frac{C\sqrt{\log(p/\beta)}\sigma(x^*(\bar{s}))}{\sqrt{n_2}}) + P(\hat{\sigma}^2(x^*(\bar{s})) > 2\sigma^2(x^*(\bar{s}))) \quad \text{since } q_{1-\beta} \leq C\sqrt{\log(p/\beta)} \\
&\leq C \exp\left(-\frac{cn_2\epsilon^2}{D_1^2\sigma^2(x^*(\bar{s}))}\right) + C \exp\left(-\frac{cn_2}{D_1^4}\right)
\end{aligned}$$

Combining the two probability bounds and noting that $p \exp\left(-\frac{cn_2^{2/3}}{D_1^{10/3}}\right)$ dominates $\exp\left(-\frac{cn_2}{D_1^4}\right)$ (because $\frac{n_2}{D_1^4} = \frac{n_2^{2/3}}{D_1^{10/3}} \cdot \left(\frac{n_2}{D_1^2}\right)^{1/3}$ and $\frac{n_2}{D_1^2} \geq 1$ can be assumed), we obtain the desired conclusion. $\square$

Proof of Corollary 1. When $\bar{H} > \gamma$, we have $\epsilon \rightarrow \bar{H} - \gamma > 0$ in Theorems 1 and 2, therefore the exponential error term with ϵ vanishes as $n_2 \rightarrow \infty$. Under the condition that $p \exp(-n_2^{1/7}) \rightarrow 0$ it is straightforward to check that other error terms also vanish. $\square$

Proof of Theorem 3. Unlike the proof of Theorem 1, we use the Bernoulli structure to derive the error bound. Note that in this case $\gamma = 1 - \alpha$. Define events

$$\begin{aligned}
E_1 &= \left\{ \hat{H}_j \geq 1 - \alpha + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for some } j = 1, \dots, p \text{ in (8)} \right\} \\
E_2 &= \left\{ H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j \text{ such that } H(x^*(s_j)) \in (\alpha, 1 - \alpha) \right\} \\
E_3 &= \left\{ \hat{H}_j < 1 - \alpha + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j \text{ such that } H(x^*(s_j)) \leq \alpha \right\}.
\end{aligned}$$

Then we have

$$\begin{aligned}
P(x^*(\hat{s}^*) \text{ is feasible for (1)}) &\geq P(E_1 \cap E_2 \cap E_3) \\
&\geq 1 - P(E_1^c) - P(E_2^c) - P(E_3^c) \\
&= P(E_2) - P(E_1^c) - P(E_3^c). \tag{EC.7}
\end{aligned}$$

We bound the three probabilities. Let $q_{1-\beta}^\alpha$ be the $1 - \beta$ quantile of $\max\{Z_j : H(x^*(s_j)) \in (\alpha, 1 - \alpha), 1 \leq j \leq p\}$ where $(Z_1, \dots, Z_p) \sim N_p(0, \hat{\Sigma})$. By stochastic dominance it is clear that $q_{1-\beta}^\alpha \leq q_{1-\beta}$ almost surely, therefore

$$\begin{aligned}
P(E_2) &\geq P\left(H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta}^\alpha}{\sqrt{n_2}} \text{ for all } j \text{ such that } H(x^*(s_j)) \in (\alpha, 1 - \alpha)\right) \\
&\geq 1 - \beta - C \left(\frac{\log^7(pn_2)}{n_2\alpha}\right)^{\frac{1}{6}}
\end{aligned}$$

by applying Theorem EC.9 to $\{h(x^*(s_j), \xi) : H(x^*(s_j)) \in (\alpha, 1 - \alpha), 1 \leq j \leq p\}$ and noticing that Assumption EC.2 is satisfied with $D_1 = \frac{C}{\sqrt{\alpha}}$ for some universal constant C .

We then bound the second probability

$$\begin{aligned}
P(E_1^c) &= P(\hat{H}_j < 1 - \alpha + \frac{q_{1-\beta}}{\sqrt{n_2}} \text{ for all } j = 1, \dots, p) \\
&\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{q_{1-\beta}}{\sqrt{n_2}}) \text{ where } \bar{j} \text{ is the index such that } H(x^*(s_{\bar{j}})) = 1 - \bar{\alpha} \\
&\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{C\sqrt{\log(p/\beta)}}{\sqrt{n_2}}) \text{ because } q_{1-\beta} \leq C \max_j \hat{\sigma}_j \sqrt{\log(p/\beta)} \leq C\sqrt{\log(p/\beta)} \\
&\leq \exp\left(-\frac{n_2 \epsilon^2}{2(\bar{\alpha}(1 - \bar{\alpha}) + \epsilon/3)}\right)
\end{aligned}$$

where in the last line we use a Bernstein's inequality for sums of bounded random variables (see equation (2.10) in Boucheron et al. (2013)). Note that this is further bounded by $\exp(-cn_2 \min\{\epsilon, \frac{\epsilon^2}{\bar{\alpha}}\})$ if $\bar{\alpha} \leq 1/2$.

The third probability can be bounded as

$$\begin{aligned}
P(E_3^c) &\leq P(\hat{H}_j \geq 1 - \alpha \text{ for some } j \text{ such that } H(x^*(s_j)) \leq \alpha) \\
&\leq \sum_{j: H(x^*(s_j)) \leq \alpha} P(\hat{H}_j \geq 1 - \alpha) \\
&\leq p \exp(-2n_2(1 - 2\alpha)^2) \leq p \exp(-cn_2) \text{ by Hoeffding's inequality.}
\end{aligned}$$

Substituting the bounds into (EC.7) leads to

$$P(x^*(\hat{s}^*) \text{ is feasible for (1)}) \leq 1 - \beta - C \left(\frac{\log^7(pn_2)}{n_2 \alpha} \right)^{\frac{1}{6}} - \exp\left(-cn_2 \min\left\{\epsilon, \frac{\epsilon^2}{\bar{\alpha}}\right\}\right) - p \exp(-cn_2).$$

It remains to show that $p \exp(-cn_2)$ is negligible relative to other error terms. Since $\alpha < 1$ it is clear that $\left(\frac{1}{n_2}\right)^{1/6} \leq \left(\frac{\log^7(pn_2)}{n_2 \alpha}\right)^{1/6}$, and we argue that $\left(\frac{1}{n_2}\right)^{1/6} \geq p \exp(-cn_2)$ can be assumed so that $p \exp(-cn_2) \leq \left(\frac{\log^7(pn_2)}{n_2 \alpha}\right)^{1/6}$. If $\left(\frac{1}{n_2}\right)^{1/6} < p \exp(-cn_2)$, then $p > \exp(cn_2) n_2^{-1/6}$, and $\frac{\log^7(pn_2)}{n_2 \alpha} \geq \frac{(cn_2)^7}{n_2 \alpha} \geq c^7 n_2^6$, hence the first error term already exceeds 1 (enlarge the universal constant C if necessary) and the error bound holds true trivially. $\square$

Proof of Theorem 4. The proof follows the one for Theorem 3, and we focus on the modifications. The events are now defined as

$$\begin{aligned}
E_1 &= \left\{ \hat{H}_j \geq 1 - \alpha + \frac{q_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}} \text{ for some } j = 1, \dots, p \text{ in (9)} \right\} \\
E_2 &= \left\{ H(x^*(s_j)) \geq \hat{H}_j - \frac{q_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j \text{ such that } H(x^*(s_j)) \in (\alpha, 1 - \alpha) \right\} \\
E_3 &= \left\{ \hat{H}_j < 1 - \alpha + \frac{q_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}} \text{ for all } j \text{ such that } H(x^*(s_j)) \leq \alpha \right\}.
\end{aligned}$$

Again we have $P(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq P(E_2) - P(E_1^c) - P(E_3^c)$.

The first probability bound becomes

$$P(E_2) \geq 1 - \beta - C \left(\left(\frac{\log^7(pn_2)}{n_2 \alpha} \right)^{\frac{1}{6}} + \frac{\log^2(pn_2)}{\sqrt{n_2 \alpha}} + p \exp(-c(n_2 \alpha)^{2/3}) \right)$$

by using the second half of Theorem EC.10 and noting that $\delta = \alpha(1 - \alpha) \geq \frac{1}{2}\alpha$ if $\alpha < \frac{1}{2}$ and $D_1 = \frac{C}{\sqrt{\alpha}}$. For the second probability we have

$$\begin{aligned}
P(E_1^c) &\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{q_{1-\beta}\hat{\sigma}_{\bar{j}}}{\sqrt{n_2}}) \text{ where } \bar{j} \text{ is the index such that } H(x^*(s_{\bar{j}})) = 1 - \bar{\alpha} \\
&\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{q_{1-\beta}t}{\sqrt{n_2}}) + P(\hat{\sigma}_{\bar{j}} > t) \text{ where } t = \sqrt{\bar{\alpha}(1 - \bar{\alpha})} + \sqrt{2\log(n_2\alpha)/n_2} \\
&\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{q_{1-\beta}t}{\sqrt{n_2}}) + \frac{1}{n_2\alpha} \\
&\quad \text{where the bound } \frac{1}{n_2\alpha} \text{ is derived from (EC.3) (see Theorem 10 in Maurer and Pontil (2009))} \\
&\leq P(\hat{H}_{\bar{j}} < 1 - \alpha + \frac{C\sqrt{(\bar{\alpha} + \log(n_2\alpha)/n_2)\log(p/\beta)}}{\sqrt{n_2}}) + \frac{1}{n_2\alpha} \text{ because } q_{1-\beta} \leq C\sqrt{\log(p/\beta)} \\
&\leq \exp\left(-\frac{n_2\epsilon^2}{2(\bar{\alpha}(1 - \bar{\alpha}) + \epsilon/3)}\right) + \frac{1}{n_2\alpha} \text{ by Bernstein's inequality.}
\end{aligned}$$

Whereas for the third probability we still have $P(E_3^c) \leq p \exp(-cn_2)$.

Finally, using a similar argument in the proof of Theorem 3, we can show that $\frac{1}{n_2\alpha}$, $p \exp(-cn_2)$, and $p \exp(-c(n_2\alpha)^{2/3})$ are all dominated by $\left(\frac{\log^7(pn_2)}{n_2\alpha}\right)^{1/6}$ when $\left(\frac{\log^7(pn_2)}{n_2\alpha}\right)^{1/6} < 1$, therefore the desired conclusion follows from combining the three probability bounds. $\square$

Proof of Corollary 2. Like Corollary 1, this is a direct consequence of the finite sample result, Theorem 3 or 4. $\square$

Appendix EC.3: Proofs of Results in Section 4

Proof of Proposition 1. Case (i): Assumption 3 follows from the Jain-Marcus theorem (see Example 2.11.13 in Van der Vaart and Wellner (1996)). Assumption 4 holds because $\sup_{x \in \mathcal{X}} |h(x, \xi)| \leq |h(\tilde{x}, \xi)| + \text{diam}(\mathcal{X})M(\xi)$, where $\text{diam}(\mathcal{X})$ denotes the (finite) diameter of $\mathcal{X}$, and finiteness of second moments of $h(\tilde{x}, \xi)$ and $M(\xi)$. Assumption 5 then follows from the Lipschitz continuity of $h(x, \xi)$ in x and an application of the dominated convergence theorem.

Case (ii): We need two results from empirical process theory to verify Assumption 3:

THEOREM EC.11 (Theorem 2.6.8 and its proof in Van der Vaart and Wellner (1996)). *If a class $\mathcal{F}$ of measurable functions satisfies:*

- i. *there exists a countable subset $\mathcal{F}_c \subseteq \mathcal{F}$ such that for every $f \in \mathcal{F}$ there exists a sequence $f_n \in \mathcal{F}_c$ such that $\lim_{n \rightarrow \infty} f_n(\xi) = f(\xi)$ for all ξ ;*
 - ii. *the envelope $\mathcal{E}(\xi) := \sup_{f \in \mathcal{F}} |f(\xi)|$ satisfies $\mathbb{E}_F[(\mathcal{E}(\xi))^2] < \infty$;*
 - iii. *$\mathcal{F}$ is a VC-subgraph class (see Section 2.6.2 of Van der Vaart and Wellner (1996)),*
- then $\mathcal{F}$ is F -Donsker.*

THEOREM EC.12 (Theorem 2.10.1 in Van der Vaart and Wellner (1996)). *If a function class $\mathcal{F}$ is F -Donsker, then any subclass $\mathcal{G} \subseteq \mathcal{F}$ is also F -Donsker.*

THEOREM EC.13 (Example 2.10.8 in Van der Vaart and Wellner (1996)). *If $\mathcal{F}$ and $\mathcal{G}$ both are uniformly bounded F -Donsker classes, then $\mathcal{F} \cdot \mathcal{G} := \{fg : f \in \mathcal{F}, g \in \mathcal{G}\}$ is also F -Donsker.*

In order to show F -Donskerness of the class of constraint functions, it suffices to show F -Donskerness for the larger function class $\tilde{\mathcal{F}} := \{\mathbf{1}(a'_k x_k \leq b_k y_k + z_k \text{ for } k = 1, \dots, K) : x_k \in \mathbb{R}^{m_k}, y_k, z_k \in \mathbb{R}, \text{ for } k = 1, \dots, K\}$ according to Theorem EC.12. Moreover, note that $\tilde{\mathcal{F}} = \tilde{\mathcal{F}}_1 \cdot \tilde{\mathcal{F}}_2 \cdots \tilde{\mathcal{F}}_K$ where each $\tilde{\mathcal{F}}_k := \{\mathbf{1}(a'_k x \leq b_k y + z) : x \in \mathbb{R}^{m_k}, y, z \in \mathbb{R}\}$, therefore by applying Theorem EC.13 recursively we see that F -Donskerness for all $\tilde{\mathcal{F}}_k$'s implies F -Donskerness of $\tilde{\mathcal{F}}$. It remains to prove F -Donskerness of each $\tilde{\mathcal{F}}_k$ using Theorem EC.11. Among the conditions of Theorem EC.11, (ii) is trivially satisfied since the family of indicator functions is uniformly bounded by 1. By writing $a'_k x \leq b_k y + z$ as $(a'_k, -b_k, -1)(x', y, z)' \leq 0$ and noting that the collection of all half-spaces on $\mathbb{R}^{m_k+2}$ has a VC dimension $m_k + 4$ (Problem 14 in Section 2.6 in Van der Vaart and Wellner (1996)), we have that $\tilde{\mathcal{F}}_k$ is a VC-subgraph class (Problem 9 in Section 2.6 in Van der Vaart and Wellner (1996)) therefore (iii) holds. To verify condition (i), consider the countable subclass $\tilde{\mathcal{F}}_k^c = \{\mathbf{1}(a'_k x \leq b_k y + z) : x \in \mathbb{Q}^{m_k}, y, z \in \mathbb{Q}\}$ where $\mathbb{Q}$ denotes the set of all rationals. Given $x_o \in \mathbb{R}^{m_k}, y_o, z_o \in \mathbb{R}$, one can pick a sequence $x_o^i \in \mathbb{Q}^{m_k}, y_o^i, z_o^i \in \mathbb{Q}$ such that $z_o^i > z_o$ for all i , $\lim_{i \rightarrow \infty} x_o^i \rightarrow x_o, \lim_{i \rightarrow \infty} y_o^i \rightarrow y_o, \lim_{i \rightarrow \infty} z_o^i \rightarrow z_o$ and

$$\lim_{i \rightarrow \infty} \frac{\|x_o^i - x_o\|_2 + |y_o^i - y_o|}{z_o^i - z_o} = 0. \quad (\text{EC.8})$$

For every fixed a_k, b_k such that $a'_k x_o < b_k y_o + z_o$ ($a'_k x_o > b_k y_o + z_o$) we have $a'_k x_o^i < b_k y_o^i + z_o^i$ ($a'_k x_o^i > b_k y_o^i + z_o^i$) for sufficiently large i because of the convergence of x_o^i, y_o^i, z_o^i to x_o, y_o, z_o . For a_k, b_k such that $a'_k x_o = b_k y_o + z_o$ we have $a'_k x_o^i \leq b_k y_o^i + z_o^i$ for sufficiently large i thanks to (EC.8). Therefore $\mathbf{1}(a'_k x_o^i \leq b_k y_o^i + z_o^i)$ converges to $\mathbf{1}(a'_k x_o \leq b_k y_o + z_o)$ pointwise as $i \rightarrow \infty$, giving rise to condition (i). Theorem EC.11 then implies that each $\tilde{\mathcal{F}}_k$ is F -Donsker.

Assumption 4 trivially holds since indicator functions are uniformly bounded by 1. It remains to prove Assumption 5. For any x, x' we write

$$\begin{aligned} & |\mathbf{1}(a'_k A_k(x) \leq b_k \text{ for } k = 1, \dots, K) - \mathbf{1}(a'_k A_k(x') \leq b_k \text{ for } k = 1, \dots, K)| \\ & \leq \sum_{k=1}^K \mathbf{1}(a'_k A_k(x) \leq b_k < a'_k A_k(x') \text{ or } a'_k A_k(x') \leq b_k < a'_k A_k(x)) \\ & \leq \sum_{k=1}^K \mathbf{1}(|a'_k A_k(x) - b_k| \leq \|a_k\|_2 \|A_k(x) - A_k(x')\|_2). \end{aligned}$$

Therefore

$$\begin{aligned} & \mathbb{E}_F[|\mathbf{1}(a'_k A_k(x) \leq b_k \text{ for } k = 1, \dots, K) - \mathbf{1}(a'_k A_k(x') \leq b_k \text{ for } k = 1, \dots, K)|^2] \\ & \leq \left(\sum_{k=1}^K \mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \|a_k\|_2 \|A_k(x) - A_k(x')\|_2) \right)^2 \end{aligned}$$

hence it suffices to show each $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \|a_k\|_2 \|A_k(x) - A_k(x')\|_2) \rightarrow 0$ as $x' \rightarrow x$. We use the bound $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \|a_k\|_2 \|A_k(x) - A_k(x')\|_2) \leq \mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \epsilon) + \mathbb{P}_F(\|a_k\|_2 \|A_k(x) - A_k(x')\|_2 > \epsilon)$ for any $\epsilon > 0$. On one hand we have $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$. To explain, if a_k has a density and $b_k \neq 0$, then $a'_k A_k(x) - b_k$ either has a density on $\mathbb{R}$ or is a point mass at b_k (when $A_k(x)$ is the zero vector), either of which implies $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \epsilon) \rightarrow 0$. Otherwise if (a_k, b_k) has a joint density, $a'_k A_k(x) - b_k$ has a density hence $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \epsilon) \rightarrow 0$ again. On the other hand, by the continuity of A_k it holds $A_k(x') \rightarrow A_k(x)$ hence $\|a_k\|_2 \|A_k(x) - A_k(x')\|_2 = o_p(1)$, leading to $\mathbb{P}_F(\|a_k\|_2 \|A_k(x) - A_k(x')\|_2 > \epsilon) \rightarrow 0$ as $x' \rightarrow x$ for each fixed ϵ . By sending ϵ to 0, we show $\mathbb{P}_F(|a'_k A_k(x) - b_k| \leq \|a_k\|_2 \|A_k(x) - A_k(x')\|_2) \rightarrow 0$. $\square$

Proof of Proposition 2. For any $s_1 < s_2$, $v(s_1) \leq v(s_2)$ follows trivially from the monotonicity property $\text{Sol}(s_2) \subseteq \text{Sol}(s_1)$. Furthermore, if $x^*(s_1)$ and $x^*(s_2)$ are the unique optimal solutions for $\text{OPT}(s_1)$ and $\text{OPT}(s_2)$ respectively and they are distinct, then we have $v(s_1) = f(x^*(s_1)) < f(x^*(s_2)) = v(s_2)$ because $x^*(s_2)$ is feasible but not optimal for $\text{OPT}(s_1)$. Otherwise if $x^*(s_1) = x^*(s_2)$ then obviously $v(s_1) = f(x^*(s_1)) = f(x^*(s_2)) = v(s_2)$. $\square$

Proof of Proposition 3. A consequence of Assumption 5 is the continuity of $H(x)$ on $\mathcal{X}$ because for every x', x it holds $|H(x') - H(x)| \leq \mathbb{E}_F[|h(x', \xi) - h(x, \xi)|] \leq \sqrt{\mathbb{E}_F[|h(x', \xi) - h(x, \xi)|^2]}$.

We prove the uniqueness of optimal solution by contradiction. Suppose there are $x_1 \neq x_2$ and both $x_1, x_2 \in \mathcal{X}_S^*$. Case (i): there are $s_1, s_2 \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ such that $x_1 = x^*(s_1), x_2 = x^*(s_2)$. In this case we must have $s_1 \neq s_2$ hence $f(x_1) \neq f(x_2)$ by Proposition 2, contradicting with the fact that both x_1, x_2 are optimal. Case (ii): there exists some $s_1 \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ such that $x_1 = x^*(s_1)$, and $x_2 \in x^*(\tilde{s}_{i^*})$ for some $1 \leq i^* \leq M-1$ but $x_2 \neq x^*(s)$ for all $s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$. Since x_2 is feasible we have $H(x_2) \geq \gamma$ on one hand. On the other hand, $H(x_2) \neq \gamma$ due to Assumption 8, therefore we must have $H(x_2) > \gamma$. We argue that it must be the case that $x_2 = x^*(\tilde{s}_{i^*} +)$. If $x_2 = x^*(\tilde{s}_{i^*} -)$ then as $s \rightarrow \tilde{s}_{i^*} -$ we must have $x^*(s) \neq x_2$ and $x^*(s) \rightarrow x_2$, therefore by the continuity of $H(x)$ there exist $s'_1 < s'_2 < \tilde{s}_{i^*}$ such that $H(x^*(s'_1)) > \gamma, H(x^*(s'_2)) > \gamma$ and $x^*(s'_1) \neq x^*(s'_2)$. For such s'_1, s'_2 we have $f(x^*(s'_1)) < f(x^*(s'_2)) \leq f(x_2)$ from Proposition 2, i.e., $x^*(s'_1)$ is a feasible solution with strictly less objective value than x_2 , contradicting with the optimality of x_2 . Hence $x_2 = x^*(\tilde{s}_{i^*} +)$ must hold. If $s_1 < \tilde{s}_{i^*}$, we argue that $f(x_1) < f(x_2)$ hence arrive at a contradiction. Note that the feasible set $\text{Sol}(s_1)$ is closed, that $\text{Sol}(s) \subseteq \text{Sol}(s_1)$ for all $s > s_1$, and that $x_2 = \lim_{s \rightarrow \tilde{s}_{i^*}} x^*(s)$ with each $x^*(s) \in \text{Sol}(s_1)$, hence $x_2 \in \text{Sol}(s_1)$. Since $x_2 \neq x^*(s_1) = x_1$ we must have $f(x_1) < f(x_2)$ by the uniqueness of $x^*(s_1)$ for $\text{OPT}(s_1)$. Otherwise if $s_1 > \tilde{s}_{i^*}$, we take an $s \in (\tilde{s}_{i^*}, s_1)$ sufficiently close to $\tilde{s}_{i^*}$ so that $x^*(s)$ is sufficiently close to x_2 and $x^*(s) \neq x_1 = x^*(s_1)$, then from Proposition 2 we have $f(x_2) \leq f(x^*(s)) < f(x_1)$, a contradiction again. Case (iii): there are $\tilde{s}_{i_1^*}, \tilde{s}_{i_2^*}$ such that $x_1 \in x^*(\tilde{s}_{i_1^*})$ and $x_2 \in x^*(\tilde{s}_{i_2^*})$, but there is no $s \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ such that $x_1 = x^*(s)$ or $x_2 = x^*(s)$. By the same argument in Case (ii), we can show that it must be the case that $x_1 = x^*(\tilde{s}_{i_1^*} +)$ and $x_2 = x^*(\tilde{s}_{i_2^*} +)$, therefore $H(x_1), H(x_2) > \gamma$. Assume $\tilde{s}_{i_1^*} < \tilde{s}_{i_2^*}$ without loss of generality, and consider an $s \in (\tilde{s}_{i_1^*}, \tilde{s}_{i_2^*})$ that

is sufficiently close to $\tilde{s}_{i_1^*}$ so that $H(x^*(s)) > \gamma$, then by Proposition 2 we have $f(x_1) = f(x^*(s)) = f(x_2)$ hence $x^*(s) \in \mathcal{X}_S^*$, and we are in Case (ii) again. This proves that $\mathcal{X}_S^*$ must be a singleton.

To show that the optimal parameter set S^* must be a closed interval, we first observe that S^* must be a closed set due to the continuity of the solution path. Let $s_l^* = \min\{s : s \in S^*\}$ and $s_u^* = \max\{s : s \in S^*\}$, then we have $S^* \subseteq [s_l^*, s_u^*]$. Case (i): both $s_l^*, s_u^* \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$. Note that $x^*(s_l^*) = x^*(s_u^*) = x_S^*$ and $v(s_l^*) = v(s_u^*)$, hence Proposition 2 then forces $x^*(s) = x_S^*$ hence $s \in S^*$ for all $s \in [s_l^*, s_u^*] - \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ because otherwise $v(s_l^*) < v(s) < v(s_u^*)$. This further implies $x^*(\tilde{s}_i-) = x^*(\tilde{s}_i+) = \{x_S^*\}$ and subsequently $\tilde{s}_i \in S^*$ for any $\tilde{s}_i \in [s_l^*, s_u^*]$. Altogether we have $[s_l^*, s_u^*] = S^*$. Case (ii): $s_l^* = \tilde{s}_{i_1^*}$ for some $1 \leq i_1^* \leq M-1$ and $s_u^* \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$. Using Proposition 2 as in Case (i), one can show that for every $s \in [s_l^*, s_u^*] - \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ we have $v(s_l^*) = v(s) = v(s_u^*)$ and $x^*(s) = x_S^*$, therefore $S^* = [s_l^*, s_u^*]$ again. Case (iii): $s_u^* = \tilde{s}_{i_2^*}$ for some $1 \leq i_2^* \leq M-1$ and $s_l^* \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$. This case resembles Case (ii) and $S^* = [s_l^*, s_u^*]$ can be shown using the same argument. Case (iv): $s_l^* = \tilde{s}_{i_1^*}, s_u^* = \tilde{s}_{i_2^*}$ for some $1 \leq i_1^* \leq i_2^* \leq M-1$. If $\tilde{s}_{i_1^*} = \tilde{s}_{i_2^*}$ then x_S^* is a singleton and the interval representation trivially holds, so we focus on the case $\tilde{s}_{i_1^*} < \tilde{s}_{i_2^*}$. We argue that $x^*(\tilde{s}_{i_1^*}+) = x_S^*$. Otherwise if $x^*(\tilde{s}_{i_1^*}-) = x_S^*$, then Assumption 8 forces $H(x^*(\tilde{s}_{i_1^*}-)) > \gamma$, and the continuity of the solution path and the constraint H imply that $H(x^*(s')) > \gamma$ for some s' sufficiently close to $\tilde{s}_{i_1^*}$ but $s' < \tilde{s}_{i_1^*}$. Note that such an s' corresponds to an objective value $f(x^*(s')) = v(s') \leq \lim_{s \rightarrow \tilde{s}_{i_1^*}-} v(s) = f(x_S^*)$ by Proposition 2, therefore $x^*(s') = x_S^*$ by the uniqueness of $\mathcal{X}_S^*$, contradicting with the definition of s_l^* . Therefore it must be the case that $x^*(\tilde{s}_{i_1^*}+) = x_S^*$. Because $H(x^*(\tilde{s}_{i_1^*}+)) > \gamma$, there exists a $\delta > 0$ so that $\tilde{s}_{i_1^*} + \delta < \tilde{s}_{i_2^*}$, $\tilde{s}_{i_1^*} + \delta \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$, and $H(x^*(s)) > \gamma$ for all $s \in (\tilde{s}_{i_1^*}, \tilde{s}_{i_1^*} + \delta]$. Since Proposition 2 implies $v(s) = f(x_S^*)$ for such s , we must have $x^*(s) = x_S^*$ hence $[s_l^*, s_l^* + \delta] \subseteq S^*$. The rest part $[s_l^* + \delta, s_u^*] \subseteq S^*$ can be shown by treating $s_l^* + \delta$ as the s_l^* in Case (iii). Altogether we still have $S^* = [s_l^*, s_u^*]$. In particular, when $v(s)$ is strictly monotonic, it is clear that there can be at most one optimal parameter hence S^* becomes a singleton. $\square$

Proof of Theorem 5. For any function class $\mathcal{G}$ and $g \in \mathcal{G}$, we write $P(g) = \mathbb{E}_F[g(\xi)]$ (or just Pg) and $P_{n_2}(g) = \frac{1}{n_2} \sum_{i=1}^{n_2} g(\xi_i)$ (or just $P_{n_2}g$), as functions from $\mathcal{G} \rightarrow \mathbb{R}$. For any function $\phi : \mathcal{G} \rightarrow \mathbb{R}$, define $\|\phi\|_{\mathcal{G}} = \sup_{g \in \mathcal{G}} |\phi(g)|$. For example $\|P_{n_2} - P\|_{\mathcal{G}}$ denotes the maximal deviation of the sample mean.

First we show uniform convergence of the standard-deviation-adjusted sample mean to the expected constraint value. The function class $\{h(x, \cdot) | x \in \mathcal{X}\}$ is F -Donsker by Assumption 3, hence is F -Glivenko-Cantelli(GC). By Lemma 2.10.14 from Van der Vaart and Wellner (1996), the squared class $\{h^2(x, \cdot) | x \in \mathcal{X}\}$ is also F -GC under Assumptions 3 and 4. Define $\mathcal{X}_S = \{x^*(s) : s \in S \setminus \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}\} \cup (\cup_{i=1}^{M-1} x^*(\tilde{s}_i))$. As sub-classes, $\mathcal{F}_{\mathcal{X}_S} := \{h(x, \cdot) | x \in \mathcal{X}_S\}$ and $\mathcal{F}_{\mathcal{X}_S}^2 := \{h^2(x, \cdot) | x \in \mathcal{X}_S\}$ are both F -GC, i.e.,

$$\|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}} \rightarrow 0 \text{ a.s.}$$

$$\|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}^2} \rightarrow 0 \text{ a.s..}$$

Letting $\hat{\sigma}^2(h) = P_{n_2}(h^2) - (P_{n_2}(h))^2$ and $\sigma^2(h) = P(h^2) - (P(h))^2$ be the sample and true variances, we have

$$\begin{aligned} \|\hat{\sigma}^2 - \sigma^2\|_{\mathcal{F}_{\mathcal{X}_S}} &\leq \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}^2} + \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}}^2 + 2\|P\|_{\mathcal{F}_{\mathcal{X}_S}} \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}} \\ &= \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}^2} + \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}}^2 + 2 \sup_{x \in \mathcal{X}_S} |H(x)| \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}} \\ &\rightarrow 0 \text{ a.s.} \end{aligned} \quad (\text{EC.9})$$

where the limit comes from the fact that $\sup_{x \in \mathcal{X}_S} |H(x)| < \infty$ because H is continuous (implied by Assumption 5) and $\mathcal{X}_S$ is compact (implied by the piecewise uniform continuity condition, i.e., Assumption 7). By Assumption 4 we have $\|\sigma^2\|_{\mathcal{F}_{\mathcal{X}_S}} = \sup_{x \in \mathcal{X}_S} \text{Var}(h(x, \xi)) \leq \mathbb{E}[\sup_{x \in \mathcal{X}} h^2(x, \xi)] < \infty$, and arrive at

$$\left\| P_{n_2} - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma} - P \right\|_{\mathcal{F}_{\mathcal{X}_S}} \leq \|P_{n_2} - P\|_{\mathcal{F}_{\mathcal{X}_S}} + \frac{z_{1-\beta}}{\sqrt{n_2}} \sqrt{\|\hat{\sigma}^2\|_{\mathcal{F}_{\mathcal{X}_S}}} \rightarrow 0 \text{ a.s..}$$

When we use a discrete mesh $\{s_1, \dots, s_p\}$, it is clear that, using the notations from Algorithm 4 and $H_j := H(x^*(s_j))$

$$\max_{1 \leq j \leq p} \left| \hat{H}_j - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma}_j - H_j \right| \leq \left\| P_{n_2} - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma} - P \right\|_{\mathcal{F}_{\mathcal{X}_S}} \rightarrow 0 \text{ a.s..} \quad (\text{EC.10})$$

Secondly, we prove convergence of the estimated solution $x^*(\hat{s}^*)$ to the optimum x_S^* . Fixing any $\epsilon > 0$, we argue that almost surely there exists a finite N and $\bar{\epsilon}_S > 0$ such that for all $n_2 \geq N$ and $\epsilon_S \leq \bar{\epsilon}_S$ it holds $\|x^*(\hat{s}^*) - x_S^*\| < \epsilon$. To proceed, define

$$\delta := \min_{x \in \mathcal{X}_S} \{f(x) - f(x_S^*) | H(x) \geq \gamma, \|x - x_S^*\| \geq \epsilon\}. \quad (\text{EC.11})$$

Since the objective f is continuous and $\{x | x \in \mathcal{X}_S, H(x) \geq \gamma, \|x - x_S^*\| \geq \epsilon\}$ is a compact set, by a compactness argument we must have $\delta > 0$. By Assumption 10, for any $\epsilon' \leq \epsilon$ there exists some $s' \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ such that $H(x^*(s')) > \gamma$ and $\|x^*(s') - x_S^*\| < \epsilon'$. By continuity of f , one can set ϵ' small enough so that $f(x^*(s')) - f(x_S^*) < \frac{\delta}{2}$. Moreover, due to the continuity of $x^*(s)$ at s' and the continuity of f and H , there exists an $\bar{\epsilon}_S > 0$ such that $\min_{s \text{ s.t. } |s-s'| \leq \bar{\epsilon}_S} H(x^*(s)) > \gamma$ and $\max_{s \text{ s.t. } |s-s'| \leq \bar{\epsilon}_S} f(x^*(s)) < f(x_S^*) + \frac{\delta}{2}$. Therefore, when the mesh size $\epsilon_S \leq \bar{\epsilon}_S$, there must exist some $s_{j'} \in \{s_1, \dots, s_p\}$ such that

$$H(x^*(s_{j'})) \geq \min_{s \text{ s.t. } |s-s'| \leq \bar{\epsilon}_S} H(x^*(s)) > \gamma \quad (\text{EC.12})$$

$$f(x^*(s_{j'})) \leq \max_{s \text{ s.t. } |s-s'| \leq \bar{\epsilon}_S} f(x^*(s)) < f(x_S^*) + \frac{\delta}{2}. \quad (\text{EC.13})$$

For the given ϵ define for $\Delta \geq 0$

$$\delta_\Delta := \min_{x \in \mathcal{X}_S} \{f(x) - f(x_S^*) | H(x) \geq \gamma - \Delta, \|x - x_S^*\| \geq \epsilon\}. \quad (\text{EC.14})$$

We argue that $\lim_{\Delta \rightarrow 0+} \delta_\Delta \rightarrow \delta$ by contradiction. Clearly δ_Δ is non-increasing in Δ and $\delta_\Delta \leq \delta$, hence the limit $\lim_{\Delta \rightarrow 0+} \delta_\Delta$ must exist and is finite. Suppose $\lim_{\Delta \rightarrow 0+} \delta_\Delta < \delta$, then there exist $\tilde{\delta} < \delta$ and a

sequence $\{x_k\}_{k=1}^\infty \subset \mathcal{X}_S$ such that $\|x_k - x_S^*\| \geq \epsilon$, $H(x_k) \geq \gamma - \Delta_k$ with $\Delta_k \rightarrow 0+$, and $f(x_k) - f(x_S^*) \leq \tilde{\delta}$. By the compactness of $\mathcal{X}_S$, there must exist a subsequence $\{x_{k_s}\}_{s=1}^\infty$ converging to some $x_\infty \in \mathcal{X}_S$, and by continuity x_∞ must satisfy $\|x_\infty - x_S^*\| \geq \epsilon$, $H(x_\infty) \geq \gamma$ and $f(x_\infty) - f(x_S^*) \leq \tilde{\delta}$. From the definition (EC.11) of δ this implies $\delta \leq \tilde{\delta}$, a contradiction. Now pick a small enough Δ' so that $\delta_{\Delta'} > \frac{\delta}{2}$. From the uniform convergence (EC.10) we know that almost surely there exists some N such that for any $n_2 \geq N$

$$\max_{1 \leq j \leq p} \left| \hat{H}_j - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma}_j - H_j \right| < \min\{\Delta', \min_{s \text{ s.t. } |s-s'| \leq \epsilon_S} H(x^*(s)) - \gamma\} \leq \min\{\Delta', H(x^*(s_{j'})) - \gamma\}. \quad (\text{EC.15})$$

where the second inequality is due to (EC.12). In particular, (EC.15) implies that for such n_2 we have $\hat{H}(x^*(s_{j'})) - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma}(x^*(s_{j'})) > H(x^*(s_{j'})) - \min\{\Delta', H(x^*(s_{j'})) - \gamma\} \geq \gamma$, therefore on one hand we must have

$$f(x^*(\hat{s}^*)) \leq f(x^*(s_{j'})) < f(x_S^*) + \frac{\delta}{2} \quad (\text{EC.16})$$

where the first inequality holds due to the way $\hat{s}^*$ is chosen and the second results from (EC.13). On the other hand it also follows from (EC.15) that

$$\begin{aligned} H(x^*(\hat{s}^*)) &> \hat{H}(x^*(\hat{s}^*)) - \frac{z_{1-\beta}}{\sqrt{n_2}} \hat{\sigma}(x^*(\hat{s}^*)) - \min\{\Delta', H(x^*(s_{j'})) - \gamma\} \\ &\geq \gamma - \min\{\Delta', H(x^*(s_{j'})) - \gamma\} \\ &\geq \gamma - \Delta'. \end{aligned} \quad (\text{EC.17})$$

The bounds (EC.16) and (EC.17) on the objective value and the constraint value at the estimated solution $x^*(\hat{s}^*)$, together with the fact that $\delta_{\Delta'} > \frac{\delta}{2}$ due to the way Δ' is chosen, imply that $\|x^*(\hat{s}^*) - x_S^*\| < \epsilon$ by the definition (EC.14) of δ_Δ . Since ϵ can be arbitrarily small, we have $\lim_{n_2 \rightarrow \infty} x^*(\hat{s}^*) = x_S^*$ a.s.. Convergence of $\hat{s}^*$ to the optimal parameter set S^* is then a consequence of the convergence of $x^*(\hat{s}^*)$ to x_S^* . Suppose $\hat{s}^*$ does not converge to S^* , then by compactness of S there exists a subsequence $\hat{s}_k^*$ converging to some $s_\infty \notin S^*$. Since the corresponding $x^*(\hat{s}_k^*) \rightarrow x_S^*$ we have either $s_\infty \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ with $x^*(s_\infty) = x_S^*$ or $s_\infty \in \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ with $x_S^* \in x^*(s_\infty)$, however in either case $s_\infty \in S^*$, a contradiction.

Then we prove the feasibility guarantees. The case $H(x_S^*) > \gamma$ is relatively straightforward. By the continuity of H and that a.s. $x^*(\hat{s}^*) \rightarrow x_S^*$ we have $H(x^*(\hat{s}^*)) \rightarrow H(x_S^*) > \gamma$ a.s.. Almost surely convergence implies convergence in probability, thus $H(x^*(\hat{s}^*)) \rightarrow H(x_S^*)$ in probability and, in particular, $P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \rightarrow 1$. If $H(x_S^*) = \gamma$ we denote by

$$\mathbb{G}_{n_2}(x) := \sqrt{n_2}(P_{n_2}(h(x, \cdot)) - P(h(x, \cdot)))$$

the empirical process indexed by the decision x and let $\hat{\sigma}^2(x), \sigma^2(x)$ represent the sample and true variance of $h(x, \xi)$, and then write

$$H(x^*(\hat{s}^*)) \geq H(x^*(\hat{s}^*)) - \left(\hat{H}(x^*(\hat{s}^*)) - z_{1-\beta} \frac{\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} - \gamma \right) \quad (\text{EC.18})$$

$$\begin{aligned}
&= \gamma + (H(x^*(\hat{s}^*)) - \hat{H}(x^*(\hat{s}^*))) + z_{1-\beta} \frac{\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \\
&= \gamma - \frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x^*(\hat{s}^*)) + z_{1-\beta} \frac{\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \\
&= \gamma - \frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + z_{1-\beta} \frac{\hat{\sigma}(x_S^*)}{\sqrt{n_2}} + E_1 + E_2
\end{aligned} \tag{EC.19}$$

where the errors

$$E_1 = \frac{1}{\sqrt{n_2}} (\mathbb{G}_{n_2}(x_S^*) - \mathbb{G}_{n_2}(x^*(\hat{s}^*))), \quad E_2 = \frac{z_{1-\beta}}{\sqrt{n_2}} (\hat{\sigma}(x^*(\hat{s}^*)) - \hat{\sigma}(x_S^*)).$$

We need to show that $E_1 = o_p(\frac{1}{\sqrt{n_2}})$, $E_2 = o_p(\frac{1}{\sqrt{n_2}})$. We deal with E_2 first. E_2 can be bounded as

$$|E_2| \leq \frac{z_{1-\beta}}{\sqrt{n_2}} (2 \|\hat{\sigma} - \sigma\|_{\mathcal{F}_{\mathcal{X}_S}} + |\sigma(x^*(\hat{s}^*)) - \sigma(x_S^*)|).$$

On one hand we have already shown that $x^*(\hat{s}^*) \rightarrow x_S^*$ a.s.. On the other hand, $\sigma^2(x)$ is continuous in x . Therefore $|\sigma(x^*(\hat{s}^*)) - \sigma(x_S^*)| \rightarrow 0$ a.s. as $n_2 \rightarrow \infty$. By uniform convergence (EC.9) and the relation $\|\hat{\sigma} - \sigma\|_{\mathcal{F}_{\mathcal{X}_S}} \leq \sqrt{\|\hat{\sigma}^2 - \sigma^2\|_{\mathcal{F}_{\mathcal{X}_S}}}$, we have $\|\hat{\sigma} - \sigma\|_{\mathcal{F}_{\mathcal{X}_S}} \rightarrow 0$ a.s.. Consequently it holds $\sqrt{n_2} |E_2| = o(1)$ a.s. and, in particular, $E_2 = o_p(\frac{1}{\sqrt{n_2}})$.

To bound the error E_1 , let $\rho(x, x') = \sqrt{\text{Var}(h(x, \xi) - h(x', \xi))}$ denote the intrinsic semimetric of the tight Gaussian process $\mathbb{G}$ indexed by $x \in \mathcal{X}_S$ with mean zero and covariance structure $\text{Cov}(\mathbb{G}(x), \mathbb{G}(x')) = \text{Cov}_F(h(x, \xi), h(x', \xi))$, and for any $\epsilon > 0$ let $\delta(\epsilon) = \sup\{\rho(x, x_S^*) | x \in \mathcal{X}_S, \|x - x_S^*\| < \epsilon\}$. Note that Assumption 5 entails $\delta(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$. We have for any $\epsilon > 0$

$$\begin{aligned}
|E_1| &= |E_1| \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| < \epsilon\} + |E_1| \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon\} \\
&\leq \sup_{x \in \mathcal{X}_S \text{ s.t. } \|x - x_S^*\| < \epsilon} \frac{1}{\sqrt{n_2}} |\mathbb{G}_{n_2}(x_S^*) - \mathbb{G}_{n_2}(x)| \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| < \epsilon\} + \infty \cdot \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon\} \\
&\quad \text{where } \infty \cdot 0 = 0 \\
&\leq \sup_{x \in \mathcal{X}_S \text{ s.t. } \|x - x_S^*\| < \epsilon} \frac{1}{\sqrt{n_2}} |\mathbb{G}_{n_2}(x_S^*) - \mathbb{G}_{n_2}(x)| + \infty \cdot \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon\} \\
&\leq \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta(\epsilon)} \frac{1}{\sqrt{n_2}} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| + \infty \cdot \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon\}.
\end{aligned}$$

We have already shown that $\|x^*(\hat{s}^*) - x_S^*\| \rightarrow 0$ a.s., hence $P_{\xi_{1:n_2}}(\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon) \rightarrow 0$ for any fixed $\epsilon > 0$. Therefore we can choose an n_2 -dependent $\epsilon := \epsilon_{n_2}$ such that both $P_{\xi_{1:n_2}}(\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon_{n_2}) \rightarrow 0$ and $\epsilon_{n_2} \rightarrow 0$ as $n_2 \rightarrow \infty$, and get

$$|E_1| \leq \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta(\epsilon_{n_2})} \frac{1}{\sqrt{n_2}} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| + \infty \cdot \mathbf{1}\{\|x^*(\hat{s}^*) - x_S^*\| \geq \epsilon_{n_2}\}. \tag{EC.20}$$

By the way ϵ_{n_2} is chosen, the second term on the right hand side of (EC.20) is of arbitrarily small order, in particular, $o_p(\frac{1}{\sqrt{n_2}})$. To control the first term, note that $\delta(\epsilon_{n_2}) \rightarrow 0$ as $n_2 \rightarrow \infty$. Since the function class

$\mathcal{F}_{\mathcal{X}_S}$ is F -Donsker, the empirical process $\mathbb{G}_{n_2}$ on $\mathcal{F}_{\mathcal{X}_S}$ is asymptotically tight, hence by Theorem 1.5.7 and Addendum 1.5.8 from Van der Vaart and Wellner (1996) $\mathbb{G}_{n_2}$ is asymptotically uniformly equicontinuous in probability with respect to the intrinsic semimetric ρ of the limit Gaussian process $\mathbb{G}$, i.e., for any $\epsilon > 0$

$$\lim_{\delta \rightarrow 0} \limsup_{n_2 \rightarrow \infty} P_{\boldsymbol{\xi}_{1:n_2}} \left(\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| > \epsilon \right) = 0. \quad (\text{EC.21})$$

Note that $\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')|$ is monotonically increasing in δ a.s. and $\delta(\epsilon_{n_2}) \rightarrow 0$, it must hold that for any fixed $\delta > 0$

$$\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta(\epsilon_{n_2})} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| \leq \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| \quad \text{a.s.}$$

when n_2 is sufficiently large, therefore for any $\epsilon > 0$ the first term in (EC.20) can be controlled as

$$\begin{aligned} & \limsup_{n_2 \rightarrow \infty} P_{\boldsymbol{\xi}_{1:n_2}} \left(\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta(\epsilon_{n_2})} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| > \epsilon \right) \\ & \leq \limsup_{n_2 \rightarrow \infty} P_{\boldsymbol{\xi}_{1:n_2}} \left(\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| > \epsilon \right). \end{aligned} \quad (\text{EC.22})$$

Due to (EC.21) the right hand side of (EC.22) can be made arbitrarily small by sending $\delta \rightarrow 0$, hence the left hand side of (EC.22) must be identical to zero. Since ϵ is arbitrary, by definition $\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \delta(\epsilon_{n_2})} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| = o_p(1)$, which in turn leads to $E_1 = o_p(\frac{1}{\sqrt{n_2}})$.

We now go back to the representation (EC.19) of $H(x^*(\hat{s}^*))$ to conclude the coverage guarantee. From (EC.19) we see that

$$\begin{aligned} & \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\boldsymbol{\xi}_{1:n_2}} (H(x^*(\hat{s}^*)) \geq \gamma) \\ & \geq \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\boldsymbol{\xi}_{1:n_2}} \left(\gamma - \frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + z_{1-\beta} \frac{\hat{\sigma}(x_S^*)}{\sqrt{n_2}} + E_1 + E_2 \geq \gamma \right) \\ & = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\boldsymbol{\xi}_{1:n_2}} \left(-\frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + z_{1-\beta} \frac{\hat{\sigma}(x_S^*)}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0 \right) \\ & = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\boldsymbol{\xi}_{1:n_2}} \left(\frac{\mathbb{G}_{n_2}(x_S^*)}{\hat{\sigma}(x_S^*)} + o_p(1) \leq z_{1-\beta} \right) \quad \text{since } \hat{\sigma}(x_S^*) \rightarrow \sigma(x_S^*) \text{ a.s. and } \sigma^2(x_S^*) > 0 \text{ (Assumption 6)} \\ & = 1 - \beta \end{aligned}$$

where in the last equality we use Slutsky's theorem to justify that $\frac{\mathbb{G}_{n_2}(x_S^*)}{\hat{\sigma}(x_S^*)} + o_p(1)$ weakly converges to the standard normal. $\square$

Proof of Theorem 6. Following the proof of Theorem 5, we see that in order to conclude the tight feasibility confidence level it suffices to show that the inequality gap of (EC.18) is of order $o_p(\frac{1}{\sqrt{n_2}})$, i.e.,

$$\gamma \leq \hat{H}(x^*(\hat{s}^*)) - z_{1-\beta} \frac{\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \leq \gamma + o_p\left(\frac{1}{\sqrt{n_2}}\right). \quad (\text{EC.23})$$

Indeed, once the second inequality in (EC.23) is shown, we can use the representation (EC.19) and apply Slutsky's theorem, like in the proof of Theorem 5, to get

$$\begin{aligned}
& \lim P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
&= \lim P_{\xi_{1:n_2}}\left(\gamma - \frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + z_{1-\beta} \frac{\hat{\sigma}(x_S^*)}{\sqrt{n_2}} + E_1 + E_2 + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq \gamma\right) \\
&= \lim P_{\xi_{1:n_2}}\left(-\frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + z_{1-\beta} \frac{\hat{\sigma}(x_S^*)}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0\right) \text{ since } E_1, E_2 = o_p\left(\frac{1}{\sqrt{n_2}}\right) \\
&= \lim P_{\xi_{1:n_2}}\left(\frac{\mathbb{G}_{n_2}(x_S^*)}{\hat{\sigma}(x_S^*)} + o_p(1) \leq z_{1-\beta}\right) \\
&= 1 - \beta.
\end{aligned}$$

Now we prove the second inequality in (EC.23). By Proposition 3 the optimal parameter set is a singleton $S^* = \{s^*\}$. Moreover, in the case $H(x_S^*) = \gamma$ Assumption 8 forces that $s^* \notin \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} \cup \{s_l, s_u\}$. Suppose $s^* \in (\tilde{s}_{i^*}, \tilde{s}_{i^*+1})$ for some $0 \leq i^* \leq M-1$ (note that $\tilde{s}_0 = s_l, \tilde{s}_M = s_u$). Assumption 7 then ensures that the parameter-to-solution mapping $x^*(\cdot)$ is uniformly continuous in some neighborhood $\mathcal{N}(s^*) \subseteq (\tilde{s}_{i^*}, \tilde{s}_{i^*+1})$ of s^* . Since $\mathcal{N}(s^*)$ is contained in a compact set, the standard deviation function $\sigma(x^*(\cdot))$ is uniformly continuous in $\mathcal{N}(s^*)$. Moreover, the semimetric $\rho(x^*(\cdot), x^*(\cdot))$ between two solutions is also uniformly continuous in $\mathcal{N}(s^*) \times \mathcal{N}(s^*)$. Therefore as $\epsilon_S \rightarrow 0$ the following holds

$$\begin{aligned}
\omega_\sigma(2\epsilon_S) &:= \sup_{s, s' \in \mathcal{N}(s^*) \text{ s.t. } \|s-s'\| < 2\epsilon_S} |\sigma(x^*(s)) - \sigma(x^*(s'))| = o(1) \\
\omega_\rho(2\epsilon_S) &:= \sup_{s, s' \in \mathcal{N}(s^*) \text{ s.t. } \|s-s'\| < 2\epsilon_S} \rho(x^*(s), x^*(s')) = o(1).
\end{aligned}$$

According to the criterion of choosing $\hat{s}^*$ we must have for every parameter value s_j either $f(x^*(s_j)) \geq f(x^*(\hat{s}^*))$ or $\hat{H}(x^*(s_j)) - z_{1-\beta} \frac{\hat{\sigma}(x^*(s_j))}{\sqrt{n_2}} < \gamma$. Therefore if $\hat{s}^* \in \mathcal{N}(s^*)$, say $\hat{s}^* = s_{j^*}^{i^*}$, and $s_{j^*}^{i^*} \in \mathcal{N}(s^*)$ as well, then because $s_{j^*}^{i^*} < s_{j^*}^{i^*}$ and the parameter-to-objective mapping $f(x^*(\cdot))$ is strictly increasing in s it must hold that

$$\hat{H}(x^*(s_{j^*}^{i^*})) - z_{1-\beta} \frac{\hat{\sigma}(x^*(s_{j^*}^{i^*}))}{\sqrt{n_2}} < \gamma. \tag{EC.24}$$

We shall use this fact to derive (EC.23). For convenience, we denote by $B(s, \epsilon) := \{s' \in \mathcal{S} \mid \|s' - s\| < \epsilon\}$ the ball of radius $\epsilon > 0$ centered at s . Because $\hat{s}^* \rightarrow s^*$ a.s. and $\epsilon_S \rightarrow 0$, it is implied that $P_{\xi_{1:n_2}}(B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)) \rightarrow 1$ as $n_2 \rightarrow \infty$. Thus we can write

$$\begin{aligned}
& \hat{H}(x^*(\hat{s}^*)) - z_{1-\beta} \frac{\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \\
& \leq \infty \cdot \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \not\subseteq \mathcal{N}(s^*)\} + \left(\hat{H}(x^*(s_{j^*}^{i^*})) - z_{1-\beta} \frac{\hat{\sigma}(x^*(s_{j^*}^{i^*}))}{\sqrt{n_2}}\right) \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)\} \text{ where } s_{j^*}^{i^*} = \hat{s}^* \\
& \leq o_p\left(\frac{1}{\sqrt{n_2}}\right) + \left(\hat{H}(x^*(s_{j^*}^{i^*})) - z_{1-\beta} \frac{\hat{\sigma}(x^*(s_{j^*}^{i^*}))}{\sqrt{n_2}}\right) \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)\} +
\end{aligned}$$

$$\begin{aligned}
& \left(\left| \hat{H}(x^*(s_{j^*}^{i*}) - \hat{H}(x^*(s_{j^*}^{i*})) \right| + \frac{z_{1-\beta}}{\sqrt{n_2}} \left| \hat{\sigma}(x^*(s_{j^*}^{i*}) - \hat{\sigma}(x^*(s_{j^*}^{i*})) \right| \right) \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)\} \\
& \leq o_p\left(\frac{1}{\sqrt{n_2}}\right) + \gamma + \left(\left| H(x^*(s_{j^*}^{i*}) - H(x^*(s_{j^*}^{i*})) \right| + \frac{1}{\sqrt{n_2}} \left| \mathbb{G}_{n_2}(x^*(s_{j^*}^{i*}) - \mathbb{G}_{n_2}(x^*(s_{j^*}^{i*})) \right| + \right. \\
& \quad \left. \frac{z_{1-\beta}}{\sqrt{n_2}} (2\|\hat{\sigma} - \sigma\|_{\mathcal{F}_{\mathcal{X}_S}} + \left| \sigma(x^*(s_{j^*}^{i*}) - \sigma(x^*(s_{j^*}^{i*})) \right|) \right) \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)\} \\
& \quad \text{where the } \gamma \text{ term comes from (EC.24)} \\
& \leq o_p\left(\frac{1}{\sqrt{n_2}}\right) + \gamma + \left(o\left(\frac{1}{\sqrt{n_2}}\right) + \frac{1}{\sqrt{n_2}} \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \omega_\rho(2\epsilon_S)} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| + \right. \\
& \quad \left. \frac{z_{1-\beta}}{\sqrt{n_2}} (2\|\hat{\sigma} - \sigma\|_{\mathcal{F}_{\mathcal{X}_S}} + \omega_\sigma(2\epsilon_S)) \right) \mathbf{1}\{B(\hat{s}^*, 2\epsilon_S) \subseteq \mathcal{N}(s^*)\} \\
& \quad \text{where the } o\left(\frac{1}{\sqrt{n_2}}\right) \text{ terms is due to condition (21)} \\
& \leq o_p\left(\frac{1}{\sqrt{n_2}}\right) + \gamma + o\left(\frac{1}{\sqrt{n_2}}\right) + \frac{1}{\sqrt{n_2}} \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \omega_\rho(2\epsilon_S)} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| + \frac{z_{1-\beta}}{\sqrt{n_2}} (o_p(1) + o(1)) \\
& = \gamma + o_p\left(\frac{1}{\sqrt{n_2}}\right) + \frac{1}{\sqrt{n_2}} \sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \omega_\rho(2\epsilon_S)} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')|.
\end{aligned}$$

Since $\omega_\rho(2\epsilon_S) \rightarrow 0$, through an argument similar to (EC.22) the asymptotically uniform equicontinuity of $\mathbb{G}_{n_2}$ results in $\sup_{x, x' \in \mathcal{X}_S \text{ s.t. } \rho(x, x') \leq \omega_\rho(2\epsilon_S)} |\mathbb{G}_{n_2}(x) - \mathbb{G}_{n_2}(x')| = o_p(1)$. This finally leads to the upper bound in (EC.23), hence concludes the theorem. $\square$

Proof of Theorem 7. We first treat the unnormalized validator (Algorithm 2). As the first step, we introduce some notations and a few auxiliary Gaussian processes. Let $S^o := S \setminus \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$ be the parameter space after excluding the pathological points $\{\tilde{s}_1, \dots, \tilde{s}_{M-1}\}$. We denote by

$$\{\overline{\mathbb{G}}(s) : s \in S^o\}$$

the Gaussian process with mean zero and covariance structure $\text{Cov}(s, s') = \text{Cov}_F(h(x^*(s), \xi), h(x^*(s'), \xi))$, and by

$$\{\overline{\mathbb{G}}'(s) : s \in S^o\}$$

the Gaussian process with mean zero and covariance structure $\text{Cov}(s, s') = \frac{1}{n_2} \sum_{i=1}^{n_2} (h(x^*(s), \xi_i) - \hat{H}(x^*(s)))(h(x^*(s'), \xi_i) - \hat{H}(x^*(s')))$ where $\hat{H}(x^*(s)) = (1/n_2) \sum_{i=1}^{n_2} h(x^*(s), \xi_i)$ is the sample mean at $x^*(s)$ and $\hat{H}(x^*(s'))$ is the sample mean at $x^*(s')$. For a generic stochastic process $\{Y(\theta) : \theta \in \Theta\}$ over some set Θ , we denote by

$$\psi_{1-\beta}(\{Y(\theta) : \theta \in \Theta\})$$

the $1 - \beta$ quantile of $\sup_{\theta \in \Theta} Y(\theta)$. We can formally express the critical value calibrated in Algorithm 2 as $q_{1-\beta} = \psi_{1-\beta}(\{\overline{\mathbb{G}}'(s) : s \in \{s_1, \dots, s_p\}\})$, and $\bar{q}_{1-\beta} = \psi_{1-\beta}(\{\overline{\mathbb{G}}(s) : s \in S^o\})$. Under Assumption 3, the Gaussian process $\overline{\mathbb{G}}$ as the weak limit of the empirical process $\{\sqrt{n_2}((1/n_2) \sum_{i=1}^{n_2} h(x^*(s), \xi_i) - H(x^*(s))) : s \in S^o\}$ is a tight Borel measurable element in $l^\infty(S^o) := \{f : f \text{ is a function } S^o \rightarrow \mathbb{R} \text{ such that } \sup_{s \in S^o} |f(s)| < \infty\}$, therefore the sample path of $\overline{\mathbb{G}}$ is uniformly continuous with respect to

the semimetric $\rho(s, s') := \sqrt{\text{Var}_F(h(x^*(s), \xi) - h(x^*(s'), \xi))}$ almost surely (Example 1.5.10 in Van der Vaart and Wellner (1996)). Note that, under Assumptions 5 and 7, on each continuous piece of the solution path this semimetric is continuous in the pair s, s' with respect to the Euclidean metric on S . In other words, almost surely the sample path of $\overline{\mathbb{G}}$ is continuous with respect to the Euclidean metric $d(s, s') := |s - s'|$ on each piece $(\tilde{s}_i, \tilde{s}_{i+1})$. Therefore, by continuity, every countable dense (w.r.t. the Euclidean metric) subset $S_c^o \subset S^o$, e.g., the set of all rational s , renders

$$\sup_{s \in S^o} \overline{\mathbb{G}}(s) = \sup_{s \in S_c^o} \overline{\mathbb{G}}(s) \text{ almost surely.}$$

Suppose $S_c^o = \{s_j^o\}_{j=1}^\infty$, then $\max_{1 \leq j \leq k} \overline{\mathbb{G}}(s_j^o)$ monotonically increases in k towards the limit $\sup_{s \in S^o} \overline{\mathbb{G}}(s)$ almost surely, and almost sure convergence implies convergence in distribution therefore $\psi_{1-\beta}(\{\overline{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\})$ monotonically increases in k towards the limit $\bar{q}_{1-\beta}$, i.e.,

$$\psi_{1-\beta}(\{\overline{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\}) \leq \bar{q}_{1-\beta}, \text{ and } \lim_{k \rightarrow \infty} \psi_{1-\beta}(\{\overline{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\}) = \bar{q}_{1-\beta}. \quad (\text{EC.25})$$

As the second step, we want to show that $q_{1-\beta}$ converges to $\bar{q}_{1-\beta}$ almost surely. Under Assumptions 3 and 4, Theorem 10.6 in Kosorok (2008) states that, for almost every realization of the data sequence $\{\xi_i\}_{i=1}^\infty$, the Gaussian process $\overline{\mathbb{G}}'$ weakly converges to $\overline{\mathbb{G}}$ as $n_2 \rightarrow \infty$. By the continuous mapping theorem, $\sup_{s \in S^o} \overline{\mathbb{G}}'(s)$ also weakly converges to $\sup_{s \in S^o} \overline{\mathbb{G}}(s)$ almost surely, therefore $\lim_{n_2 \rightarrow \infty} \psi_{1-\beta}(\{\overline{\mathbb{G}}'(s) : s \in S^o\}) = \bar{q}_{1-\beta}$ almost surely. It is obvious that $q_{1-\beta} \leq \psi_{1-\beta}(\{\overline{\mathbb{G}}'(s) : s \in S^o\})$, hence we have established that $\limsup_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} \leq \bar{q}_{1-\beta}$. To show the other direction, we exploit the separability (EC.25) of $\overline{\mathbb{G}}$. For each i , let $s_{j_i} \in \{s_1, \dots, s_p\}$ be such that $s_{j_i} \rightarrow s_i^o$ as $\epsilon_S \rightarrow 0$. Consider two more Gaussian processes $\{\overline{\mathbb{G}}'(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}$ and $\{\overline{\mathbb{G}}(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}$. For a fixed k , let $\hat{\Sigma}', \hat{\Sigma}$ be the covariance matrices of $\{\overline{\mathbb{G}}'(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}$ and $\{\overline{\mathbb{G}}(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}$ respectively, and let Σ be the covariance matrix of $\{\overline{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\}$. Assumption 5 and the convergence of each s_{j_i} to s_i^o ensure that $\hat{\Sigma} \rightarrow \Sigma$ as $\epsilon_S \rightarrow 0$. To argue that $\hat{\Sigma}' - \hat{\Sigma} \rightarrow \mathbf{0} \in \mathbb{R}^{k \times k}$, where $\mathbf{0}$ denotes the $k \times k$ matrix with zero entries, we need the F-Glivenko-Contelli property of the product class $\mathcal{F} \cdot \mathcal{F} := \{f(\cdot) = h(x, \cdot)h(x', \cdot) : x, x' \in \mathcal{X}\}$. F-Donskerness implies F-Glivenko-Contelli, therefore $\mathcal{F} = \{h(x, \cdot) : x \in \mathcal{X}\}$ is F-Glivenko-Contelli under Assumption 3, which together with Assumption 4 forces the product class $\mathcal{F} \cdot \mathcal{F}$ to be F-Glivenko-Contelli by statement (ii) of Corollary 9.27 from Kosorok (2008). As a result, we have

$$\sup_{x, x' \in \mathcal{X}} \left| \frac{1}{n_2} \sum_{i=1}^{n_2} (h(x, \xi_i) - \hat{H}(x))(h(x', \xi_i) - \hat{H}(x')) - \text{Cov}_F(h(x, \xi), h(x', \xi)) \right| \rightarrow 0 \text{ as } n_2 \rightarrow \infty$$

almost surely, where $\hat{H}(x) = (1/n_2) \sum_{i=1}^{n_2} h(x, \xi_i)$ and $\hat{H}(x')$ is similar. In particular $\hat{\Sigma}' - \hat{\Sigma} \rightarrow \mathbf{0} \in \mathbb{R}^{k \times k}$ as desired. Combining this with the convergence of $\hat{\Sigma}$ to Σ , we conclude that $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} \hat{\Sigma}' = \Sigma$ almost surely. Since the distribution of a zero mean multivariate Gaussian is uniquely determined by its covariance matrix, we must have $\{\overline{\mathbb{G}}'(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}$ weakly converges to $\{\overline{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\}$,

and hence $\psi_{1-\beta}(\{\bar{\mathbb{G}}'(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}) \rightarrow \psi_{1-\beta}(\{\bar{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\})$ almost surely. Note that $\psi_{1-\beta}(\{\bar{\mathbb{G}}'(s) : s \in \{s_{j_1}, \dots, s_{j_k}\}\}) \leq \psi_{1-\beta}(\{\bar{\mathbb{G}}'(s) : s \in \{s_1, \dots, s_p\}\}) = q_{1-\beta}$, hence

$$\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} \geq \psi_{1-\beta}(\{\bar{\mathbb{G}}(s) : s \in \{s_1^o, \dots, s_k^o\}\}) \text{ for each } k.$$

This together with (EC.25) gives $\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} \geq \bar{q}_{1-\beta}$. Altogether we have $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} = \bar{q}_{1-\beta}$ almost surely.

The rest of the proof closely follows that of Theorem 5. We only highlight some modifications. First, each occurrence of $\frac{z_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}}$, $\frac{z_{1-\beta}\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}}$ and $\frac{z_{1-\beta}\hat{\sigma}(x_S^*)}{\sqrt{n_2}}$ shall be replaced by $\frac{q_{1-\beta}}{\sqrt{n_2}}$. Second, the second error E_2 in (EC.19) is no longer present, and the series of inequalities in the last paragraph become

$$\begin{aligned} & \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\ & \geq \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}\left(-\frac{1}{\sqrt{n_2}}\mathbb{G}_{n_2}(x_S^*) + \frac{q_{1-\beta}}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0\right) \\ & = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}\left(-\frac{1}{\sqrt{n_2}}\mathbb{G}_{n_2}(x_S^*) + \frac{\bar{q}_{1-\beta}}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0\right) \text{ since } q_{1-\beta} \rightarrow \bar{q}_{1-\beta} \text{ a.s.} \\ & = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}\left(\frac{\mathbb{G}_{n_2}(x_S^*)}{\sigma(x_S^*)} + o_p(1) \leq \frac{\bar{q}_{1-\beta}}{\sigma(x_S^*)}\right) \\ & = \Phi\left(\frac{\bar{q}_{1-\beta}}{\sigma(x_S^*)}\right) \text{ by Slutsky's theorem.} \end{aligned}$$

This completes the proof for Algorithm 2.

Now we prove the results for Algorithm 3 by a similar argument. Consider the Gaussian process

$$\{\tilde{\mathbb{G}}(s) : s \in S^o\}$$

with mean zero and covariance structure $\text{Cov}(s, s') = \text{Cov}_F(h(x^*(s), \xi), h(x^*(s'), \xi)) / (\sigma(x^*(s))\sigma(x^*(s')))$, and the Gaussian process

$$\{\tilde{\mathbb{G}}'(s) : s \in S^o\}$$

with mean zero and covariance structure $\text{Cov}(s, s') = \left[\frac{1}{n_2} \sum_{i=1}^{n_2} (h(x^*(s), \xi_i) - \hat{H}(x^*(s)))(h(x^*(s'), \xi_i) - \hat{H}(x^*(s')))\right] / (\hat{\sigma}(x^*(s))\hat{\sigma}(x^*(s')))$ where $\hat{H}(x^*(s)) = (1/n_2) \sum_{i=1}^{n_2} h(x^*(s), \xi_i)$, $\hat{\sigma}^2(x^*(s)) = (1/n_2) \sum_{i=1}^{n_2} (h(x^*(s), \xi_i) - \hat{H}(x^*(s)))^2$ and $\hat{H}(x^*(s'))$, $\hat{\sigma}^2(x^*(s'))$ are similarly defined. We have $\tilde{q}_{1-\beta} = \psi_{1-\beta}(\{\tilde{\mathbb{G}}(s) : s \in S^o\})$, and $q_{1-\beta} = \psi_{1-\beta}(\{\tilde{\mathbb{G}}'(s) : s \in \{s_1, \dots, s_p\}\}) \leq \psi_{1-\beta}(\{\tilde{\mathbb{G}}'(s) : s \in S^o\})$. Under the depicted conditions, Lemma 3 from Lam (2019) states that, for almost every realization of the data sequence $\{\xi_i\}_{i=1}^\infty$, the Gaussian process $\tilde{\mathbb{G}}'$ weakly converges to $\tilde{\mathbb{G}}$, so we have $\limsup_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} \leq \limsup_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} \psi_{1-\beta}(\{\tilde{\mathbb{G}}'(s) : s \in S^o\}) = \tilde{q}_{1-\beta}$ almost surely. By a similar argument based on the separability of $\tilde{\mathbb{G}}$ and the uniform convergence of covariance as in the case of Algorithm 2, we can show the other direction $\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} \geq \tilde{q}_{1-\beta}$ and thereby conclude that $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} q_{1-\beta} = \tilde{q}_{1-\beta}$ almost surely. The rest of the proof for Algorithm 3 also follows that of

Theorem 5, but with each occurrence of $z_{1-\beta}$ replaced by $q_{1-\beta}$. The display in the last paragraph should be modified to be

$$\begin{aligned}
& \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
& \geq \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}} \left(-\frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + \frac{q_{1-\beta} \hat{\sigma}(x_S^*)}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0 \right) \\
& = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}} \left(-\frac{1}{\sqrt{n_2}} \mathbb{G}_{n_2}(x_S^*) + \frac{\tilde{q}_{1-\beta} \sigma(x_S^*)}{\sqrt{n_2}} + o_p\left(\frac{1}{\sqrt{n_2}}\right) \geq 0 \right) \\
& \quad \text{since } q_{1-\beta} \rightarrow \tilde{q}_{1-\beta} \text{ and } \hat{\sigma}(x_S^*) \rightarrow \sigma(x_S^*) > 0 \text{ a.s.} \\
& = \liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}} \left(\frac{\mathbb{G}_{n_2}(x_S^*)}{\sigma(x_S^*)} + o_p(1) \leq \tilde{q}_{1-\beta} \right) \\
& = \Phi(\tilde{q}_{1-\beta}) \text{ by Slutsky's theorem.}
\end{aligned}$$

Lastly, by stochastic dominance of the supremum of the Gaussian process $\mathbb{G}$ or $\tilde{\mathbb{G}}$ over each of its marginal Gaussian component, it is straightforward that $\bar{q}_{1-\beta} \geq z_{1-\beta} \sigma(x_S^*)$ and that $\tilde{q}_{1-\beta} \geq z_{1-\beta}$, therefore both $\Phi\left(\frac{\bar{q}_{1-\beta}}{\sigma(x_S^*)}\right)$ and $\Phi(\tilde{q}_{1-\beta})$ are at least $1 - \beta$. $\square$

Appendix EC.4: Proofs of Results in Section 5

We first provide a lemma on the continuity of the solution path $x^*(s)$:

LEMMA EC.3. *Suppose the formulation $OPT(s)$ satisfies Assumptions 9 and 11-13. If $\hat{\mathcal{F}}(s) = \{x : g_t(x, s) \leq 0, t = 1, \dots, T\}$ for some finite T where each g_t is jointly continuous in x, s and convex in x for every fixed s , the objective $f(x)$ is continuous, and $OPT(s) := \min\{f(x) : x \in \mathcal{X} \cap \hat{\mathcal{F}}(s)\}$ has a unique solution $x^*(s)$ for all $s \in [\underline{s}, \bar{s}]$, then the solution path $x^*(s)$ is continuous on $[\underline{s}, \bar{s}]$.*

Proof of Lemma EC.3. The lemma is an application of Proposition 4.4 from Bonnans and Shapiro (2013). Based on the discussion following Proposition 4.4 in Bonnans and Shapiro (2013), we argue one by one that $OPT(s)$ satisfies assumptions (i)-(iv) of Proposition 4.4. Assumption (i): The objective $f(x)$ is continuous and independent of s hence it's jointly continuous in x, s . Assumption (ii): The constraints of $OPT(s)$ can be formulated as $(f_1(x, s), \dots, f_R(x, s), w'_1 x - z_1, \dots, w'_L x - z_L, g_1(x, s), \dots, g_T(x, s)) \in [0, +\infty)^{R+L+T}$, where the left hand side is a vector of continuous functions and the right hand size is a closed convex cone. Assumption (iii) is implied by our Assumptions 9 and 13. Assumption (iv): Since $OPT(s)$ is convex, our Assumptions 9 and 12 ensure Slater's condition for $OPT(s)$ for all $s \in [s_l, s_u]$, and Slater's condition implies Robinson's constraint qualification, a sufficient condition for assumption (iv). Therefore the set-valued mapping $\mathcal{X}^*(s) := \{x \in \mathcal{X} \cup \hat{\mathcal{F}}(s) : f(x) = \min_{x \in \mathcal{X} \cup \hat{\mathcal{F}}(s)} f(x)\}$ is upper semicontinuous at every $s \in [\underline{s}, \bar{s}]$. When the optimal solution $x^*(s)$ for $OPT(s)$ is unique, upper semicontinuity implies continuity, hence $x^*(s)$ is continuous on $[\underline{s}, \bar{s}]$. $\square$

The second lemma we present concerns the uniqueness of $x^*(s)$ for linear objectives:

LEMMA EC.4. *Consider an optimization problem in the form of $\min c'x$ subject to $f_r(x) \leq 0$ for $r = 1, \dots, R$ and $Ax \leq b$ for $A = [a_1, \dots, a_L]' \in \mathbb{R}^{L \times d}$ and $b \in \mathbb{R}^L$ where each f_r is continuous and convex and c is a non-zero vector. For each f_r and any two solutions $x_1 \neq x_2$ such that $f_r(x_1) = f_r(x_2) = 0$, assume $f_r(\theta x_1 + (1 - \theta)x_2) < 0$ for any $\theta \in (0, 1)$. If any $k \leq d - 1$ rows of A does not satisfy the SCI condition, then the optimal solution must be unique whenever one exists.*

Proof of Lemma EC.4. Suppose there are two optimal solutions x_1, x_2 . By convexity any solution in the form $\theta x_1 + (1 - \theta)x_2$ for $\theta \in [0, 1]$ is also optimal, and because of the condition on f_r we can assume that $f_r(x_1) < 0, f_r(x_2) < 0$ for all $r = 1, \dots, R$. Therefore, only the linear constraints can be binding on the line segment $\theta x_1 + (1 - \theta)x_2, \theta \in [0, 1]$. Let $A_o x \leq b_o$ be the binding linear constraints on the segment where A_o consists of rows of A and b_o contains the corresponding components of b , then it is clear that solution of the form $\theta x_1 + (1 - \theta)x_2$ is optimal for the linear program $\min c'x$ subject to $A_o x \leq b_o$. Since $A_o x_1 = A_o x_2 = b_o$, we have $A_o(x_2 - x_1) = 0$ hence the rank of A_o is at most $d - 1$. Now consider the dual $\min b_o'y$ subject to $A_o'y = -c, y \geq 0$. Since the rank of A_o is at most $d - 1$, by removing linearly dependent rows, the constraint $A_o'y = -c$ can be simplified to $\tilde{A}_o'y = -\tilde{c}$ where $\tilde{A}_o'$ has at most $d - 1$ linearly independent rows. Let y^* be an optimal basic feasible solution of the dual with the simplified constraint $\tilde{A}_o'y = -\tilde{c}$, then y^* has at most $d - 1$ non-zero (positive) components. However as a feasible solution y^* has to satisfy $A_o'y^* = -c$ therefore the SCI condition holds for the rows of A corresponding to the positive components of y^* , leading to a contradiction. $\square$

Proof of Theorem 9. We only need to verify the conditions of Lemma EC.3. In both cases (i) and (ii), $\hat{\mathcal{F}}(s) = \{x : \gamma + s - \frac{1}{n} \sum_{i=1}^n h(x, \xi_i) \leq 0\}$ and $\gamma + s - \frac{1}{n} \sum_{i=1}^n h(x, \xi_i)$ is obviously jointly continuous in x, s and convex in x , and also $f(x)$ is continuous. Therefore, it only remains to check uniqueness of $x^*(s)$ in order to apply Lemma EC.3.

In case (i), the strict convexity of $f(x)$ forces the solution $x^*(s)$ to be unique. In case (ii), we first treat the case when h is linear in x . We first note that for such h the SAA takes the form $-\left(\frac{1}{n} \sum_{i=1}^n A(\xi_i)\right)'x \leq \frac{1}{n} \sum_{i=1}^n b(\xi_i) - \gamma - s$. Therefore each constraint of $OPT(s)$ is either linear or strictly convex, and thanks to Lemma EC.4 it remains to show that the SCI condition is not satisfied for each $s \in S$. For any $k \leq d - 2$ rows $\{w_{l(1)}, \dots, w_{l(k)}\}$ of W and the coefficient vector $-\frac{1}{n} \sum_{i=1}^n A(\xi_i)$, we want to show that the SCI condition does not hold for $\{w_{l(1)}, \dots, w_{l(k)}, -\frac{1}{n} \sum_{i=1}^n A(\xi_i)\}$. Suppose SCI does hold, then we have the representation $-\frac{1}{n} \sum_{i=1}^n A(\xi_i) = \sum_{j=1}^k \lambda_j w_{l(j)} + \lambda_c c$, i.e., $-\frac{1}{n} \sum_{i=1}^n A(\xi_i)$ lies in the subspace of dimension spanned by $\{w_{l(1)}, \dots, w_{l(k)}, c\}$. However, $-\frac{1}{n} \sum_{i=1}^n A(\xi_i)$ has a density hence lies in any given subspace of dimension $\leq d - 1$ with probability zero. Therefore almost surely SCI does not hold for $\{w_{l(1)}, \dots, w_{l(k)}, -\frac{1}{n} \sum_{i=1}^n A(\xi_i)\}$. If only linear coefficients from $Wx \leq z$ are considered, SCI condition is again not satisfied by the condition imposed. Therefore almost surely SCI is not satisfied for $OPT(s)$. By noting that the SCI condition is independent of s since s is on the right hand side, we conclude that almost

surely SCI is not satisfied for all $s \in S$. When $h(x, \xi)$ is strictly concave in x , Lemma EC.4 can be directly applied to show the uniqueness of $x^*(s)$. $\square$

Proof of Theorem 11. We first argue that the constraint function $g(x, s) = \inf \left\{ \sum_{i=1}^n w_i h(x, \xi_i) : \sum_{i=1}^n \frac{1}{n} \phi(nw_i) \leq s, \sum_{i=1}^n w_i = 1, w_i \geq 0 \text{ for all } i \right\}$ is jointly continuous in x, s . Viewing both x, s as parameters of the optimization problem defining $g(x, s)$, one can easily check that the assumptions of Proposition 4.4 from Bonnans and Shapiro (2013) are satisfied, hence $g(x, s)$ as the optimal value of the optimization problem is continuous in the parameters x, s . It is also obvious that $g(x, s)$ is concave in x for every s because of its representation as the minimum of a family of concave functions. By Lemma EC.3 it remains to show the uniqueness of $x^*(s)$.

Case (i) follows from the strict convexity of f as in Theorem 9. In case (ii), we would like to show that the constraint function $g(x, s)$ is strictly concave in x . Indeed, due to compactness an optimal weight vector $\mathbf{w}^*$ must exist for the minimization problem defining $g(x, s)$. Consider $x_1 \neq x_2$ and $\theta \in (0, 1)$, and let $\mathbf{w}^*$ be the minimizing weight vector that gives the worst-case value $g(\theta x_1 + (1 - \theta)x_2, s)$ at the solution $\theta x_1 + (1 - \theta)x_2$. Then because of the strict concavity of $h(x, \xi)$ in x , we have

$$\begin{aligned} g(\theta x_1 + (1 - \theta)x_2, s) &= \sum_{i=1}^n w_i^* h(\theta x_1 + (1 - \theta)x_2, \xi_i) \\ &> \sum_{i=1}^n w_i^* (\theta h(x_1, \xi_i) + (1 - \theta)h(x_2, \xi_i)) \\ &\geq \theta g(x_1, s) + (1 - \theta)g(x_2, s). \end{aligned}$$

Therefore $g(x, s)$ is strictly concave in x , and uniqueness of $x^*(s)$ follows from the SCI condition not being satisfied and applying Lemma EC.4. In case (iii), the strict concavity of $g(x, s)$ can be shown as follows. Due to the strict convexity of ϕ , for each decision x the minimizing weight vector $\mathbf{w}^*$ not only exists but also is unique. Let $x_1 \neq x_2$, then by the condition there must be some $\theta' \in [0, 1)$ such that $\widehat{\text{Corr}}(x_1, \theta' x_1 + (1 - \theta')x_2) \neq 1$, and let $\mathbf{w}^{1*}, \mathbf{w}^{\theta'*}$ be the respective minimizing weight vectors. We argue that $\mathbf{w}^{1*} \neq \mathbf{w}^{\theta'*}$. The condition $\lim_{x \rightarrow 0+} \phi(x) = +\infty$ ensures positive components of the minimizing weight vector, hence the optimality condition entails $h(x_1, \xi_i) = \lambda_1 \phi'(nw_i^{1*}) - \beta_1$ and $h(\theta' x_1 + (1 - \theta')x_2, \xi_i) = \lambda_{\theta'} \phi'(nw_i^{\theta'*}) - \beta_{\theta'}$ for all $i = 1, \dots, n$ and some constants $\lambda_1, \beta_1, \lambda_{\theta'}, \beta_{\theta'}$. Therefore the empirical correlation between $h(x_1, \xi)$ and $h(\theta' x_1 + (1 - \theta')x_2, \xi)$ takes the form

$$\widehat{\text{Corr}}(x_1, \theta' x_1 + (1 - \theta')x_2) = \frac{(1/n) \sum_{i=1}^n (\phi'(nw_i^{1*}) - \bar{\phi}_1')(\phi'(nw_i^{\theta'*}) - \bar{\phi}_{\theta'}')}{\sqrt{(1/n) \sum_{i=1}^n (\phi'(nw_i^{1*}) - \bar{\phi}_1')^2 (1/n) \sum_{i=1}^n (\phi'(nw_i^{\theta'*}) - \bar{\phi}_{\theta'}')^2}}$$

where $\bar{\phi}_1' = (1/n) \sum_{i=1}^n \phi'(nw_i^{1*})$, $\bar{\phi}_{\theta'}' = (1/n) \sum_{i=1}^n \phi'(nw_i^{\theta'*})$. If $\mathbf{w}^{1*} = \mathbf{w}^{\theta'*}$, we have $\widehat{\text{Corr}}(x_1, \theta' x_1 + (1 - \theta')x_2) = 1$, a contradiction. Therefore, if $\theta' > 0$, we have $g(\theta' x_1 + (1 - \theta')x_2, s) = \sum_{i=1}^n w_i^{\theta'*} h(\theta' x_1 + (1 - \theta')x_2, \xi_i) \geq \theta' \sum_{i=1}^n w_i^{\theta'*} h(x_1, \xi_i) + (1 - \theta') \sum_{i=1}^n w_i^{\theta'*} h(x_2, \xi_i) > \theta' g(x_1, s) + (1 - \theta')g(x_2, s)$, hence

$g(\theta x_1 + (1 - \theta)x_2, s) > \theta g(x_1, s) + (1 - \theta)g(x_2, s)$ for all $\theta \in (0, 1)$ by the (non-strict) concavity of $g(x, s)$ in x . Otherwise, if $\theta' = 0$, i.e., $\widehat{\text{Corr}}(x_1, x_2) \neq 1$, then by continuity there exists a small enough $\theta > 0$ such that $\widehat{\text{Corr}}(x_1, \theta x_1 + (1 - \theta)x_2) \neq 1$, hence things reduce to the previous case. $\square$

Proof of Theorem 12. Strong duality results from Gao and Kleywegt (2016) or Blanchet et al. (2016) show that the constraint function takes the form

$$g(x, s) := \inf \left\{ \mathbb{E}_G[h(x, \xi)] : d_p(G, F_n) \leq s \right\} = \sup_{\lambda \geq 0} \left\{ \frac{1}{n} \sum_{i=1}^n \inf_{\xi \in \Xi} (h(x, \xi) + \lambda \|\xi - \xi_i\|^p) - \lambda s^p \right\}.$$

We would like to show that $g(x, s)$ is jointly continuous in x, s . Let $a(x, \lambda) = \frac{1}{n} \sum_{i=1}^n \inf_{\xi \in \Xi} (h(x, \xi) + \lambda \|\xi - \xi_i\|^p)$. It is clear that $a(x, \lambda) - \lambda s^p \leq \frac{1}{n} \sum_{i=1}^n h(x, \xi_i) - \lambda s^p \leq \frac{1}{n} \sum_{i=1}^n h(x, \xi_i) - \lambda s_i^p$ by taking $\xi = \xi_i$ in each infimum, and that $a(x, 0) = \inf_{\xi \in \Xi} h(x, \xi)$. For each x choose $\bar{\lambda}(x)$ so that $\frac{1}{n} \sum_{i=1}^n h(x, \xi_i) - \bar{\lambda}(x)s_i^p = \inf_{\xi \in \Xi} h(x, \xi)$. Since $h(x, \xi)$ is jointly continuous in x, ξ and Ξ is compact, $h(x, \xi)$ is uniformly continuous in x, ξ on $[x_o - \delta, x_o + \delta] \times \Xi$ for given x_o and $0 < \delta < \infty$. This uniform continuity implies that the infimum $\inf_{\xi \in \Xi} h(x, \xi)$ is continuous in x because $|\inf_{\xi \in \Xi} h(x, \xi) - \inf_{\xi \in \Xi} h(x_o, \xi)| \leq \sup_{\xi \in \Xi} |h(x, \xi) - h(x_o, \xi)| \rightarrow 0$ as $x \rightarrow x_o$. Therefore $\bar{\lambda}(x)$ is continuous in x and

$$g(x, s) = \sup_{0 \leq \lambda \leq \bar{\lambda}(x)} (a(x, \lambda) - \lambda s^p) = \sup_{0 \leq \lambda \leq \bar{\lambda}(x)} \left\{ \frac{1}{n} \sum_{i=1}^n \inf_{\xi \in \Xi} (h(x, \xi) + \lambda \|\xi - \xi_i\|^p) - \lambda s^p \right\}.$$

Since Ξ is compact, by an argument similar to the one used to prove the continuity of $\inf_{\xi \in \Xi} h(x, \xi)$ we see that $a(x, \lambda)$ is jointly continuous in x, λ . Continuity of $a(x, \lambda)$ and $\bar{\lambda}(x)$ leads to the joint continuity of $g(x, s)$ in x, s . To explain, for a fixed x and some $\delta > 0$, define $\bar{\lambda}_\delta := \sup_{x' \text{ s.t. } \|x' - x\|_2 \leq \delta} \bar{\lambda}(x')$, so for all x', s' such that $\|x' - x\|_2 \leq \delta$ and $|s' - s| \leq \delta$ we have

$$\begin{aligned} |g(x', s') - g(x, s)| &= \left| \sup_{0 \leq \lambda \leq \bar{\lambda}_\delta} (a(x', \lambda) - \lambda s'^p) - \sup_{0 \leq \lambda \leq \bar{\lambda}_\delta} (a(x, \lambda) - \lambda s^p) \right| \\ &\leq \sup_{0 \leq \lambda \leq \bar{\lambda}_\delta} |a(x', \lambda) - a(x, \lambda)| + \bar{\lambda}_\delta |s'^p - s^p| \rightarrow 0 \text{ as } x' \rightarrow x, s' \rightarrow s \end{aligned}$$

where the limit holds because $a(x, \lambda)$ is uniformly continuous on the compact set $\{x' : \|x' - x\| \leq \delta\} \times [0, \bar{\lambda}_\delta]$. Concavity of $g(x, s)$ in x holds because for any probability measure G the expectation $\mathbb{E}_G[h(x, \xi)]$ is concave in x and the infimum operation preserves concavity.

In order to utilize Lemma EC.3, it remains to prove uniqueness of $x^*(s)$ for all $s \in S$. In case (i) uniqueness trivially follows from strict convexity of f . In case (ii), we first establish a result concerning the existence of the worst-case distribution:

LEMMA EC.5. *Under the same conditions of Theorem 12, if $g(x, s) = a(x, \lambda^*) - \lambda^* s^p$ for some $\lambda^* > 0$, then there exists a distribution G^* that belongs to the Wasserstein ball and that achieves the worst-case expectation, i.e., $g(x, s) = \mathbb{E}_{G^*}[h(x, \xi)]$.*

Proof of Lemma EC.5. This is a direct consequence of Corollary 1 in Gao and Kleywegt (2016). Note that, since Ξ is compact and $h(x, \xi)$ is continuous, for each decision x the quantity $a(x, \lambda)$ is finite for all $\lambda \geq 0$. Corollary 1 from Gao and Kleywegt (2016) then entails the existence of the worst-case distribution if there exists a dual maximizer $\lambda^* > 0$. $\square$

Consider $x_1 \neq x_2$. If there exists some $\theta' \in (0, 1)$ such that $g(\theta'x_1 + (1 - \theta')x_2, s) = a(\theta'x_1 + (1 - \theta')x_2, \lambda') - \lambda's^p$ for some $\lambda' > 0$, then there exists some distribution $G_{\theta'}$ in the Wasserstein ball generating the worst-case expectation $g(\theta'x_1 + (1 - \theta')x_2, s)$. The strict concavity of h then implies

$$\begin{aligned} g(\theta'x_1 + (1 - \theta')x_2, s) &= \mathbb{E}_{G_{\theta'}}[h(\theta'x_1 + (1 - \theta')x_2, \xi)] \\ &> \mathbb{E}_{G_{\theta'}}[\theta'h(x_1, \xi) + (1 - \theta')h(x_2, \xi)] \\ &\geq \theta'g(x_1, s) + (1 - \theta')g(x_2, s). \end{aligned}$$

Since $g(x, s)$ is (non-strictly) concave in x , the above strict inequality at a certain θ' extends to all other θ , i.e., $g(\theta x_1 + (1 - \theta)x_2, s) > \theta g(x_1, s) + (1 - \theta)g(x_2, s)$ for all $\theta \in (0, 1)$. Otherwise if $g(\theta x_1 + (1 - \theta)x_2, s) = a(\theta x_1 + (1 - \theta)x_2, 0) > a(\theta x_1 + (1 - \theta)x_2, \lambda) - \lambda s^p$ for all $\theta \in (0, 1)$ and $\lambda > 0$, since $a(x, 0) = \inf_{\xi \in \Xi} h(x, \xi)$ we still have the strict concavity of $g(\theta x_1 + (1 - \theta)x_2, s)$ in θ . Therefore, according to Lemma EC.4, the solution $x^*(s)$ is unique. $\square$

Proof of Theorem 13. We first show that each $g_i(x, s) := \sup_{(\mu, \Sigma) \in \mathcal{U}_i(s)} \mu'x + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x' \Sigma x}$ is jointly continuous in x and s . For a fixed pair x_o, s_o and an arbitrary pair x, s , we write

$$\begin{aligned} &|g_i(x, s) - g_i(x_o, s_o)| \\ &\leq |g_i(x_o, s) - g_i(x_o, s_o)| + |g_i(x, s) - g_i(x_o, s)| \\ &\leq |g_i(x_o, s) - g_i(x_o, s_o)| + \sup_{(\mu, \Sigma) \in \mathcal{U}_i(s)} |\mu'x + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x' \Sigma x} - (\mu'x_o + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma x_o})| \\ &\leq |g_i(x_o, s) - g_i(x_o, s_o)| + \sup_{(\mu, \Sigma) \in \mathcal{U}_i(s_u)} |\mu'x + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x' \Sigma x} - (\mu'x_o + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma x_o})| \\ &\quad \text{where } s_u \text{ is the maximal value for } s. \end{aligned} \tag{EC.26}$$

Note that $\mu'x + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x' \Sigma x}$ as a function jointly in μ, Σ, x is continuous, and hence by the compactness of $\mathcal{U}_i(s_u)$ is uniformly continuous for $(\mu, \Sigma) \in \mathcal{U}_i(s_u)$ and x in some neighborhood of x_o . Uniform continuity implies that the second term in (EC.26) vanishes as $x \rightarrow x_o$. It remains to show that the first term in (EC.26) also vanishes, i.e., $g_i(x_o, s) \rightarrow g_i(x_o, s_o)$, as $s \rightarrow s_o$. We first show that as $s \rightarrow s_o$

$$\sup_{(\mu_s, \Sigma_s) \in \mathcal{U}_i(s)} \inf_{(\mu_{s_o}, \Sigma_{s_o}) \in \mathcal{U}_i(s_o)} (\|\mu_s - \mu_{s_o}\|_2 + \|\Sigma_s - \Sigma_{s_o}\|_2) \rightarrow 0, \tag{EC.27}$$

$$\sup_{(\mu_{s_o}, \Sigma_{s_o}) \in \mathcal{U}_i(s_o)} \inf_{(\mu_s, \Sigma_s) \in \mathcal{U}_i(s)} (\|\mu_s - \mu_{s_o}\|_2 + \|\Sigma_s - \Sigma_{s_o}\|_2) \rightarrow 0. \tag{EC.28}$$

We prove (EC.27) by contradiction. Suppose there exists $(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)$ and $s_k \rightarrow s_o$ such that $\inf_{(\mu_{s_o}, \Sigma_{s_o}) \in \mathcal{U}_i(s_o)} (\|\mu_{s_k} - \mu_{s_o}\|_2 + \|\Sigma_{s_k} - \Sigma_{s_o}\|_2) > \epsilon$ for some $\epsilon > 0$. Note that all $(\mu_{s_k}, \Sigma_{s_k})$ lie

in the compact set $\mathcal{U}_i(s_u)$, hence there is a subsequence converging to some $(\mu_\infty, \Sigma_\infty)$ such that $\inf_{(\mu_{s_o}, \Sigma_{s_o}) \in \mathcal{U}_i(s_o)} (\|\mu_\infty - \mu_{s_o}\|_2 + \|\Sigma_\infty - \Sigma_{s_o}\|_2) \geq \epsilon$, i.e., $(\mu_\infty, \Sigma_\infty) \notin \mathcal{U}_i(s_o)$. Since $\cap_{s > s_o} \mathcal{U}_i(s) = \mathcal{U}_i(s_o)$ and $\mathcal{U}_i(s)$ is non-decreasing in s , there exists some $\delta > 0$ such that $(\mu_\infty, \Sigma_\infty) \notin \mathcal{U}_i(s)$ for all $s \leq s_o + \delta$, a contradiction with the convergence to $(\mu_\infty, \Sigma_\infty)$. To show (EC.28), suppose there exists $(\mu_k, \Sigma_k) \in \mathcal{U}_i(s_o)$ and $s_k \rightarrow s_o$ such that

$$\inf_{(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)} (\|\mu_k - \mu_{s_k}\|_2 + \|\Sigma_k - \Sigma_{s_k}\|_2) > \epsilon \quad (\text{EC.29})$$

for some $\epsilon > 0$. By compactness, assume without loss of generality that (μ_k, Σ_k) converges to some limit $(\mu_\infty, \Sigma_\infty) \in \mathcal{U}_i(s_o)$. However, the condition $\overline{\cup_{s < s_o} \mathcal{U}_i(s)} = \mathcal{U}_i(s_o)$ ensures that $\inf_{(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)} (\|\mu_\infty - \mu_{s_k}\|_2 + \|\Sigma_\infty - \Sigma_{s_k}\|_2) \rightarrow 0$ as $s_k \rightarrow s_o$, which further entails that

$$\begin{aligned} & \inf_{(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)} (\|\mu_k - \mu_{s_k}\|_2 + \|\Sigma_k - \Sigma_{s_k}\|_2) \\ & \leq \inf_{(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)} (\|\mu_\infty - \mu_{s_k}\|_2 + \|\Sigma_\infty - \Sigma_{s_k}\|_2 + \|\mu_\infty - \mu_k\|_2 + \|\Sigma_\infty - \Sigma_k\|_2) \\ & \leq \inf_{(\mu_{s_k}, \Sigma_{s_k}) \in \mathcal{U}_i(s_k)} (\|\mu_\infty - \mu_{s_k}\|_2 + \|\Sigma_\infty - \Sigma_{s_k}\|_2) + \|\mu_\infty - \mu_k\|_2 + \|\Sigma_\infty - \Sigma_k\|_2 \\ & \rightarrow 0 \end{aligned}$$

a contradiction with (EC.29). This proves (EC.28). Now we use (EC.27) and (EC.28) to conclude $g_i(x_o, s) \rightarrow g_i(x_o, s_o)$ as $s \rightarrow s_o$. Since $\mathcal{U}_i(s_o)$ is compact, there exists an $(\mu_{s_o}^*, \Sigma_{s_o}^*) \in \mathcal{U}_i(s_o)$ such that $g_i(x_o, s_o) = \mu_{s_o}^{*'} x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_{s_o}^* x_o}$. (EC.28) entails that there exists some $(\mu_s, \Sigma_s) \in \mathcal{U}_i(s)$ for each s such that $(\mu_s, \Sigma_s) \rightarrow (\mu_{s_o}^*, \Sigma_{s_o}^*)$, therefore $\liminf_{s \rightarrow s_o} g_i(x_o, s) \geq \liminf_{s \rightarrow s_o} \mu_s' x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_s x_o} = g_i(x_o, s_o)$. On the other hand, for each s , compactness of $\mathcal{U}_i(s)$ implies the existence of some $(\mu_s^*, \Sigma_s^*) \in \mathcal{U}_i(s)$ such that $g_i(x_o, s) = \mu_s^{*'} x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_s^* x_o}$. (EC.27) then implies that there exists corresponding $(\mu_s^o, \Sigma_s^o) \in \mathcal{U}_i(s_o)$ such that $\|\mu_s^o - \mu_s^*\| \rightarrow 0$ and $\|\Sigma_s^o - \Sigma_s^*\| \rightarrow 0$ as $s \rightarrow s_o$. Since $\mu_s' x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_s x_o}$ as a function of (μ, Σ) is uniformly continuous on $\mathcal{U}_i(s_u)$, we have $\limsup_{s \rightarrow s_o} g_i(x_o, s) = \limsup_{s \rightarrow s_o} \mu_s^{*'} x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_s^* x_o} = \limsup_{s \rightarrow s_o} \mu_s^{o'} x_o + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_o' \Sigma_s^o x_o} \leq g_i(x_o, s_o)$. Altogether we have shown that $g_i(x_o, s) \rightarrow g_i(x_o, s_o)$, hence g_i is jointly continuous in x, s .

Secondly, we show the uniqueness of $x^*(s)$ so that the desired result follows from applying Lemma EC.3. Note that the supremum of a family of convex functions is still convex, therefore each g_i is convex in x . In case (i), strictly convexity of f automatically forces uniqueness of $x^*(s)$. In case (ii), we prove uniqueness by either condition (3) or condition (4). Consider $x_1 \neq x_2$ such that $g_i(x_1, s) = g_i(x_2, s) = b_i$, and $x_\theta := (1 - \theta)x_1 + \theta x_2$ for some $\theta \in (0, 1)$. Note that it is impossible that $x_1 = cx_2$ or $x_2 = cx_1$ for some $c \geq 0$ and $c \neq 1$, because otherwise $g_i(x_1, s) = cg_i(x_2, s) = cb_i \neq b_i$ or $g_i(x_2, s) = cg_i(x_1, s) = cb_i \neq b_i$. Let $(\mu_\theta, \Sigma_\theta) \in \mathcal{U}_i(s)$ be such that $g_i(x_\theta, s) = \mu_\theta' x_\theta + \sqrt{\frac{1-\alpha_i}{\alpha_i}} \sqrt{x_\theta' \Sigma_\theta x_\theta}$. Under condition (3), Σ_θ is automatically positive definite. Under condition (4), Σ_θ can be taken to be Σ_s because $x_\theta' \Sigma_s x_\theta - x_\theta' \Sigma_\theta x_\theta = x_\theta' (\Sigma_s - \Sigma_\theta) x_\theta \geq 0$,

where the last inequality is due to $\Sigma_\theta \preceq \Sigma_s$. That is, in either case, Σ_θ can be taken to be positive definite. We then follow the proof of Theorem 15 to show that

$$\begin{aligned} g_i(x_\theta, s) &< (1 - \theta)(\mu_\theta' x_1 + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x_1' \Sigma_\theta x_1}) + \theta(\mu_\theta' x_2 + \sqrt{\frac{1 - \alpha_i}{\alpha_i}} \sqrt{x_2' \Sigma_\theta x_2}) \\ &\leq (1 - \theta)g_i(x_1, s) + \theta g_i(x_2, s) \end{aligned}$$

and to conclude uniqueness of $x^*(s)$ for each s using Lemma EC.4. $\square$

Proof of Theorem 14. We first transform the infinitely constrained robust counterpart into finitely many constraints. Note that, since each uncertainty set $\mathcal{U}_i(s)$ is a bounded polytope, in each robust constraint $\max_{a_i \in \mathcal{U}_i(s)} a_i' x \leq b_i$ the maximum is attained at a vertex of $\mathcal{U}_i(s)$. The set of vertices of $\mathcal{U}_i(s)$ takes the form

$$\mathcal{V}_i(s) = \left\{ \widetilde{\mathcal{W}}_i^{-1} z_i + s \widetilde{\mathcal{W}}_i^{-1} e_i : \begin{array}{l} \widetilde{\mathcal{W}}_i \in \mathbb{R}^{d \times d} \text{ is an invertible submatrix of } \mathcal{W}_i \\ \mathcal{W}_i \widetilde{\mathcal{W}}_i^{-1} z_i - z_i \leq s(e_i - \mathcal{W}_i \widetilde{\mathcal{W}}_i^{-1} e_i) \end{array} \right\}$$

where the second condition ensures that $\widetilde{\mathcal{W}}_i^{-1} z_i + s \widetilde{\mathcal{W}}_i^{-1} e_i \in \mathcal{U}_i(s)$. The robust counterpart then becomes $v_i' x \leq b_i, v_i \in \mathcal{V}_i(s)$ for all $i = 1, \dots, K$. We make two important observations for $\mathcal{V}_i(s)$. First, the number of elements in $\mathcal{V}_i(s)$ is no more than the number of square submatrices of $\mathcal{W}_i$ which is finite. Second, the right hand side of $\mathcal{W}_i \widetilde{\mathcal{W}}_i^{-1} z_i - z_i \leq s(e_i - \mathcal{W}_i \widetilde{\mathcal{W}}_i^{-1} e_i)$ is linear in s hence the system of inequalities are valid for s in some interval of the form $(-\infty, u], [l, -\infty)$ or $[l, u]$, therefore the set of bases corresponding to vertices in $\mathcal{V}_i(s)$ changes at only finitely many s values. That is, there are $s_l = s'_0 \leq s'_1 < \dots < s'_{q-1} < s'_q = s_u$ such that, for each $1 \leq j \leq q$, there exist submatrices $\widetilde{\mathcal{W}}_{i,1}, \widetilde{\mathcal{W}}_{i,2}, \dots, \widetilde{\mathcal{W}}_{i,k_{i,j}}$ of each $\mathcal{W}_i$ such that the polyhedral RO $OPT(s)$ for all $s \in [s'_{j-1}, s'_j]$ can be simply expressed as

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & c'x \\ \text{subject to} \quad & (\widetilde{\mathcal{W}}_{i,l}^{-1} z_i + s \widetilde{\mathcal{W}}_{i,l}^{-1} e_i)' x \leq b_i \text{ for all } i = 1, \dots, K, l = 1, \dots, k_{i,j}. \end{aligned}$$

The uniqueness of $x^*(s)$ is relatively straightforward to justify. The above representation of the RO and an application of Lemma EC.4 suggest that, under the imposed conditions regarding satisfaction of the SCI condition, the solution $x^*(s)$ can be non-unique at only finitely many s values.

We now prove piecewise uniform continuity of the solution path. If at parameter value $\tilde{s}$ the solution $x^*(\tilde{s})$ is not unique, we call it a *non-unique point*. Between every two consecutive non-unique points $\tilde{s}_j < \tilde{s}_{j+1}$, $x^*(s)$ is unique hence is continuous in $(\tilde{s}_j, \tilde{s}_{j+1})$ due to Lemma EC.3. To show that $x^*(s)$ is actually uniformly continuous, it is sufficient and necessary to demonstrate that, as s approaches some non-unique point $\tilde{s}$, left and right limits $\lim_{s \rightarrow \tilde{s}^-} x^*(s), \lim_{s \rightarrow \tilde{s}^+} x^*(s)$ exist. Without loss of generality, we focus on left limit. Toward this goal, we first derive a convenient formula of the optimal solution $x^*(s)$ for s in a sufficiently small neighborhood $[\tilde{s} - \delta, \tilde{s})$ of $\tilde{s}$. As shown in the first step, for sufficiently small δ the reformulation $OPT(s)$ takes the form

$$\begin{aligned} \min_x \quad & c'x \\ \text{subject to} \quad & (\widetilde{\mathcal{W}}_{i,l}^{-1} z_i + s \widetilde{\mathcal{W}}_{i,l}^{-1} e_i)' x \leq b_i \text{ for all } i = 1, \dots, K, l = 1, \dots, k_i \\ & Wx \leq z \end{aligned}$$

for all $s \in [\tilde{s} - \delta, \tilde{s}]$. For convenience, we rewrite the above parametric program in a more compact form

$$\begin{aligned} \min_x \quad & c'x \\ \text{subject to} \quad & (A + s\Delta)x \leq b \end{aligned} \tag{EC.30}$$

where the matrix A contains all $\widetilde{\mathcal{W}}_{i,l}^{-1}z_i$'s and W as its rows, and the right hand side b has all the corresponding b_i 's and z as its components, whereas the perturbation matrix Δ consists of all the $\widetilde{\mathcal{W}}_{i,l}^{-1}e_i$'s (and zero entries for the W part of A). Note again that $x^*(s)$ is the unique optimal solution of (EC.30) for all $s \in [\tilde{s} - \delta, \tilde{s}]$. The dual of (EC.30) takes the form

$$\begin{aligned} \max_y \quad & b'y \\ \text{subject to} \quad & (A + s\Delta)'y = c \\ & y \leq 0 \end{aligned} \tag{EC.31}$$

By the theory of simplex method, for the dual (EC.31) there exists some basis $A_\beta + s\Delta_\beta$, where β is a subset of size d of $\{1, 2, \dots, \sum_{i=1}^K k_i + L\}$ and A_β, Δ_β denote the submatrices formed by the corresponding rows of A, Δ , that gives rise to the optimal solution $y_\beta^*(s) = (A'_\beta + s\Delta'_\beta)^{-1}c$ to (EC.31) (other components of $y^*(s)$ are all zero). Moreover, the corresponding primal optimal solution to (EC.30) is $x^*(s) = (A_\beta + s\Delta_\beta)^{-1}b_\beta$. By statement (ii) in Lemma 1 from Freund (1985), this optimal basis β for (EC.31) can change for only finitely many times as the parameter s varies, therefore by choosing a small enough δ this basis β remain the same one for all $s \in [\tilde{s} - \delta, \tilde{s}]$. That is, the unique optimal solution $x^*(s) = (A_\beta + s\Delta_\beta)^{-1}b_\beta$ for all $s \in [\tilde{s} - \delta, \tilde{s}]$ and some basis β . Given this convenient formula, we now establish existence of the left limit. Case (i): $A_\beta + \tilde{s}\Delta_\beta$ is invertible. In this case the inverse $(A_\beta + s\Delta_\beta)^{-1}$ must be continuous in s at the non-unique point $\tilde{s}$, hence the left limit $\lim_{s \rightarrow \tilde{s}^-} x^*(s) = (A_\beta + \tilde{s}\Delta_\beta)^{-1}b_\beta$. Case (ii): b_β is the zero vector. This case is trivial because $x^*(s)$ is also the zero vector hence the left limit exists and is the zero vector. Case (iii): $A_\beta + \tilde{s}\Delta_\beta$ is singular and b_β is a non-zero vector. Note that Assumption 13 implies that the solution path $\{x^*(s) : s \in \mathcal{S}\}$ is confined within a bounded region, and we shall use this key information to conclude this case. For convenience we reparametrize the solution path as $s_o = (s - \tilde{s} + \delta)^{-1}$ and $x_o^*(s_o) := x^*(s)$ for $s \in (\tilde{s} - \delta, \tilde{s})$. Letting $A_\beta^o = A_\beta + (\tilde{s} - \delta)\Delta_\beta$, we can express the reparametrization $x_o^*(s_o)$ as

$$x_o^*(s_o) = s_o((A_\beta^o)^{-1}\Delta_\beta + s_o I_d)^{-1}(A_\beta^o)^{-1}b_\beta \text{ for } s_o \in (\delta^{-1}, +\infty)$$

and our goal is to show $\lim_{s_o \rightarrow \delta^{-1}+} x_o^*(s_o)$ exists. The matrix $(A_\beta^o)^{-1}\Delta_\beta$ admits a Jordan decomposition $(A_\beta^o)^{-1}\Delta_\beta = P^{-1}JP$, where P is an invertible matrix with complex entries and J is the Jordan normal form with the diagonal structure

$$J = \begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_T \end{pmatrix}, \text{ with each diagonal block } J_t = \begin{pmatrix} \lambda_t & 1 & & \\ & \lambda_t & 1 & \\ & & \ddots & \\ & & & \lambda_t & 1 \\ & & & & \lambda_t \end{pmatrix}$$

where each λ_t is an eigenvalue of $(A_\beta^o)^{-1}\Delta_\beta$. With the Jordan decomposition, the reparametrized solution path takes the form

$$x_o^*(s_o) = s_o P^{-1} (J + s_o I_d)^{-1} P (A_\beta^o)^{-1} b_\beta.$$

Moreover, the inverse $(J + s_o I_d)^{-1}$ can be expressed as

$$(J + s_o I_d)^{-1} = \begin{pmatrix} (J_1 + s_o I)^{-1} & & & \\ & (J_2 + s_o I)^{-1} & & \\ & & \ddots & \\ & & & (J_T + s_o I)^{-1} \end{pmatrix}$$

where each diagonal block, if $J_t \in \mathbb{R}^{d_t \times d_t}$, has the form

$$(J_t + s_o I)^{-1} = \begin{pmatrix} (\lambda_t + s_o)^{-1} & -(\lambda_t + s_o)^{-2} & \cdots & & (-1)^{d_t-1} (\lambda_t + s_o)^{-d_t} \\ & (\lambda_t + s_o)^{-1} & -(\lambda_t + s_o)^{-2} & \cdots & \\ & & \ddots & \ddots & \vdots \\ & & & (\lambda_t + s_o)^{-1} & -(\lambda_t + s_o)^{-2} \\ & & & & (\lambda_t + s_o)^{-1} \end{pmatrix}. \quad (\text{EC.32})$$

If we let $(P(A_\beta^o)^{-1}b_\beta)_t$ be the vector of length d_t consisting of the $(1 + \sum_{i=1}^{t-1} d_i)$ -th to $(\sum_{i=1}^t d_i)$ -th components of $P(A_\beta^o)^{-1}b_\beta$, then

$$x_o^*(s_o) = s_o P^{-1} \begin{pmatrix} (J_1 + s_o I)^{-1} (P(A_\beta^o)^{-1}b_\beta)_1 \\ (J_2 + s_o I)^{-1} (P(A_\beta^o)^{-1}b_\beta)_2 \\ \vdots \\ (J_T + s_o I)^{-1} (P(A_\beta^o)^{-1}b_\beta)_T \end{pmatrix}.$$

We argue that $(P(A_\beta^o)^{-1}b_\beta)_t$ must be the zero vector for all t such that $\lambda_t = -\delta^{-1}$. Note that, since $A_\beta + \tilde{s}\Delta_\beta$ is singular, some λ_t must be $-\delta^{-1}$. Consider a Jordan block J_t with $\lambda_t = -\delta^{-1}$. From the form (EC.32) of the inverse, one can check that $\|(J_t + s_o I)^{-1}v\|_2 \rightarrow \infty$ as $s_o \rightarrow \delta^{-1}$ for any given non-zero vector v . However, the solution $x^*(s)$, hence each $(J_t + s_o I)^{-1}(P(A_\beta^o)^{-1}b_\beta)_t$, is confined to a bounded region, therefore $(P(A_\beta^o)^{-1}b_\beta)_t$ must be zero if $\lambda_t = -\delta^{-1}$. For those blocks with $\lambda_t \neq -\delta^{-1}$, the inverse $(J_t + s_o I)^{-1}$ is continuous in s_o at $s_o = \delta^{-1}$. Altogether, each block $(J_t + s_o I)^{-1}(P(A_\beta^o)^{-1}b_\beta)_t$ is either constantly zero or continuous at $s_o = \delta^{-1}$, therefore $x_o^*(s_o)$ has right limit at $s_o = \delta^{-1}$. This proves the existence of left limit of $x^*(s)$ at $s = \tilde{s}$. $\square$

Proof of Theorem 15. The continuity of the second-order cone constraints in x, s and its convexity in x are straightforward. We only focus on the uniqueness of $x^*(s)$. In case (i) uniqueness trivially follows from strict convexity of f . In case (ii), we want to show for each cone constraint that for any x_1, x_2 such that $\mu'_i x_1 + s \|\Sigma_i x_1\|_2 = \mu'_i x_2 + s \|\Sigma_i x_2\|_2 = b_i$ we must have $\mu'_i x_\theta + s \|\Sigma_i x_\theta\|_2 < b_i$ for any $x_\theta = \theta x_1 + (1-\theta)x_2$ where $\theta \in (0, 1)$. First of all, there exists no $c \geq 0, c \neq 1$ such that $x_1 = cx_2$ or $x_2 = cx_1$ because otherwise $\mu'_i x_1 + s \|\Sigma_i x_1\|_2 = c(\mu'_i x_2 + s \|\Sigma_i x_2\|_2) = cb_i \neq b_i$. Second, if there exists some $c < 0$ such that $x_1 = cx_2$ or $x_2 = cx_1$, then $\mu'_i x_\theta + s \|\Sigma_i x_\theta\|_2$ is piecewise linear in θ and has two pieces with different slopes,

therefore $\mu'_i x_\theta + s \|\Sigma_i x_\theta\|_2 < b_i$ for all $\theta \in (0, 1)$. Finally, if x_1 and x_2 are not parallel, then it is easy to verify that $\|\Sigma_i x_\theta\|_2$ is strictly convex in θ by examining its second order derivative, therefore we have $\mu'_i x_\theta + s \|\Sigma_i x_\theta\|_2 < b_i$ again. Together with the SCI condition not being satisfied, we can use Lemma EC.4 to conclude the uniqueness of $x^*(s)$ for all $s \in S$. Lemma EC.3 then implies the desired conclusion. $\square$

Appendix EC.5: Finite Sample Performance Guarantees for Univariate Gaussian Validator

This section provides finite-sample errors regarding the performance guarantees presented in Theorem 5, focusing on two general classes of constraints: differentiable stochastic constraints (Section EC.5.1) and linear chance constraints (Section EC.5.2).

EC.5.1. Differentiable Constraints

In order to derive finite-sample errors, we need stronger versions of Assumptions 3-5 and 10. Assumption 4 is replaced by boundedness of the fourth order moment:

ASSUMPTION EC.4. $m_4 := (\mathbb{E}_F[\sup_{x \in \mathcal{X}} |h(x, \xi) - H(x)|^4])^{\frac{1}{4}} < \infty$.

The L_2 -continuity condition for the constraint function h in Assumption 5 is now strengthened to a differentiability condition:

ASSUMPTION EC.5. *The random function $h(\cdot, \xi)$ is continuously differentiable on $\mathcal{X}$ for almost every $\xi \in \Xi$, and $\nabla H(x) = \mathbb{E}_F[\nabla h(x, \xi)]$. Assume $\bar{\rho} := \sup_{x \in \mathcal{X}} \rho(\text{Cov}_F(\nabla h(x, \xi))) < \infty$ where $\rho(\cdot)$ denotes the largest eigenvalue of a matrix.*

Note that, in the presence of Assumption 4, Assumption EC.5 implies Assumption 5 through the dominated convergence theorem. When the gradient has a square integrable envelope, i.e., $\mathbb{E}_F[\sup_{x \in \mathcal{X}} \|\nabla h(x, \xi)\|^2] < \infty$, and the decision space $\mathcal{X}$ is compact, Assumption EC.5 also implies Assumption 3.

We then assume uniqueness of the optimal parameter, and local differentiability of the solution path and the expectation constraint:

ASSUMPTION EC.6 (Unique optimal parameter and local differentiability). *The optimal parameter is unique, i.e., $S^* = \{s^*\}$, and $H(x_S^*) = \gamma$ at the optimal solution $x_S^* = x^*(s^*)$. Moreover, $H(x)$ is continuously differentiable in a neighborhood of x_S^* , and the parameter-to-solution mapping $x^*(s)$ is continuously differentiable in a neighborhood of s^* . There exists a $\delta > 0$ such that for all $s \in [s^* - \delta, s^* + \delta] \subseteq S$ it holds $\frac{1}{2} \leq \nabla H(x^*(s))' \nabla x^*(s) / \nabla H(x_S^*)' \nabla x^*(s^*) \leq 2$ and $\|\nabla x^*(s)\|_2 / \|\nabla x^*(s^*)\|_2 \leq 2$, and that for all $s \leq s^* - \delta$ it holds $H(x^*(s)) \leq H(x^*(s^* - \delta))$.*

We have the following finite-sample performance bounds for Algorithm 4:

THEOREM EC.14 (Finite-sample feasibility guarantee with univariate Gaussian validator).

Suppose Assumptions 2, 6-9 and EC.4-EC.6 hold, and $\{s_1, \dots, s_p\} \cap \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} = \emptyset$. Recall the mesh size $\epsilon_S = \sup_{s \in S} \inf_{1 \leq j \leq p} |s - s_j|$. Denote by $c^* := \nabla H(x_S^*)' \nabla x^*(s^*) / \|\nabla x^*(s^*)\|_2$, and by C some universal constant. For any $t > 0$ such that

$$2\epsilon_S < \text{err}(p, n_2, t) := \frac{4(1 + z_{1-\beta})\mathbf{m}_4}{c^* \|\nabla x^*(s^*)\|_2} \sqrt{\frac{t \log p}{n_2}} < \frac{\delta}{2}$$

it holds for the parameter $\hat{s}^*$ output by Algorithm 4 that

$$P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\text{err}(p, n_2, t)) \leq \frac{C}{t}.$$

If

$$2\epsilon_S < \frac{4(1 + z_{1-\beta})\mathbf{m}_4}{c^* \|\nabla x^*(s^*)\|_2} \cdot \frac{(\log p)^{1/4}}{n_2^{3/8}} < \frac{\delta}{2}$$

we have

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (1)}) \geq 1 - \beta - C(1 + z_{1-\beta})^2 \left(\frac{\mathbf{m}_4}{\sigma(x_S^*)}\right)^3 \left(1 + \frac{\sqrt{\rho}}{c^*}\right)^{\frac{2}{3}} \left(\frac{(\log p)^2}{n_2}\right)^{\frac{1}{4}}. \quad (\text{EC.33})$$

Proof of Theorem EC.14. First we present a lemma concerning moment inequalities for the maximal deviation of sample means:

LEMMA EC.6. *Let $\mathcal{G}$ be function class of finite cardinality, and $G(\xi) := \max_{g \in \mathcal{G}} |g(\xi)|$ be the envelope function. Suppose $\xi_1, \dots, \xi_n$ are i.i.d. observations from a common distribution F , then for any $k \geq 1$ we have*

$$\sqrt{n} \left(\mathbb{E} \left[\max_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n g(\xi_i) - \mathbb{E}_F[g(\xi)] \right|^k \right] \right)^{1/k} \leq C \sqrt{1 + \log |\mathcal{G}|} \left(\mathbb{E}_F[(G(\xi))^{\tilde{k}}] \right)^{1/\tilde{k}}$$

where $\tilde{k} = \max(2, k)$, the constant C only depends on k , and $|\mathcal{G}|$ denotes the cardinality of $\mathcal{G}$.

Proof of Lemma EC.6. This is a direct consequence of Theorem 2.14.1 from Van der Vaart and Wellner (1996). To apply that theorem, note that the covering number of the function class $\mathcal{G}$ is at most $|\mathcal{G}|$, hence its entropy integral is at most $\sqrt{1 + \log |\mathcal{G}|}$. $\square$

We use Lemma EC.6 to derive tail bounds for various maximal deviations. Denote by $H_j = H(x^*(s_j))$, and $\sigma_j^2 = \sigma^2(x^*(s_j))$ for convenience. Applying Lemma EC.6 to $\{h(x^*(s_j), \cdot) - H_j : j = 1, \dots, p\}$ with $k = 4$ gives

$$\begin{aligned} n_2^2 \mathbb{E}_{\xi_{1:n_2}} \left[\left(\max_j |\hat{H}_j - H_j| \right)^4 \right] &\leq C(\log p)^2 \mathbb{E}_F \left[\left(\max_j |h(x^*(s_j), \xi) - H_j| \right)^4 \right] \\ &\leq C(\log p)^2 \mathbf{m}_4^4 \end{aligned}$$

where C is a universal constant (because k is fixed at 4) and $\mathbb{E}_{\xi_{1:n_2}}$ denotes the expectation conditioned on Phase one data and with respect to Phase two data. Similarly applying the lemma to the squared class $\{(h(x^*(s_j), \cdot) - H_j)^2 - \sigma_j^2 : j = 1, \dots, p\}$ with $k = 2$ gives

$$\begin{aligned} & n_2 \mathbb{E}_{\xi_{1:n_2}} \left[\left(\max_j \left| \frac{1}{n_2} \sum_{i=1}^{n_2} (h(x^*(s_j), \xi_i) - H_j)^2 - \sigma_j^2 \right| \right)^2 \right] \\ & \leq C \log p \mathbb{E}_F \left[\left(\max_j |h(x^*(s_j), \xi) - H_j|^2 - \sigma_j^2 \right)^2 \right] \\ & \leq C \log p \mathbf{m}_4^4. \end{aligned}$$

By Markov's inequality, for any $t_1 > 0$ we have

$$\max_{j=1, \dots, p} \left| \hat{H}_j - H_j \right| \leq \frac{\mathbf{m}_4 t_1}{\sqrt{n_2}}$$

with probability at least $1 - C(\log p)^2/t_1^4$ and

$$\max_{j=1, \dots, p} |\hat{\sigma}_j^2 - \sigma_j^2| \leq \max_j \left| \frac{1}{n_2} \sum_{i=1}^{n_2} (h(x^*(s_j), \xi_i) - H_j)^2 - \sigma_j^2 \right| + \max_{j=1, \dots, p} (\hat{H}_j - H_j)^2 \leq \frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} \quad (\text{EC.34})$$

with probability at least

$$1 - \frac{C(\log p)^2}{t_1^4} - \frac{C(\log p)}{t_1^2}.$$

Note that, when the upper bound (EC.34) holds, $\max_j \hat{\sigma}_j^2 \leq \max_j \sigma_j^2 + \frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} \leq \mathbf{m}_4^2 + \frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2}$.

Therefore for any $t_1 > 0$

$$\begin{aligned} \max_{j=1, \dots, p} \left| \hat{H}_j - z_{1-\beta} \frac{\hat{\sigma}_j}{\sqrt{n_2}} - H_j \right| & \leq \frac{\mathbf{m}_4 t_1}{\sqrt{n_2}} + z_{1-\beta} \sqrt{\frac{\mathbf{m}_4^2}{n_2} + \frac{\mathbf{m}_4^2 t_1}{n_2^{3/2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2^2}} \\ & \leq (1 + z_{1-\beta}) \mathbf{m}_4 \frac{1 + t_1}{\sqrt{n_2}} \leq 2(1 + z_{1-\beta}) \mathbf{m}_4 \frac{t_1}{\sqrt{n_2}} \end{aligned} \quad (\text{EC.35})$$

for all $t_1 \geq 1$ with probability at least

$$1 - \frac{C(\log p)^2}{t_1^4} - \frac{C(\log p)}{t_1^2}.$$

For every constant $\epsilon < \delta$, the solution path $x^*(s)$ is differentiable for $s \in [s^* - \epsilon, s^* + \epsilon]$. Therefore for any s_j such that $|s_j - s^*| \leq \epsilon$, by differentiability we have

$$\begin{aligned} |[h(x^*(s_j), \xi) - H_j] - [h(x_s^*, \xi) - H(x_s^*)]| & = \left| \int_{s^*}^{s_j} (\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s) ds \right| \\ & \leq \int_{s^* - \epsilon}^{s^* + \epsilon} |(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)| ds. \end{aligned}$$

The right hand side of the above inequality serves as an envelope function of the function class $\{[h(x^*(s_j), \xi) - H_j] - [h(x_s^*, \xi) - H(x_s^*)] : |s_j - s^*| \leq \epsilon\}$. Assumption EC.5 entails that

$\mathbb{E}_F[|(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)|^2] \leq \bar{\rho} \|\nabla x^*(s)\|_2^2$ for all s , therefore by Jensen's inequality (or Minkowski's integral inequality)

$$\begin{aligned} & \mathbb{E}_F \left[\left(\int_{s^*-\epsilon}^{s^*+\epsilon} |(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)| ds \right)^2 \right] \\ & \leq \left(\int_{s^*-\epsilon}^{s^*+\epsilon} \sqrt{\mathbb{E}_F[|(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)|^2]} ds \right)^2 \\ & \leq \bar{\rho} \left(\int_{s^*-\epsilon}^{s^*+\epsilon} \|\nabla x^*(s)\|_2 ds \right)^2 \\ & \leq 16\bar{\rho} \|\nabla x^*(s^*)\|_2^2 \epsilon^2 \end{aligned}$$

an upper bound for the second moment of the envelope. Now applying Lemma EC.6 with $k = 2$ to $\{[h(x^*(s_j), \xi) - H_j] - [h(x_S^*, \xi) - H(x_S^*)] : |s_j - s^*| \leq \epsilon\}$, and noting that the cardinality does not exceed p , we have

$$n_2 \mathbb{E}_{\mathbf{\xi}_{1:n_2}} \left[\max_{j: |s_j - s^*| \leq \epsilon} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))|^2 \right] \leq C(\log p) \bar{\rho} \|\nabla x^*(s^*)\|_2^2 \epsilon^2$$

which implies through Markov's inequality that for every $t_2 > 0$

$$\max_{j: |s_j - s^*| \leq \epsilon} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| \leq \frac{t_2}{\sqrt{n_2}} \quad (\text{EC.36})$$

with probability at least

$$1 - \frac{C(\log p) \bar{\rho} \|\nabla x^*(s^*)\|_2^2 \epsilon^2}{t_2^2}.$$

Deviation inequalities (EC.34), (EC.35) and (EC.36) are the key elements for establishing finite sample error bounds. Lastly, we also need a bound characterizing the modulus of continuity of the variance $\sigma^2(x^*(s))$. For every s_j such that $|s_j - s^*| \leq \epsilon$

$$\begin{aligned} |\sigma_j^2 - \sigma^2(x_S^*)| &= |\mathbb{E}_F[(h(x^*(s_j), \xi) - H_j)^2] - \mathbb{E}_F[(h(x_S^*, \xi) - H(x_S^*))^2]| \\ &= \left| \mathbb{E}_F \left[\int_{s^*}^{s_j} 2(h(x^*(s), \xi) - H(x^*(s))) (\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s) ds \right] \right| \\ &\leq \mathbb{E}_F \left[\int_{s^*}^{s_j} 2|h(x^*(s), \xi) - H(x^*(s))| |(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)| ds \right] \\ &\leq \mathbb{E}_F \left[\int_{s^*-\epsilon}^{s^*+\epsilon} 2|h(x^*(s), \xi) - H(x^*(s))| |(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)| ds \right] \\ &= \int_{s^*-\epsilon}^{s^*+\epsilon} 2\mathbb{E}_F[|h(x^*(s), \xi) - H(x^*(s))| |(\nabla h(x^*(s), \xi) - \nabla H(x^*(s)))' \nabla x^*(s)|] ds \\ &\quad \text{by Fubini's theorem} \\ &\leq \int_{s^*-\epsilon}^{s^*+\epsilon} 2\sigma(x^*(s)) \sqrt{\bar{\rho}} \|\nabla x^*(s)\|_2 ds \quad \text{by Cauchy Schwartz inequality} \\ &\leq 8\mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon. \end{aligned}$$

That is, for all $\epsilon < \delta$

$$\max_{j: |s_j - s^*| \leq \epsilon} |\sigma_j^2 - \sigma^2(x_S^*)| \leq 8\mathbf{m}_4 \sqrt{\rho} \|\nabla x^*(s^*)\|_2 \epsilon. \quad (\text{EC.37})$$

We first show the deviation inequality for $\hat{s}^*$. If (EC.35) happens, and t_1 is such that

$$2\epsilon_S < \epsilon(t_1, n_2) := \frac{4(1 + z_{1-\beta})\mathbf{m}_4}{c^* \|\nabla x^*(s^*)\|_2} \cdot \frac{t_1}{\sqrt{n_2}} < \frac{\delta}{2} \quad (\text{EC.38})$$

we want to show that $|\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2)$. By Assumption EC.6, for any $s \in (s^*, s^* + \delta]$ the constraint value $H(x^*(s)) \geq \gamma + \frac{s-s^*}{2} \nabla H(x^*(s^*))' \nabla x^*(s^*) = \gamma + \frac{s-s^*}{2} c^* \|\nabla x^*(s^*)\|_2$, and similarly $H(x^*(s)) \leq \gamma + \frac{s-s^*}{2} c^* \|\nabla x^*(s^*)\|_2$ for all $s \in [s^* - \delta, s^*)$. Therefore $H_j > \gamma + 2(1 + z_{1-\beta})\mathbf{m}_4 \frac{t_1}{\sqrt{n_2}}$ for all $s_j \in (s^* + \epsilon(t_1, n_2), s^* + \delta]$ and $H_j < \gamma - 2(1 + z_{1-\beta})\mathbf{m}_4 \frac{t_1}{\sqrt{n_2}}$ for all $s_j \in [s^* - \delta, s^* - \epsilon(t_1, n_2))$. Under the condition that $2\epsilon_S < \epsilon(t_1, n_2)$ there must be some $\bar{j}$ for which $s_{\bar{j}} \in (s^* + \epsilon(t_1, n_2), s^* + 2\epsilon(t_1, n_2)) \subset (s^* + \epsilon(t_1, n_2), s^* + \delta]$ and hence $\hat{H}_{\bar{j}} - z_{1-\beta} \frac{\hat{\sigma}_{\bar{j}}}{\sqrt{n_2}} > \gamma$ on one hand. On the other hand the solution path has a derivative $\nabla x^*(s)$ that is non-zero in $[s^* - \delta, s^* + \delta]$ hence the parameter-to-objective mapping $v(s)$ strictly increases in s in the same interval. Therefore the picked parameter $\hat{s}^* \leq s_{\bar{j}}$. Similarly, there exists some $\underline{j}$ such that $s_{\underline{j}} \in (s^* - 2\epsilon(t_1, n_2), s^* - \epsilon(t_1, n_2)) \subset (s^* - \delta, s^* - \epsilon(t_1, n_2))$ and $H_{\underline{j}} < \gamma - 2(1 + z_{1-\beta})\mathbf{m}_4 \frac{t_1}{\sqrt{n_2}}$. Since $H(x^*(s^* - \delta)) \geq H(x^*(s))$ for all $s \leq s^* - \delta$, we have for all $s_j \leq s_{\underline{j}}$ that $H_j \leq H_{\underline{j}}$ and $\hat{H}_j - z_{1-\beta} \frac{\hat{\sigma}_j}{\sqrt{n_2}} < \gamma$, therefore $\hat{s}^* \leq s_{\underline{j}}$ is impossible. That is, it must be the case that $\hat{s}^* \in (s_{\underline{j}}, s_{\bar{j}}] \subset [s^* - 2\epsilon(t_1, n_2), s^* + 2\epsilon(t_1, n_2)]$. This gives the deviation inequality

$$P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\epsilon(t_1, n_2)) \leq C \left(\frac{(\log p)^2}{t_1^4} + \frac{\log p}{t_1^2} \right) \leq \frac{C \log p}{t_1^2}$$

provided that (EC.38) holds. Since the above bound becomes trivial when $(\log p)/t_1^2 \geq 1$, hence we can assume $(\log p)/t_1^2 < 1$ without loss of generality (and enlarge the universal constant C if necessary) to get

$$P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\epsilon(t_1, n_2)) \leq \frac{C \log p}{t_1^2}. \quad (\text{EC.39})$$

Now we derive the finite sample error for the feasibility confidence level. Using the same notation $\epsilon(t_1, n_2)$, we write

$$\begin{aligned} & P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\ & \geq P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2)) \\ & \geq P_{\xi_{1:n_2}}(\hat{H}(x^*(\hat{s}^*)) - H(x^*(\hat{s}^*)) - \frac{z_{1-\beta}\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \leq \hat{H}(x^*(\hat{s}^*)) - \frac{z_{1-\beta}\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} - \gamma, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2)) \\ & \geq P_{\xi_{1:n_2}}(\hat{H}(x^*(\hat{s}^*)) - H(x^*(\hat{s}^*)) - \frac{z_{1-\beta}\hat{\sigma}(x^*(\hat{s}^*))}{\sqrt{n_2}} \leq 0, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2)) \\ & = P_{\xi_{1:n_2}}(\hat{H}(x_S^*) - H(x_S^*) - \frac{z_{1-\beta}\sigma(x_S^*)}{\sqrt{n_2}} + \Delta_H + \Delta_\sigma \leq 0, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2)) \\ & \quad \text{where } \Delta_H = (\hat{H}(x^*(\hat{s}^*)) - H(x^*(\hat{s}^*))) - (\hat{H}(x_S^*) - H(x_S^*)), \Delta_\sigma = (z_{1-\beta}/\sqrt{n_2})(\sigma(x_S^*) - \hat{\sigma}(x^*(\hat{s}^*))) \end{aligned}$$

$$\begin{aligned}
&\geq P_{\xi_{1:n_2}} \left(\hat{H}(x_S^*) - H(x_S^*) - \frac{z_{1-\beta} \sigma(x_S^*)}{\sqrt{n_2}} + \max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| + \right. \\
&\quad \left. \frac{z_{1-\beta}}{\sqrt{n_2}} \max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| \leq 0, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2) \right) \\
&\geq P_{\xi_{1:n_2}} \left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} + \frac{\sqrt{n_2}}{\sigma(x_S^*)} \max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| + \right. \\
&\quad \left. \frac{z_{1-\beta}}{\sigma(x_S^*)} \max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| \leq z_{1-\beta}, |\hat{s}^* - s^*| \leq 2\epsilon(t_1, n_2) \right).
\end{aligned}$$

It follows from (EC.34) and (EC.37) that

$$\max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\sigma^2(x_S^*) - \hat{\sigma}_j^2| \leq \frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2)$$

with probability at least $1 - C(\log p)/t_1^2 - C(\log p)^2/t_1^4$. If $\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \leq \sigma^2(x_S^*)/4$, it follows from mean value theorem that with at least the same probability

$$\max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| \leq \frac{1}{\sigma(x_S^*)} \left(\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right). \quad (\text{EC.40})$$

For now we assume $\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \leq \sigma^2(x_S^*)/4$ holds so that the bound (EC.40) is valid. Later on we shall show that this is without loss of generality. We proceed as

$$\begin{aligned}
&P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
&\geq P_{\xi_{1:n_2}} \left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} + \frac{t_2}{\sigma(x_S^*)} + \frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) \leq z_{1-\beta} \right) \\
&\quad - P_{\xi_{1:n_2}} \left(\max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| > \frac{t_2}{\sqrt{n_2}} \right) - P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\epsilon(t_1, n_2)) \\
&\quad - P_{\xi_{1:n_2}} \left(\max_{j: |s_j - s^*| \leq 2\epsilon(t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| > \frac{1}{\sigma(x_S^*)} \left(\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) \right) \\
&\geq P_{\xi_{1:n_2}} \left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} \leq z_{1-\beta} - \frac{t_2}{\sigma(x_S^*)} - \frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathfrak{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathfrak{m}_4^2 t_1^2}{n_2} + 16\mathfrak{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) \right) \\
&\quad - \frac{C(\log p) \bar{\rho} \|\nabla x^*(s^*)\|_2^2 (\epsilon(t_1, n_2))^2}{t_2^2} - \frac{C \log p}{t_1^2}
\end{aligned}$$

by (EC.36), (EC.39) and (EC.40).

To deal with the first probability term, we recall the Berry-Esseen theorem. There exists some universal constant C such that

$$\sup_{t \in \mathbb{R}} |P_{\xi_{1:n_2}} \left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} \leq t \right) - \Phi(t)| \leq \frac{C \mathbb{E}_F[|h(x_S^*, \xi) - H(x_S^*)|^3]}{\sigma^3(x_S^*) \sqrt{n_2}}$$

where Φ is the cumulative distribution function for the standard normal. Noting that $\mathbb{E}_F[|h(x_S^*, \xi) - H(x_S^*)|^3] \leq \mathfrak{m}_4^3$ and that Φ has a bounded derivative, we further bound the confidence level as

$$P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma)$$

$$\begin{aligned}
&\geq \Phi\left(z_{1-\beta} - \frac{t_2}{\sigma(x_S^*)} - \frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} + 16\mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) \right) - \frac{C\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} \\
&\quad - \frac{C(\log p)\bar{\rho} \|\nabla x^*(s^*)\|_2^2 (\epsilon(t_1, n_2))^2}{t_2^2} - \frac{C \log p}{t_1^2} \\
&\geq 1 - \beta - C\left(\frac{t_2}{\sigma(x_S^*)} + \frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} + 16\mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) \right) - \frac{C\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} \\
&\quad - \frac{C(\log p)\bar{\rho} \|\nabla x^*(s^*)\|_2^2 (\epsilon(t_1, n_2))^2}{t_2^2} - \frac{C \log p}{t_1^2}.
\end{aligned}$$

Arranging terms gives

$$\begin{aligned}
&1 - \beta - P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
&\leq C\left(\frac{t_2}{\sigma(x_S^*)} + \frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} + \mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} \right. \\
&\quad \left. + \frac{(\log p)\bar{\rho} \|\nabla x^*(s^*)\|_2^2 (\epsilon(t_1, n_2))^2}{t_2^2} + \frac{\log p}{t_1^2} \right) \\
&\leq C\left(\frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} + \mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} \right. \\
&\quad \left. + \frac{((\log p)\bar{\rho})^{1/3} \|\nabla x^*(s^*)\|_2^{2/3} (\epsilon(t_1, n_2))^{2/3}}{\sigma^{2/3}(x_S^*)} + \frac{\log p}{t_1^2} \right) \tag{EC.41} \\
&\quad \text{by minimizing the bound over } t_2 \\
&\leq C\left(\frac{z_{1-\beta}}{\sigma^2(x_S^*)} \left(\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \right) + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} \right. \\
&\quad \left. + \frac{((\log p)\bar{\rho})^{1/3} \|\nabla x^*(s^*)\|_2^{2/3} (\epsilon(t_1, n_2))^{2/3}}{\sigma^{2/3}(x_S^*)} + \frac{\log p}{t_1^2} \right)
\end{aligned}$$

where in the last inequality we leave out the terms $\frac{\mathbf{m}_4^2 t_1^2}{n_2}$ because when $\frac{t_1}{\sqrt{n_2}} \leq 1$ it holds that $\frac{t_1^2}{n_2} \leq \frac{t_1}{\sqrt{n_2}}$ hence the former can be absorbed into the latter. Previously we assume that $\frac{\mathbf{m}_4^2 t_1}{\sqrt{n_2}} + \frac{\mathbf{m}_4^2 t_1^2}{n_2} + 16\mathbf{m}_4 \sqrt{\bar{\rho}} \|\nabla x^*(s^*)\|_2 \epsilon(t_1, n_2) \leq \sigma^2(x_S^*)/4$. This is without loss of generality, because otherwise the first error term in (EC.41) is of constant order which makes the upper bound trivial. Now expanding the $\epsilon(t_1, n_2)$ we further bound the error as follows

$$\begin{aligned}
&1 - \beta - P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
&\leq C\left(\frac{z_{1-\beta}\mathbf{m}_4^2}{\sigma^2(x_S^*)} \left(\frac{t_1}{\sqrt{n_2}} + \left[\frac{\bar{\rho}(1+z_{1-\beta})^2}{c^{*2}n_2} \right]^{1/2} t_1 \right) + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} + \left[\frac{(\log p)\bar{\rho}(1+z_{1-\beta})^2\mathbf{m}_4^2}{\sigma^2(x_S^*)c^{*2}n_2} \right]^{1/3} t_1^{2/3} + \frac{\log p}{t_1^2} \right) \\
&\leq C\left(\frac{z_{1-\beta}\mathbf{m}_4^2}{\sigma^2(x_S^*)} \left(1 + \frac{\sqrt{\bar{\rho}}(1+z_{1-\beta})}{c^*} \right) \frac{t_1}{\sqrt{n_2}} + (\log p)^{1/3} \left[\frac{\mathbf{m}_4}{\sigma(x_S^*)} \left(1 + \frac{\sqrt{\bar{\rho}}(1+z_{1-\beta})}{c^*} \right) \frac{t_1}{\sqrt{n_2}} \right]^{2/3} \right. \\
&\quad \left. + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} + \frac{\log p}{t_1^2} \right) \\
&\leq C\left(\left(\frac{z_{1-\beta}\mathbf{m}_4}{\sigma(x_S^*)} + (\log p)^{1/3}\right) \left[\frac{\mathbf{m}_4}{\sigma(x_S^*)} \left(1 + \frac{\sqrt{\bar{\rho}}(1+z_{1-\beta})}{c^*} \right) \frac{t_1}{\sqrt{n_2}} \right]^{2/3} + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*)\sqrt{n_2}} + \frac{\log p}{t_1^2} \right) \\
&\quad \text{since it can be assumed } \frac{\mathbf{m}_4}{\sigma(x_S^*)} \left(1 + \frac{\sqrt{\bar{\rho}}(1+z_{1-\beta})}{c^*} \right) \frac{t_1}{\sqrt{n_2}} \leq 1
\end{aligned}$$

$$\begin{aligned}
&\leq C \left(\frac{(1+z_{1-\beta})^{5/3} (\log p)^{1/3} \mathbf{m}_4^3}{\sigma^3(x_S^*) n_2^{1/3}} \left(1 + \frac{\sqrt{\bar{\rho}}}{c^*}\right)^{2/3} t_1^{2/3} + \frac{\log p}{t_1^2} + \frac{\mathbf{m}_4^3}{\sigma^3(x_S^*) \sqrt{n_2}} \right) \quad \text{since } \mathbf{m}_4 \geq \sigma(x_S^*) \\
&\leq C \left(\frac{(1+z_{1-\beta})^{5/3} (\log p)^{1/3} \mathbf{m}_4^3}{\sigma^3(x_S^*) n_2^{1/3}} \left(1 + \frac{\sqrt{\bar{\rho}}}{c^*}\right)^{2/3} t_1^{2/3} + \frac{\log p}{t_1^2} \right) \quad (\text{EC.42})
\end{aligned}$$

where in the last inequality we drop the last term since it's dominated by the first when $t_1 \geq 1$. Note that (EC.42) holds only under the condition (EC.38). It is straightforward to see that the bound (EC.42) is minimized at

$$t_1^* := \frac{(\log p)^{1/4} \sigma^{9/8}(x_S^*) n_2^{1/8}}{(1+z_{1-\beta})^{5/8} \mathbf{m}_4^{9/8} (1 + \sqrt{\bar{\rho}}/c^*)^{1/4}}$$

by equating the two error terms. Consider $\tilde{t}_1 := (\log p)^{1/4} n_2^{1/8}$. Since $\tilde{t}_1 = t_1^* (1+z_{1-\beta})^{5/8} \left(\frac{\mathbf{m}_4}{\sigma(x_S^*)}\right)^{9/8} (1 + \sqrt{\bar{\rho}}/c^*)^{1/4} > t_1^*$, the first term dominates at $t_1 = \tilde{t}_1$. Therefore when (EC.38) is satisfied at $t_1 = \tilde{t}_1$, we have

$$1 - \beta - P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \leq C(1+z_{1-\beta})^{\frac{5}{3}} \left(\frac{\mathbf{m}_4}{\sigma(x_S^*)}\right)^3 \left(1 + \frac{\sqrt{\bar{\rho}}}{c^*}\right)^{\frac{2}{3}} \left(\frac{(\log p)^2}{n_2}\right)^{\frac{1}{4}}.$$

The desired bound is obtained by replacing $\frac{5}{3}$ with 2 as the exponent of $1+z_{1-\beta}$. $\square$

EC.5.2. Linear Chance Constraints

Consider linear chance constraints in the form of $\mathbb{P}_F(a'_k x \leq b_k \text{ for } k = 1, \dots, K) \geq 1 - \alpha$. We assume the following isotropy condition:

ASSUMPTION EC.7 (Isotropy). *There exist constants D_2, D_3 such that for all unit vector $\nu \in \mathbb{R}^d$ and all $a_k, 1 \leq k \leq K$, the random variable $a'_k \nu$ has a sub-Gaussian norm at most D_2 , i.e., $\mathbb{E}[\exp((\frac{a'_k x}{D_2})^2)] \leq 2$, and has a density bounded above by D_3 . Each b_k is a non-zero constant.*

This assumption stipulates that each a_k has variability of constant order in all directions, and it trivially holds when each a_k is standard Gaussian.

We have the following finite-sample performance bounds for linear chance constraints:

THEOREM EC.15 (Finite-sample chance constraint feasibility guarantee with univariate Gaussian validator).

Consider (12) with a linear chance constraint $h(x, \xi) = \mathbf{1}(a'_k x \leq b_k \text{ for } k = 1, \dots, K)$ and $0 < \alpha < \frac{1}{2}$. Suppose Assumptions 2, 7-9, and EC.6-EC.7 hold, and $\{s_1, \dots, s_p\} \cap \{\tilde{s}_1, \dots, \tilde{s}_{M-1}\} = \emptyset$. Recall the notations ϵ_S and c^ from Theorem EC.14. For any $t > 0$ such that*

$$2\epsilon_S < \text{err}(p, n_2, t) := \frac{6(1+z_{1-\beta})}{c^* \|\nabla x^*(s^*)\|_2} \left(\sqrt{\frac{\alpha \log(4pt)}{n_2}} + \frac{\log(4pt)}{n_2} \right) < \frac{\delta}{2}$$

it holds for the parameter $\hat{s}^$ output by Algorithm 4 that*

$$P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\text{err}(p, n_2, t)) \leq \frac{1}{t}.$$

If

$$2\epsilon_S < \frac{6(1+z_{1-\beta})}{c^* \|\nabla x^*(s^*)\|_2} \left(\sqrt{\frac{\alpha \log(pn_2)}{n_2}} + \frac{\log(pn_2)}{n_2} \right) < \frac{\delta}{2}$$

we have

$$P_{\xi_{1:n_2}}(x^*(\hat{s}^*) \text{ is feasible for (12)}) \geq 1 - \beta - C(1 + z_{1-\beta})^2(1 + \sqrt{\tilde{C}K}(\log(\max\{3, \frac{n_2}{\tilde{C}}\})))^{\frac{1}{4}} \left(\frac{(\log(pn_2))^3}{\alpha n_2}\right)^{\frac{1}{4}} \quad (\text{EC.43})$$

where C is a universal constant and

$$\tilde{C} = \frac{D_2^2 D_3 \sqrt{\log(2K/\alpha)}}{c^* \min_{1 \leq k \leq K} |b_k|}.$$

To get a sense of the effect of the dimension d on the finite-sample error (EC.43), suppose that $D_2, D_3, K, \{b_k, k = 1, \dots, K\}$ are all numbers of constant order and we focus on the number c^* . The latter is the derivative of the satisfaction probability $P(x^*(s))$ with respect to the parameter s when the solution path is reparameterized to move at a unit speed. Therefore a proxy for the finite-sample performance of Algorithm 4 is the sensitivity of the satisfaction probability along the direction of the solution path. The more sensitive it is, the better is the finite-sample performance. Note that this sensitivity does not explicitly depend on the dimension.

Here we provide the proof Theorem EC.15:

Proof of Theorem EC.15. The proof follows the same line of argument as that of Theorem EC.14, but uses a different set of deviation inequalities tailored to bounded random variables. To avoid repetition, we focus on the derivation of these deviation inequalities.

We need the following concentration inequalities for the sample mean and sample variance:

LEMMA EC.7 (Adapted from Maurer and Pontil (2009)). *Let $X_i, i = 1, \dots, n$ be i.i.d. $[0, 1]$ -valued random variables, $\sigma^2 = \text{Var}(X_1)$, and $\hat{\sigma}^2$ be the sample variance. Then we have for every $\epsilon \in (0, 1)$ that*

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_1]\right| > \sqrt{\frac{2\sigma^2 \log(2/\epsilon)}{n}} + \frac{\log(2/\epsilon)}{3n}\right) \leq \epsilon$$

and

$$P\left(|\hat{\sigma} - \sigma| > \sqrt{\frac{2 \log(2/\epsilon)}{n-1}}\right) \leq \epsilon.$$

Proof of Lemma EC.7. Theorem 3 in Maurer and Pontil (2009) gives the following Bennett's inequality

$$P\left(\frac{1}{n} \sum_{i=1}^n X_i < \mathbb{E}[X_1] - \left(\sqrt{\frac{2\sigma^2 \log(1/\epsilon)}{n}} + \frac{\log(1/\epsilon)}{3n}\right)\right) \leq \epsilon.$$

Applying the above inequality to $1 - X_i, i = 1, \dots, n$ and noting that $\text{Var}(1 - X_1) = \text{Var}(X_1)$, we have

$$P\left(\frac{1}{n} \sum_{i=1}^n X_i > \mathbb{E}[X_1] + \left(\sqrt{\frac{2\sigma^2 \log(1/\epsilon)}{n}} + \frac{\log(1/\epsilon)}{3n}\right)\right) \leq \epsilon.$$

The first inequality in the lemma then comes from a union bound. The second inequality in the lemma is a direct consequence of Theorem 10 from Maurer and Pontil (2009). $\square$

Let $H_j := \mathbb{P}_F(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K)$ be the satisfaction probability at $x^*(s_j)$, and $\sigma_j^2 := H_j(1 - H_j)$ be the variance. Applying Lemma EC.7 to each $\mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K)$ gives

$$|\hat{H}_j - H_j| \leq \sqrt{\frac{2H_j(1 - H_j) \log(2/t_1)}{n_2}} + \frac{\log(2/t_1)}{3n_2}$$

with probability at least $1 - t_1$, and

$$|\hat{\sigma}_j - \sigma_j| \leq \sqrt{\frac{2 \log(2/t_1)}{n_2 - 1}}$$

with probability at least $1 - t_1$. Using a union bound, we have

$$|\hat{H}_j - H_j| \leq \sqrt{\frac{2H_j(1 - H_j) \log(2p/t_1)}{n_2}} + \frac{\log(2p/t_1)}{3n_2} \text{ for all } j = 1, \dots, p \quad (\text{EC.44})$$

with probability at least $1 - t_1$, and that

$$|\hat{\sigma}_j - \sigma_j| \leq \sqrt{\frac{2 \log(2p/t_1)}{n_2 - 1}} \text{ for all } j = 1, \dots, p \quad (\text{EC.45})$$

with probability at least $1 - t_1$. When (EC.45) happen, we also have

$$\hat{\sigma}_j \leq \sigma_j + |\hat{\sigma}_j - \sigma_j| \leq \sqrt{H_j(1 - H_j)} + \sqrt{\frac{2 \log(2p/t_1)}{n_2 - 1}} \text{ for all } j = 1, \dots, p.$$

Together with (EC.44), we can conclude that, with probability at least $1 - 2t_1$, for all $j = 1, \dots, p$

$$\begin{aligned} |\hat{H}_j - \frac{z_{1-\beta} \hat{\sigma}_j}{\sqrt{n_2}} - H_j| &\leq \frac{z_{1-\beta}}{\sqrt{n_2}} \left(\sqrt{H_j(1 - H_j)} + \sqrt{\frac{2 \log(2p/t_1)}{n_2 - 1}} \right) + \sqrt{\frac{2H_j(1 - H_j) \log(2p/t_1)}{n_2}} + \frac{\log(2p/t_1)}{3n_2} \\ &\leq 2(z_{1-\beta} + \sqrt{\log(2p/t_1)}) \left(\sqrt{\frac{H_j(1 - H_j)}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) \\ &\leq 2(1 + z_{1-\beta}) \sqrt{\log(2p/t_1)} \left(\sqrt{\frac{H_j(1 - H_j)}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) \end{aligned} \quad (\text{EC.46})$$

if we assume that $p \geq 2$ so that $\log(2p/t_1) > 1$.

Deviation bounds (EC.45) and (EC.46) are CCP counterparts of (EC.34) and (EC.35). Now we try to derive the CCP counterpart of (EC.36). For any $\epsilon < \delta$ and every parameter value $s_j \in [s^* - \epsilon, s^* + \epsilon]$ we have by differentiability

$$|a'_k x^*(s_j) - a'_k x^*(s^*)| = \left| \int_{s^*}^{s_j} a'_k \nabla x^*(s) ds \right| \leq \int_{s^*}^{s_j} |a'_k \nabla x^*(s)| ds \leq \eta_k(\epsilon) := \int_{s^* - \epsilon}^{s^* + \epsilon} |a'_k \nabla x^*(s)| ds.$$

Note that the sub-Gaussian norm $\|\cdot\|_{\psi_2} : \{X \text{ is a random variable} : \|X\|_{\psi_2} < \infty\} \rightarrow \mathbb{R}$ is a convex mapping, therefore by Jensen's inequality

$$\|\eta_k(\epsilon)\|_{\psi_2} \leq \int_{s^* - \epsilon}^{s^* + \epsilon} \|a'_k \nabla x^*(s)\|_{\psi_2} ds \leq \int_{s^* - \epsilon}^{s^* + \epsilon} D_2 \|\nabla x^*(s)\|_2 ds \leq 4D_2 \|\nabla x^*(s^*)\|_2 \epsilon.$$

With the above bound of $\eta_k(\epsilon)$, we want to quantify the closeness of the linear chance constraint at the solutions $x^*(s_j)$ and x_S^* . We apply a union bound to obtain

$$\begin{aligned}
& |\mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K) - \mathbf{1}(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K)| \\
& \leq \sum_{k=1}^K |\mathbf{1}(a'_k x^*(s_j) \leq b_k) - \mathbf{1}(a'_k x_S^* \leq b_k)| \\
& \leq \sum_{k=1}^K \mathbf{1}(a'_k x^*(s_j) \leq b_k < a'_k x_S^* \text{ or } a'_k x^*(s_j) > b_k \geq a'_k x_S^*) \\
& \leq \sum_{k=1}^K \mathbf{1}(a'_k x_S^* - \eta_k(\epsilon) \leq b_k < a'_k x_S^* \text{ or } a'_k x_S^* + \eta_k(\epsilon) > b_k \geq a'_k x_S^*) \\
& \leq \sum_{k=1}^K \mathbf{1}(|a'_k x_S^* - b_k| \leq \eta_k(\epsilon)).
\end{aligned}$$

Noting that difference of two indicator functions takes values in $\{-1, 0, 1\}$, we have

$$\begin{aligned}
& \mathbb{E}_F[(\mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K) - \mathbf{1}(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K))^2] \\
& = \mathbb{E}_F[\mathbf{1}(\mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K) - \mathbf{1}(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K))] \\
& \leq \sum_{k=1}^K \mathbb{P}_F(|a'_k x_S^* - b_k| \leq \eta_k(\epsilon)). \tag{EC.47}
\end{aligned}$$

In order to derive an upper bound for each of the K probabilities above, we first need a lower bound for $\|x_S^*\|_2$. If there are some $\tilde{k} \in \{1, 2, \dots, K\}$ such that $b_{\tilde{k}} < 0$, then

$$1 - \alpha = \mathbb{P}_F(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K) \leq \mathbb{P}_F(a'_{\tilde{k}} x_S^* \leq b_{\tilde{k}}) \leq 2 \exp\left(-\frac{\min_k |b_k|^2}{D_2^2 \|x_S^*\|_2^2}\right)$$

where in the last inequality Assumption EC.7 is used. This forces $\|x_S^*\|_2 \geq \frac{\min_k |b_k|}{D_2 \sqrt{\log(2/1-\alpha)}}$. Otherwise if all $b_k > 0$ then

$$\alpha = \mathbb{P}_F\left(\max_{k=1, \dots, K} a'_k x_S^* - b_k > 0\right) \leq \sum_{k=1}^K \mathbb{P}_F(a'_k x_S^* > b_k) \leq 2K \exp\left(-\frac{\min_k |b_k|^2}{D_2^2 \|x_S^*\|_2^2}\right)$$

which forces $\|x_S^*\|_2 \geq \frac{\min_k |b_k|}{D_2 \sqrt{\log(2K/\alpha)}}$. When $\alpha > 1/2$, the second lower bound dominates hence $\|x_S^*\|_2 \geq \frac{\min_k |b_k|}{D_2 \sqrt{\log(2K/\alpha)}}$ always holds. Now we go back to (EC.47) and notice that for each k and every $c > 0$

$$\begin{aligned}
\mathbb{P}_F(|a'_k x_S^* - b_k| \leq \eta_k(\epsilon)) & \leq \mathbb{P}_F(\eta_k(\epsilon) > \epsilon) + \mathbb{P}_F(|a'_k x_S^* - b_k| \leq \epsilon) \\
& \leq \mathbb{P}_F(\eta_k(\epsilon) > c) + \mathbb{P}_F(|a'_k x_S^* - b_k| \leq c) \\
& \leq 2 \exp\left(-\frac{c^2}{16D_2^2 \|\nabla x^*(s^*)\|_2^2 \epsilon^2}\right) + \frac{2D_3 c}{\|x_S^*\|_2} \\
& \leq 2 \exp\left(-\frac{c^2}{16D_2^2 \|\nabla x^*(s^*)\|_2^2 \epsilon^2}\right) + \frac{2D_2 D_3 \sqrt{\log(2K/\alpha)} c}{\min_k |b_k|}.
\end{aligned}$$

With $c = 4D_2 \|\nabla x^*(s^*)\|_2 \epsilon \cdot \sqrt{\log \left(\max \left\{ e, \frac{\min_k |b_k|}{D_2^2 D_3 \|\nabla x^*(s^*)\|_2 \sqrt{\log(2K/\alpha)\epsilon}} \right\} \right)}$, the above bound gives

$$\mathbb{P}_F(|a'_k x_S^* - b_k| \leq \eta_k(\epsilon)) \leq 10\tilde{\epsilon} \sqrt{\log \left(\max \left\{ e, \frac{1}{\tilde{\epsilon}} \right\} \right)}$$

where $\tilde{\epsilon} := \frac{D_2^2 D_3 \|\nabla x^*(s^*)\|_2 \sqrt{\log(2K/\alpha)\epsilon}}{\min_k |b_k|}$. From the union bound (EC.47) it follows that

$$\begin{aligned} & \mathbb{E}_F \left[\left(\mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K) - \mathbf{1}(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K) \right)^2 \right] \\ & \leq \sigma_\epsilon^2 := 10K\tilde{\epsilon} \sqrt{\log \left(\max \left\{ e, \frac{1}{\tilde{\epsilon}} \right\} \right)} \end{aligned} \quad (\text{EC.48})$$

for all $s_j \in [s^* - \epsilon, s^* + \epsilon]$. In particular, σ_ϵ^2 is a valid upper bound for the variance of each $\tilde{h}(x^*(s_j), \xi) := \mathbf{1}(a'_k x^*(s_j) \leq b_k \text{ for all } k = 1, \dots, K) - \mathbf{1}(a'_k x_S^* \leq b_k \text{ for all } k = 1, \dots, K)$ since the second moment always upper bounds the variance. Note that $(\tilde{h}(x^*(s_j), \xi) + 1)/2$ is $[0, 1]$ -valued, hence applying Lemma EC.7 to $(\tilde{h}(x^*(s_j), \xi) + 1)/2$ reveals that for all $s_j \in [s^* - \epsilon, s^* + \epsilon]$

$$|\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| \leq \sqrt{\frac{2\sigma_\epsilon^2 \log(2/t_2)}{n_2}} + \frac{2\log(2/t_2)}{3n_2}$$

with probability at least $1 - t_2$, therefore

$$\max_{j: |s_j - s^*| \leq \epsilon} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| \leq \sqrt{\frac{2\sigma_\epsilon^2 \log(2p/t_2)}{n_2}} + \frac{2\log(2p/t_2)}{3n_2} \quad (\text{EC.49})$$

with probability at least $1 - t_2$, a counterpart of (EC.36).

As in the proof of Theorem EC.14, we first derive the deviation inequality for $\hat{s}^*$. Based on (EC.46), we would like to find $\underline{H}$ such that, with high probability, for all $H_j \leq \underline{H}$ we have $\hat{H}_j - \frac{z_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} < 1 - \alpha$, as well as $\overline{H}$ such that all $H_j \geq \overline{H}$ satisfies $\hat{H}_j - \frac{z_{1-\beta}\hat{\sigma}_j}{\sqrt{n_2}} > 1 - \alpha$. Given the bound (EC.46), it suffices to $\underline{H}$ and $\overline{H}$ such that

$$H - 2(1 + z_{1-\beta})\sqrt{\log(2p/t_1)} \left(\sqrt{\frac{H(1-H)}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) > 1 - \alpha, \text{ for all } H \geq \overline{H} \quad (\text{EC.50})$$

$$H + 2(1 + z_{1-\beta})\sqrt{\log(2p/t_1)} \left(\sqrt{\frac{H(1-H)}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) < 1 - \alpha, \text{ for all } H \leq \underline{H}. \quad (\text{EC.51})$$

For (EC.50), since we must have $\overline{H} > 1 - \alpha$, it holds that $H(1-H) < \alpha(1-\alpha) < \alpha$ for all $H \geq \overline{H}$. Therefore $\overline{H} := 1 - \alpha + 2(1 + z_{1-\beta})\sqrt{\log(2p/t_1)} \left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right)$ satisfies (EC.50). For (EC.51), since the left hand side is monotonic in H , we only need to find a $\underline{H}$ for which the inequality in (EC.51) holds true. If

$$(1 + z_{1-\beta})\sqrt{\log(2p/t_1)} \left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) < \frac{\sqrt{2}}{4}\alpha \quad (\text{EC.52})$$

then one can verify that $\underline{H} := 1 - \alpha - 2\sqrt{2}(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2})$ satisfies (EC.51) by noting that $\underline{H} > 1 - 2\alpha$ and hence $\underline{H}(1 - \underline{H}) < 2\alpha$. In order for (EC.52) to hold, we consider p, t_1, n_2 satisfying the following counterpart of (EC.38)

$$2\epsilon_s < \epsilon(p, t_1, n_2) := \frac{6(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}}{c^* \|\nabla x^*(s^*)\|_2} \left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) < \frac{\delta}{2}. \quad (\text{EC.53})$$

We explain why (EC.53) implies (EC.52). Assumption EC.6 stipulates that $1 \geq H(x^*(s^* + \delta)) \geq H(x_S^*) + \frac{1}{2}\nabla H(x_S^*)'\nabla x^*(s^*)\delta = 1 - \alpha + \frac{1}{2}c^* \|\nabla x^*(s^*)\|_2 \delta$, which leads to $c^* \|\nabla x^*(s^*)\|_2 \delta \leq 2\alpha$. The second inequality in (EC.53) then gives $(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}) < \alpha/6 < \sqrt{2}\alpha/4$. Similar to the proof of Theorem EC.14, when (EC.53) holds and (EC.46) happens, we must have $\hat{s}^* \in [s^* - 2\epsilon(p, t_1, n_2), s^* + 2\epsilon(p, t_1, n_2)]$. Therefore under the condition (EC.53)

$$P_{\xi_{1:n_2}}(|\hat{s}^* - s^*| > 2\epsilon(p, t_1, n_2)) \leq 2t_1. \quad (\text{EC.54})$$

Now we proceed to deal with the finite sample confidence error. Following the same steps of bounding the feasibility confidence level, we have

$$\begin{aligned} & P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\ & \geq P_{\xi_{1:n_2}}\left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} + \frac{\sqrt{n_2}}{\sigma(x_S^*)} \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\hat{H}_j - H_j - (\hat{H}(x_S^*) - H(x_S^*))| + \right. \\ & \quad \left. \frac{z_{1-\beta}}{\sigma(x_S^*)} \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| \leq z_{1-\beta}, |\hat{s}^* - s^*| \leq 2\epsilon(p, t_1, n_2)\right). \end{aligned}$$

We bound the deviation of sample standard deviation as follows

$$\begin{aligned} \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| & \leq \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sigma(x_S^*) - \sigma_j| + \max_{j=1, \dots, p} |\sigma_j - \hat{\sigma}_j| \\ & \leq \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sqrt{\alpha(1-\alpha)} - \sqrt{H_j(1-H_j)}| + \max_{j=1, \dots, p} |\sigma_j - \hat{\sigma}_j|. \end{aligned}$$

The second error is taken care of by (EC.45). To bound the first error, we note that by Assumption EC.6 we have $\max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |H_j - (1 - \alpha)| \leq 2c^* \|\nabla x^*(s^*)\|_2 \cdot 2\epsilon(p, t_1, n_2) = 24(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2})$. Therefore if $24(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}) \leq \alpha/2$, it follows from applying mean value theorem that

$$\max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sqrt{\alpha(1-\alpha)} - \sqrt{H_j(1-H_j)}| \leq \frac{1}{\sqrt{\alpha}} 24(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}).$$

Similar to (EC.40), we can argue that $24(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}) \leq \alpha/2$ can be assumed without loss of generality so that the above bound can be assumed to hold. Together with (EC.45), we have

$$\begin{aligned} & \max_{j: |s_j - s^*| \leq 2\epsilon(p, t_1, n_2)} |\sigma(x_S^*) - \hat{\sigma}_j| \\ & \leq \frac{24}{\sqrt{\alpha}} (1 + z_{1-\beta})\sqrt{\log(2p/t_1)} \left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2} \right) + \sqrt{\frac{2\log(2p/t_1)}{n_2 - 1}} \text{ for all } j = 1, \dots, p \end{aligned} \quad (\text{EC.55})$$

with probability at least $1 - t_1$. Now we can further bound the confidence level

$$\begin{aligned}
& P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
& \geq P_{\xi_{1:n_2}}\left(\frac{\sqrt{n_2}(\hat{H}(x_S^*) - H(x_S^*))}{\sigma(x_S^*)} + \frac{1}{\sigma(x_S^*)}(\sqrt{2\sigma_{2\epsilon(p,t_1,n_2)}^2 \log(2p/t_2)} + \frac{2\log(2p/t_2)}{3\sqrt{n_2}})\right. \\
& \quad \left. + \frac{24z_{1-\beta}}{\sigma(x_S^*)\sqrt{\alpha}}(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}\left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}\right) + \frac{z_{1-\beta}}{\sigma(x_S^*)}\sqrt{\frac{2\log(2p/t_1)}{n_2 - 1}} \leq z_{1-\beta}\right) \\
& \quad - t_2 - 3t_1 \text{ by (EC.49), (EC.54) and (EC.55).}
\end{aligned}$$

Like in the proof of Theorem EC.14, applying Berry-Esseen theorem to the first probability on the right hand side and absorbing various constants into the universal constant C give

$$\begin{aligned}
& 1 - \beta - P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
& \leq C\left(\frac{1}{\sqrt{\alpha n_2}} + \frac{1}{\sqrt{\alpha}}(\sqrt{2\sigma_{2\epsilon(p,t_1,n_2)}^2 \log(2p/t_2)} + \frac{2\log(2p/t_2)}{3\sqrt{n_2}})\right. \\
& \quad \left. + \frac{z_{1-\beta}}{\alpha}(1 + z_{1-\beta})\sqrt{\log(2p/t_1)}\left(\sqrt{\frac{\alpha}{n_2}} + \frac{\sqrt{\log(2p/t_1)}}{n_2}\right) + \frac{z_{1-\beta}}{\sqrt{\alpha}}\sqrt{\frac{2\log(2p/t_1)}{n_2}} + t_1 + t_2\right) \\
& \leq C\left(\frac{1}{\sqrt{\alpha}}(\sqrt{\sigma_{2\epsilon(p,t_1,n_2)}^2 \log(2p/t_2)} + \frac{\log(2p/t_2)}{\sqrt{n_2}}) + (1 + z_{1-\beta})^2\sqrt{\frac{\log(2p/t_1)}{\alpha n_2}} + t_1 + t_2\right) \\
& \quad \text{by keeping dominant terms only} \\
& \leq C\left(\frac{1}{\sqrt{\alpha}}(\sqrt{\sigma_{2\epsilon(p,1/n_2,n_2)}^2 \log(pn_2)} + \frac{\log(pn_2)}{\sqrt{n_2}}) + (1 + z_{1-\beta})^2\sqrt{\frac{\log(pn_2)}{\alpha n_2}} + \frac{1}{n_2}\right) \\
& \quad \text{by taking } t_1 = t_2 = \frac{2}{n_2} \\
& \leq C\left(\sqrt{\frac{\log(pn_2)}{\alpha}}\sigma_{2\epsilon(p,2/n_2,n_2)} + (1 + z_{1-\beta})^2\frac{\log(pn_2)}{\sqrt{\alpha n_2}}\right) \\
& \leq C\left(\sqrt{\frac{\log(pn_2)}{\alpha}}\sigma_{2\epsilon(p,2/n_2,n_2)} + (1 + z_{1-\beta})^2\frac{\log(pn_2)}{\sqrt{\alpha n_2}}\right).
\end{aligned}$$

It remains to bound the $\sigma_{2\epsilon(p,2/n_2,n_2)}$ term which by the definition (EC.48) can be expressed as

$$\sigma_{2\epsilon(p,2/n_2,n_2)} = \sqrt{20K\tilde{\epsilon}(p,n_2)}\left(\log(\max\{e, \frac{1}{2\tilde{\epsilon}(p,n_2)}\})\right)^{\frac{1}{4}}$$

where

$$\tilde{\epsilon}(p,n_2) = \frac{6(1 + z_{1-\beta})D_2^2 D_3 \sqrt{\log(2K/\alpha)}}{c^* \min_k |b_k|} \left(\sqrt{\frac{\alpha \log(pn_2)}{n_2}} + \frac{\log(pn_2)}{n_2}\right).$$

Note that $\frac{1}{2\tilde{\epsilon}(p,n_2)} \leq \frac{c^* \min_k |b_k| n_2}{D_2^2 D_3 \sqrt{\log(2K/\alpha)}} = \frac{n_2}{\tilde{C}}$, hence using this upper bound in the logarithm we have

$$\begin{aligned}
\sigma_{2\epsilon(p,2/n_2,n_2)} & \leq 11\sqrt{(1 + z_{1-\beta})K\tilde{C}}\left(\log(\max\{e, \frac{n_2}{\tilde{C}}\})\right)^{\frac{1}{4}}\left(\sqrt{\frac{\alpha \log(pn_2)}{n_2}} + \frac{\log(pn_2)}{n_2}\right)^{\frac{1}{2}} \\
& \leq 11\sqrt{(1 + z_{1-\beta})K\tilde{C}}\left(\log(\max\{e, \frac{n_2}{\tilde{C}}\})\right)^{\frac{1}{4}}\left[\left(\frac{\alpha \log(pn_2)}{n_2}\right)^{\frac{1}{4}} + \left(\frac{\log(pn_2)}{n_2}\right)^{\frac{1}{2}}\right]
\end{aligned}$$

where the second inequality follows because $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for any $a, b \geq 0$. Substituting $\sigma_{2\epsilon(p,2/n_2,n_2)}$ with its upper bound gives

$$\begin{aligned}
& 1 - \beta - P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \\
& \leq C \left(\left[(1 + z_{1-\beta})^2 + \sqrt{(1 + z_{1-\beta})K\tilde{C}} (\log(\max\{e, \frac{n_2}{\tilde{C}}\}))^{\frac{1}{4}} \right] \frac{\log(pn_2)}{\sqrt{\alpha n_2}} \right. \\
& \quad \left. + \sqrt{(1 + z_{1-\beta})K\tilde{C}} (\log(\max\{e, \frac{n_2}{\tilde{C}}\}))^{\frac{1}{4}} \frac{(\log(pn_2))^{3/4}}{(\alpha n_2)^{1/4}} \right) \\
& \leq C(1 + z_{1-\beta})^2 (1 + \sqrt{\tilde{C}K} (\log(\max\{e, \frac{n_2}{\tilde{C}}\}))^{\frac{1}{4}}) \left(\frac{(\log(pn_2))^{3/4}}{(\alpha n_2)^{1/4}} + \frac{\log(pn_2)}{\sqrt{\alpha n_2}} \right) \\
& \leq C(1 + z_{1-\beta})^2 (1 + \sqrt{\tilde{C}K} (\log(\max\{e, \frac{n_2}{\tilde{C}}\}))^{\frac{1}{4}}) \frac{(\log(pn_2))^{3/4}}{(\alpha n_2)^{1/4}}
\end{aligned}$$

where the last inequality follows because $\frac{(\log(pn_2))^{3/4}}{(\alpha n_2)^{1/4}} \leq \frac{\log(pn_2)}{\sqrt{\alpha n_2}}$ if $\frac{(\log(pn_2))^{3/4}}{(\alpha n_2)^{1/4}} \leq 1$. Note again that this bound is valid when (EC.53) is satisfied at $t_1 = \frac{2}{n_2}$. Replacing e , the base of the natural logarithm, with 3 gives the desired bound. $\square$

Appendix EC.6: Applying Univariate Gaussian Validator to Formulations with Multidimensional Conservativeness Parameters

We consider the case of multidimensional conservativeness parameter, i.e., $S \subset \mathbb{R}^q$ for some $q \geq 2$, and present the asymptotic performance guarantees of the univariate Gaussian validator. We assume the following counterpart of Assumption 7:

ASSUMPTION EC.8 (Piecewise uniformly continuous solution curve). *The parameter space $S \subset \mathbb{R}^q$ is compact, and there exist M connected and open subsets $S_1, \dots, S_M$ of S such that (i) $S_i \cap S_{i'} = \emptyset$ for all $i \neq i'$; (ii) $m(\cup_{i=1}^M S_i) = m(S)$ where $m(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}^q$; and (iii) for each $i = 1, \dots, M$, the optimal solution $x^*(s)$ of $OPT(s)$ exists and is unique for all $s \in S_i$, and $x^*(s)$ as a function of s is uniformly continuous on S_i .*

Similar to the case of scalar parameter, the solution curve $x^*(s)$ on each piece S_i can be continuously extended to the closure $\overline{S_i} := \cap_{S' \text{ is closed}, S_i \subseteq S'} S'$ under this piecewise uniform continuity assumption. Specifically, for every parameter value $s \in S \setminus \cup_{i=1}^M S_i$, we define the extended parameter-to-solution mapping to be

$$x^*(s) := \left\{ \lim_{s' \in S_i, s' \rightarrow s} x^*(s') : s \in \overline{S_i}, i = 1, \dots, M \right\}.$$

Accordingly, the optimal solution set and optimal parameter set associated with the solution path are defined as

$$\mathcal{X}_S^* := \operatorname{argmin}\{f(x) : H(x) \geq \gamma, x = x^*(s) \text{ for } s \in \cup_{i=1}^M S_i \text{ or } x \in x^*(s) \text{ for some } s \in S \setminus \cup_{i=1}^M S_i\} \quad (\text{EC.56})$$

and

$$S^* := \{s \in \cup_{i=1}^M S_i : x^*(s) \in \mathcal{X}_S^*\} \cup \{s \in S \setminus \cup_{i=1}^M S_i : x^*(s) \cap \mathcal{X}_S^* \neq \emptyset\}.$$

We also assume uniqueness of the optimal solution:

ASSUMPTION EC.9 (Unique optimal solution). *The optimal solution set $\mathcal{X}_S^*$ defined in (EC.56) is a singleton $\{x_S^*\}$.*

Note that in the case of scalar s , uniqueness of the optimal solution is a consequence (Proposition 3) of several more elementary assumptions among which monotonicity of the robust feasible set with respect to the parameter (Assumption 9) plays the key role. However, such notion of monotonicity does not completely carry to the multidimensional case. For example, one may have a formulation $OPT(s)$ such that the robust feasible set satisfies $\text{Sol}(s) \subseteq \text{Sol}(s')$ whenever $s' \leq s$ component-wise, but $\text{Sol}(s)$ and $\text{Sol}(s')$ are in general not comparable.

We also assume the following counterpart of Assumption 10:

ASSUMPTION EC.10. *For every $\epsilon > 0$ there exists an $s \in \cup_{i=1}^M S_i$ such that $H(x^*(s)) > \gamma$ and $\|x^*(s) - x_S^*\|_2 < \epsilon$, where x_S^* is the unique optimal solution from Assumption EC.9.*

We then have the following asymptotic performance guarantees for Algorithm 4:

THEOREM EC.16 (Asymptotic joint feasibility+optimality guarantee). *Suppose Assumptions 2-6 hold for (1). Also suppose that Assumptions EC.8-EC.10 hold for the formulation $OPT(s)$, and that $\{s_1, \dots, s_p\} \subseteq \cup_{i=1}^M S_i$. Denote by $\epsilon_S := \sup_{s \in S} \inf_{j=1, \dots, p} \|s - s_j\|_2$ the mesh size, and by x_S^* be the unique optimal solution from Assumption EC.9. Conditional on Phase one, as Phase two data size $n_2 \rightarrow \infty$, we have for the output of Algorithm 4 that (i) $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} x^*(\hat{s}^*) = x_S^*$ and $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} d(\hat{s}^*, S^*) = 0$ almost surely; and (ii) $\liminf_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) \geq 1 - \beta$ if $H(x_S^*) = \gamma$, and $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0} P_{\xi_{1:n_2}}(H(x^*(\hat{s}^*)) \geq \gamma) = 1$ if $H(x_S^*) > \gamma$.*

Proof of Theorem EC.16. The proof is the same as that of Theorem 5 with straightforward modifications. In particular, $\{\tilde{s}_1, \dots, \tilde{s}_M\}$ shall be replaced by $S \setminus \cup_{i=1}^M S_i$ and the solution set $\mathcal{X}_S$ is now defined as $\mathcal{X}_S := \{x^*(s) : s \in \cup_{i=1}^M S_i\} \cup (\cup_{s \in S \setminus \cup_{i=1}^M S_i} x^*(s))$. $\square$

In order to establish an asymptotically tight feasibility confidence level like in Theorem 6, we further assume uniqueness of the optimal parameter:

ASSUMPTION EC.11 (Unique optimal parameter). *The optimal parameter set S^* is a singleton $\{s^*\}$, and $s^* \in S_{i^*}$ for some $i^* = 1, \dots, M$.*

We then have the following guarantee:

THEOREM EC.17 (Asymptotically tight feasibility guarantee). *In addition to the conditions of Theorem EC.16, suppose Assumption EC.11 holds. Suppose also that the parameter-to-objective mapping $v(s)$ satisfies $v(s) < v(s')$ whenever $s < s'$ component-wise and that $H(x_S^*) = \gamma$. For each $j = 1, \dots, p$, let*

$$\tilde{j} := \arg \min_{j'} \{\|s_j - s_{j'}\|_2 : s_{j'} < s_j \text{ component-wise, } s_{j'} \text{ lies on the same piece as } s_j\}$$

and if there is no such feasible j' simply let $\tilde{j} := j$. If the mesh is such that

$$\max_{j=1, \dots, p} |H(x^*(s_j)) - H(x^*(s_{\tilde{j}}))| = o\left(\frac{1}{\sqrt{n_2}}\right) \quad (\text{EC.57})$$

then we have for the output of Algorithm 4 that $\lim_{n_2 \rightarrow \infty, \epsilon_S \rightarrow 0 \text{ s.t. (EC.57) holds}} P_{\xi_{1:n_2}}(H(x^(\hat{s}^*)) \geq \gamma) = 1 - \beta$.*

Proof of Theorem EC.17. The proof follows exactly that of Theorem 6 with straightforward modifications. For example, when bounding $\hat{H}(x^*(\hat{s}^*)) - z_{1-\beta} \frac{\hat{\sigma}(\hat{s}^*)}{\sqrt{n_2}}$ in the proof of Theorem 6 we replace the parameter value $s_{j^*}^{i^*}$ output by the algorithm with $s_{j^*-1}^{i^*}$ and use the condition (21), whereas now we shall replace the output parameter value $\hat{s}^* = s_{j^*}$ with $s_{\tilde{j}^*}$ and then use (EC.57) to obtain the same bound. $\square$